

\* For each  $N=1$  we have 4 or  $2^2$  supercharges

$Q_\alpha, \bar{Q}_{\dot{\alpha}} \rightarrow Q_1, Q_2, \bar{Q}_{\dot{1}}, \bar{Q}_{\dot{2}}$

For extended susy we have  $2^{N+2}$  supercharges.

$N=2$ : 8 supercharges

- The maximum dimension is  $D=6$ : (1,0) sugra, since  $Q_\alpha$  are symplectic Maj-Weyl w/ one chirality.

## Global Susy: Gauge Mult. for $D=6$ :

- In  $D=6$ ,  $N=2$  theories are similar to  $N=1$  theories in  $D=4$ .

The physical fields are the vect. mult. in the WZ Gauge:

$\left\{ \begin{array}{l} A_\mu^I : \text{Gauge Vect.} \\ \lambda^{iI} : \text{"gaugini"} \text{ in } \mathbb{P} \\ (+ A_{\mu X}) \end{array} \right.$

$\Theta^A \rightarrow \Theta^I$

The extra  $i=1,2$  index reflects that they are symplectic Weyl spinors

Off-shell  $A_\mu$  has  $D-1$  D.O.F. So here it has 5. Spinors in  $D=6$  have  $2^{[D/2]} = 2^3$  D.O.F.

| $D=6$          | off-shell | on shell |
|----------------|-----------|----------|
| $A_\mu$        | <u>5</u>  | <u>4</u> |
| $\lambda^{iI}$ | <u>8</u>  | <u>4</u> |

( $\rightarrow$  for  $\lambda^{iI}$  it has  $2^{[D/2]}$  and on shell

$$D \mapsto D-1 \quad 5 = 2(2) + 2$$

$$\rightarrow 2^{[D/2]} \rightarrow 2^2 = \underline{4} \quad \checkmark$$

$\Rightarrow$  Off-shell we need 5 Bosonic D.O.F. in an Aux field.

- We shall call these Bosonic D.O.F.  $\boxed{\vec{\Psi}^I}$   
 $\vec{\Psi}^I \sim D^4$

Their values after field E.O.M will relate to a Quaternionic structure of isometries of the matter sector induced by the R-symmetry i.e. a gauging of  $\chi T$ .

- For  $N=2$  the R-symmetry group is  $SU(2)_R$ . This agrees w/ the fact that  $\vec{\Psi}^I$  is a triplet in  $SU(2)_R$  i.e. the  $\underline{3}$  of  $SU(2)_R$ .  
 $\square$  Since  $sl(3) \cong sl(2) \oplus A_1 \oplus \mathbb{C}$

- We should expect that the transformation laws can be guessed.

### Aside: Symplectic Majorana Spinors:

- For  $t_1=2$ :  $D=6, 7$  we have symplectic Maj.  
 $\rightarrow$  Even # of spinors  $\chi^i$ ,  $i=1, \dots, 2k$  which satisfy a 'reality condition' containing  $\varepsilon^{ij}$   
 $\varepsilon^{ij} \varepsilon_{jk} = \delta^i_k$   $B = iC\gamma^0$

(Symplectic Maj. condition):  $\boxed{\chi^i = \varepsilon^{ij} (\chi^j)^c = \varepsilon^{ij} B^{\dagger} (\chi^j)^*$

In 5D:  $\boxed{\bar{\Psi}^i \chi_i \equiv \bar{\Psi}^i \chi^j \varepsilon_{ji}}$

this is purely imaginary while  $\bar{\Psi}^i \chi_i \chi_i$  real

Recall that  $(\chi^*)^c = (-1)^{D/2+1} \chi^*$ . With this, in  $D=6$  the symplectic Majorana condition is compatible w/ chirality.

(Symplectic Majorana-Weyl spinors)  $(P_L \chi^i, P_R \chi^i)$

$$P_L \chi^i = \epsilon^{ij} P_L (\chi^j)^c$$

$$\cdot P_L (\chi^i)^c = \epsilon^{ij} P_L \chi^j \Rightarrow P_L \chi^i = \epsilon^{ij} P_L (\chi^j)^c$$

So  $\chi^i$  are chiral and Weyl:

$$P_L \chi^i = \chi^i$$

\* We find, the spinors are symplectic Majorana-Weyl:

The bosonics and susy parameters  $\epsilon^i$ :

$$\lambda^i = P_L \lambda^i$$

$$\bar{\lambda}^i = \bar{\lambda}^i P_R$$

$$\lambda_i := (\lambda^i)^c = \lambda^j \epsilon_{ji}$$

$$\bar{\lambda}_i = -(\lambda_i)^c = \epsilon^{ij} \lambda_j$$

This holds for the susy parameter  $\epsilon^i$  as well:

$$\epsilon^i = P_L \epsilon^i$$

$$\bar{\epsilon}^i = \bar{\epsilon}^i P_R$$

$$\epsilon_i = (\epsilon^i)^c = \epsilon^j \epsilon_{ji}$$

$$\bar{\epsilon}_i = -(\epsilon_i)^c = \epsilon^{ij} \epsilon_j$$

Note for any spinor:  $(\lambda^c)^c = (B^{-1} \lambda^*)^c = (B^{-1})^c \lambda^{*c} = (B^{-1})^c B^{-1} \lambda = B^{-1} (B^*)^T B^{-1} \lambda = B^{-1} (B^{-1})^T \lambda$

Since  $B^* B = -t_1 \mathbb{1} \Rightarrow (B^{-1})^* (B^{-1}) = -t_1 \mathbb{1}$

$$(\lambda^c)^c = -t_1 \lambda = \lambda \text{ in } D=6$$

So this is where comes from.

Aside:  $SU(2)$  triplets:  $A_1: 0 \quad 2$

Take two fundamental (spinor) reps of  $SU(2)$ , the so called doublet.

$$A^i B_i = -A_i B^i$$

Let  $A^{ij}$  be an antisymmetric tensor. Then this

$$\square \Rightarrow \dim(\Theta) = \underline{3} = \frac{3 \cdot 2}{2}$$

Because all fully anti-symm tensors of  $SU(2)$  must be proportional to  $\epsilon^{ij}$

$$A^{ij} = \lambda \epsilon^{ij}$$

Multiplying this by  $\epsilon_{ja}$ .

$$A^i_k = \lambda \delta^i_k$$

Letting  $i=k$ :  $A^k_k = 2\lambda \Rightarrow \lambda = \frac{1}{2} A^k_k$

Thus, any  $\Omega^2$  element can be written as

$$A^{ij} = \frac{1}{2} \epsilon^{ij} A^k_k$$

We'd like to convert the  $\mathfrak{su}(2)$  of  $SU(2)$ :  $\square$  into a matrix rep of  $SU(2)$ . We use the following:

$$\vec{\sigma} = (\sigma^1, \sigma^2, \sigma^3)$$

$$\vec{T}_i := i \vec{\sigma}_i$$

$$\text{so } (T_A)^i_j = (\sigma_A)^i_j$$

Thus due to the fact that  $i\vec{\sigma}$  form a basis of  $\mathfrak{su}(2)$ , which is three dimensional. Thus, we would expect that, as an  $SU(2)$  matrix it is to <sup>be</sup> expanded in the basis

$$Y = Y^I \vec{T}_I$$

In this way, we find

$$\psi = (\psi)_i^j \equiv \psi(\bar{\psi})_i^j = \psi^\alpha (\sigma_1)_\alpha^j + i\psi^\alpha (\sigma_2)_\alpha^j + i\psi^\alpha (\sigma_3)_\alpha^j$$

$$\vec{Y} = -\frac{1}{2} \bar{\psi}_i^j \psi_j^i, \quad \lambda_i^j \psi_j^i = -2 \vec{\lambda} \cdot \vec{\psi}$$

$$\bar{\psi}^{ij} = \epsilon^{ik} \bar{\psi}_k^j = \bar{\psi}^{ji} = (i\sigma_3, -11, -i\sigma_1) = (\bar{\psi}_{ij})^*$$

\* w/ indices at equal "height" they are a basis of 2x2 matrices:  $\psi_{ij}$ : (Real-Symmetric)

$$\psi^{ij} = \bar{\psi}^{ij} \cdot \vec{\psi} = (\psi_{ij})^* \quad \parallel \quad \vec{\psi} = \frac{1}{2} \bar{\psi}^{ij} \psi_{ij} = \frac{1}{2} \bar{\psi}_{ij} \psi^{ij}$$

Back to D=6, N=2: Susy is fixed by the index structure of the fields. Numerical coeffs are found by checking the susy algebra:

$$\delta A_m^I = \frac{1}{2} \bar{\epsilon}^i \gamma_m \lambda_i^I$$

$$\delta \lambda^{iI} = -\frac{1}{4} \gamma^{\mu\nu} F_{\mu\nu}^I \epsilon^i - \psi^{ijI} \epsilon_j$$

$$\delta \psi^{ijI} = -\frac{1}{2} \bar{\epsilon}^{(i} \not{D} \lambda^{j)I}$$

$$\left( \begin{array}{l} D=6, N=2 \\ \text{SUSY} \end{array} \right)$$

$i$ : Susy

$I$ : Gauge

For the global case these realise the form  $[\delta(\epsilon_1), \delta(\epsilon_2)] = \delta_{\text{gauge}}(\xi^\mu)$ ,  $\xi^\mu = \frac{1}{2} \bar{\epsilon}_2^i \gamma^\mu \epsilon_{1i}$

The invariant action is given by:

$$S = \int d^6x \delta_{\text{FS}} \left( -\frac{1}{4} F_{\mu\nu}^I F^{\mu\nu I} - \frac{1}{2} \bar{\lambda}^{iI} \not{D} \lambda_i^I + 2 \bar{\psi}^I \not{\psi}^I \right)$$

Here  $\delta_{\text{FS}}$  is the Cartan-Killing metric for

the Gauge group.

\* For  $D=6$ , we have  $(\gamma_m)_{\alpha\beta} \gamma^m_{rs} = 0$ , this is needed in the non-abelian case.

$$2\vec{\gamma} \cdot \vec{z} = \gamma^i z_i$$

### Gauge Multiplets in $D=5$ :

These are the dim reduction of the  $D=6$  theory  
-> The fermions are again symplectic, however there is no chirality since the spacetime dim is odd.  
The new ingredient is that the Gauge  $A_\mu$  in 6 gives rise to a vect. and a scalar

$M^6 = M^5 \times S^1$ : In the non-Ab. case we

$$\text{have } \partial^\mu F_{\mu\nu} = \partial_{\mu\nu} A_\mu - \partial_\mu \partial_\nu A_\mu = 0$$

Fourier Expand  $A_\mu$  on  $M^5 \times S^1$  so that

$$(x, y) \sim (x, y)$$

$$A_\mu(x, y) = \sum_k e^{iky/L} A_\mu(k)(x)$$

$$A_D(x, y) = \sum_k e^{iky/L} A_D(k)(x)$$

Gauge condition:  $A_D(k) = 0 \quad \forall k \neq 0$

This condition uniquely fixes the Fourier modes  $A_\mu(k)$ ,  $k \neq 0$

$$A_\mu \mapsto A_\mu + \partial_\mu \Phi$$

$$A_D(x, y) = A_{D0}(x) \text{ scalar field.}$$

We define these  $k=0$ -mode scalars to be

$$A_{D0} \equiv \sigma I$$

## Dim. red of Superfields: $6 \rightarrow 5$

$$D=6 \quad t_0 = t_3 = -1$$

$$t_1 = t_2 = 1$$

$$D=5 \quad t_2 = t_3 = -1$$

$$t_0 = t_1 = 1$$

Equations Required:

\* Maj. Dirac relation:  $\bar{\lambda} \gamma_{m_1 \dots m_r} \chi = \text{tr } \bar{\chi} \delta_{m_1 \dots m_r} \lambda$

$$\chi = \Gamma^{(r_1)} \Gamma^{(r_2)} \dots \Gamma^{(r_p)} \lambda$$

$$\bar{\chi} = t_0^p t_{r_1} t_{r_2} \dots t_{r_p} \bar{\lambda} \Gamma^{(r_p)} \dots \Gamma^{(r_1)}$$

$$(\gamma_m)^c = B^{-1} \gamma_m^* B = (-t_0 t_1) \gamma_m$$

$$(\bar{\chi} M \lambda)^c = (\bar{\chi} M \lambda)^c = (-t_0 t_1) \bar{\chi}^c M^c \lambda^c$$

$\rightarrow$  For Left Handed:  $e_6^i \rightarrow e_5^i$   
 Right Handed:  $\gamma^A \rightarrow -i \gamma^A$   
 $\bar{\gamma}^A \rightarrow i \gamma^A$

(i)

This comes from the fact that in 5D  $\gamma^A \chi_A$  is purely imaginary

•  $\gamma^* \lambda^i \rightarrow i \lambda^i$  (Left Handed)  
 $\gamma^* \gamma^A \rightarrow -i \gamma^A$  (Right Handed)

(ii)

$\therefore$  The susy algebra in  $D=5$  is given by

$$\delta A_m^I = \frac{1}{2} \bar{\epsilon}^i \gamma_m \lambda_i^I$$

$$\delta A_6^I = \frac{1}{2} \bar{\epsilon}^i \gamma_6^* \lambda_i^I \stackrel{(ii)}{=} \frac{1}{2} i \bar{\epsilon}^i \lambda_i^I$$

$$\delta \lambda^i{}^I = -\frac{1}{a} \tilde{\gamma}^{mn} \tilde{F}_{mn}^I \epsilon^i - \gamma^{ij}{}^I \epsilon_j$$

$$\gamma^{\tilde{m}\tilde{n}} \tilde{F}_{\tilde{m}\tilde{n}}^I \rightarrow \gamma^{\rho\nu} F_{\rho\nu}^I + \gamma^{\mu\rho} F_{\mu\rho}^I + \gamma^{\mu\nu} F_{\mu\nu}^I$$

$$= 2 \gamma^{\rho\mu} F_{\rho\mu}^I + \gamma^{\mu\nu} F_{\mu\nu}^I$$

$$= 2 \gamma^{\mu\nu} \partial_\mu A_\nu^I - \partial_\mu A_\nu^I + \gamma^{\mu\nu} F_{\mu\nu}^I$$

$$= (\delta^\mu \delta^\nu - \gamma^\mu \gamma^\nu) (-\partial_\mu \delta^\perp) + \gamma^{\mu\nu} F_{\mu\nu}^\perp$$

$$= -2 \gamma^\mu \partial_\mu \delta^\perp + \gamma^{\mu\nu} F_{\mu\nu}^\perp$$

But  $\gamma^\mu \delta^\perp \rightarrow i \epsilon^\mu$

$$-\frac{1}{4} (-2 \partial \delta^\perp + \gamma^{\mu\nu} + \gamma^{\mu\nu} F_{\mu\nu}^\perp) \epsilon^i$$

This is not the whole story. In  $D=6$  we

$$\text{had } [\delta(\epsilon_1), \delta(\epsilon_2)] = \delta_{\text{gct}}(\xi^\mu)$$

$$\text{However } \delta_{\text{gct}}(\xi^\mu) = \delta_{\text{gct}}(\xi) - \delta(\xi^\mu B_\mu)$$

$$\delta_{\text{gct}}(\xi) \phi = \xi^\mu \partial_\mu \phi$$

so

$$[\delta(\epsilon_1), \delta(\epsilon_2)] = \delta_{\text{gct}}(\xi) = \underline{\xi^\mu A_\mu^\perp}$$

$$\rightarrow \xi^0 A_0^\perp + \xi^\mu A_\mu^\perp$$

$$= \frac{1}{2} \epsilon_2^i \gamma^{(i)} \epsilon_{1i} \sigma^I T_I + \xi^\mu A_\mu^\perp$$

$$= \frac{1}{2} i \epsilon_2^i \epsilon_{1i} \sigma^I \quad + \xi^\mu A_\mu^\perp$$

$\stackrel{\text{"}}{-\theta^F}$

$$[\delta(\epsilon_1), \delta(\epsilon_2)]^{(1)} = \delta_{\text{gct}}(\xi) + \delta_\theta(\theta^I)$$

where

$$\delta_\theta(\theta^I) := -\frac{1}{2} i \epsilon_2^i \epsilon_{1i} \sigma^I = -\frac{1}{2} i \epsilon_2^i \epsilon_{1i} \sigma^I \epsilon_{ij}$$

This extra  $\sigma^I$  term is then added to  $\delta \sigma^F \rightarrow \delta \sigma^F$  (Not sure if this is true it's simply the only place I can see this term coming from).

$$\text{Recall } \gamma^{(i)} = \frac{1}{2} \bar{\epsilon}^{(i)} \cdot \bar{\gamma}, \quad \bar{\gamma} = \frac{1}{2} \bar{\epsilon}_{ij} \gamma^{(j)}$$

$$\delta \gamma^{(ij)I} = -\frac{1}{2} \bar{\epsilon}^{(i)} \delta \lambda^{(j)I}$$

$$= -\frac{1}{2} \bar{\epsilon}^{(i)} \gamma^{(i)} (\partial_{(i)} - A_{(i)}) \lambda^{(j)I}$$

$$= -\frac{1}{2} \bar{\epsilon}^{(i)} \delta \lambda^{(j)I} - \frac{1}{2} \bar{\epsilon}^{(i)} \gamma_{(i)}^{(i)} (A_{(i)} \lambda^{(j)I})$$

Here we have  $A$  in the adjoint rep of the Gauge group so

$$A^I T_I \lambda = A^{\pm} (f_{\pm})^J{}_K \lambda^{J|K}$$

$$\rightarrow -\frac{1}{2} \bar{\epsilon}^{ci} \sigma^J f_{JK}^I \lambda^{j|K}$$

Note the minus sign is from

$$(T_A \phi)^i{}_j \equiv - (t_A)^i{}_j \phi^j$$

$$\delta Y^{i|J} = -\frac{1}{2} (\bar{\epsilon}^{ci} \not{D} \lambda^{j|K})^I + \epsilon^i \sigma^J f_{JK}^{\pm} \lambda^{j|K}$$

$$\begin{aligned} \delta \vec{\psi}^I &= -\frac{1}{4} \vec{\epsilon}_{ij} \bar{\epsilon}^i [\not{D} \lambda^{j|I} + i f_{JK}^{\pm} \sigma^J \lambda^{j|K}] \\ &= -\frac{1}{4} \vec{\epsilon}_{ij} \bar{\epsilon}^i [\not{D} \delta_K^I + i f_{JK}^{\pm} \sigma^J] \lambda^{j|K} \end{aligned}$$

Putting this all together:

$$\begin{aligned} \delta A_\mu^I &= \frac{1}{2} \bar{\epsilon} \gamma_\mu \lambda_i^I & \delta \lambda^{i|I} &= -\frac{1}{4} \gamma^{\mu\nu} F_{\mu\nu}^I \epsilon^i - \frac{1}{2} \not{D} \sigma^I \epsilon^i - \gamma^{i|J} \epsilon_J \\ \delta \sigma^I &= \frac{1}{2} i \bar{\epsilon}^i \lambda_i^I & \delta \vec{\psi} &= -\frac{1}{4} \vec{\epsilon}_{ij} \bar{\epsilon}^i (\not{D} \delta_K^I + i f_{JK}^{\pm} \sigma^J) \lambda^{j|K} \end{aligned}$$

$$[\delta(G_1), \delta(G_2)] = \delta_{\text{SUSY}}(\vec{Z}) + \delta_O(\Theta^I)$$

$$\Theta^I \equiv -\frac{1}{2} \sigma^I i \bar{\epsilon}_2^i \epsilon_{1i} = -\frac{1}{2} i \sigma^I \bar{\epsilon}_2^i \epsilon_{1i}^{\bar{2}} \epsilon_{ji}$$

### 5D-Central Charge:

We note that the  $\delta_O(\Theta^I)$  appears to be a central charge term. For general SUSY:

$$[\delta(G_1), \delta(G_2)] = \frac{1}{2} (\bar{\epsilon}_2 \gamma^\mu \epsilon_1 + \bar{\epsilon}_1 \gamma^\mu \epsilon_2) P_\mu + \frac{1}{2} \bar{\epsilon}_2^i \epsilon_1^j \epsilon_{ij} \vec{Z} + \frac{1}{2} \bar{\epsilon}_2 \epsilon_1 \epsilon_{ij} \epsilon^{ij} \vec{Z}$$

$$\therefore \vec{Z} = i \sigma^I T_I$$

Well,  $\sigma^I T_I$  is not known, a priori, to be 'central'

in that it commutes with the rest of the algebra.

→ For field Equations solns we find  $\sigma^I \in \mathbb{R}$  and hence it is, for  $D=5$  sugra, a central charge.

• We find that a theory w/ no central charges in  $D=6$  gives rise to a theory w/ central charges added to the susy.

Claim: No such central charge could exist for a 6D theory.

proof: Assume  $\exists$  central charges in 6D, then

$$\sum \bar{E}_i^i E_j^j \epsilon_{ji}$$

$$\bar{E}_{(1)}^\alpha E_{(1)}^\alpha = \bar{E}_{(1)}^\alpha E^{(1)\beta} C_{\beta\alpha}, \quad C^T = C = \text{sym}$$

so  $\epsilon_{ij}$  cannot be used to sum over  $\bar{E}^i E_j$ .

\* We present the Chern-Simons terms which will play an important role:

$$A \wedge F \wedge F$$

In 5D we can construct this using  $A_\mu$  as a 1-form  $\Rightarrow A^{(1)} \wedge F^{(2)} \wedge F^{(2)} = 5\text{-form}$

Because  $A^I, A^J, F^K$  are symmetric in  $ISU$  we introduce the CS term w/ fully symmetric tensor  $C_{ISU}$

$$C_{ISU} A^I \wedge F^J \wedge F^K$$

$$\rightarrow \text{Tr}(A \wedge F \wedge F) = A^I \wedge F^J \wedge F^K \text{Tr}(\bar{\psi}_I \gamma_5 \psi_J)$$

Require  $\boxed{f_{ICS}^m (C_{KL})_m = 0}$

Tensor Multiplets: These occur w/ Funn fields  
In 5D -  $A_m$ :  $F_{\mu\nu} \rightarrow *F^{(2)} = G_{\mu\nu\lambda}^{(3)} = dT_{\mu\nu}^{(2)}$   
So we also have 2-forms  $T_{\mu\nu}$ . These occur in the 5D theories.

$\rightarrow$  In 4D  $*T_{\mu\nu}$  is dual to a scalar so we can ignore these terms.

Thus a tensor mult. contains higher order tensors dual to the Gauge fields.

$A_m$ : Adjoint  $\square$

$T_{[mn]}$ :  $\square$

The Complete Lagrangian is:  $2(\bar{\psi}^I \psi^J) \gamma_{JK}$

$$\mathcal{L}_{5D} = C_{IJK} \left[ -\frac{1}{4} F_{\mu\nu}^I F^{\mu\nu J} - \frac{1}{2} \bar{\lambda}^I \not{D} \lambda^J - \frac{1}{2} D_\mu \sigma^I D^\mu \sigma^J + \right. \\ \left. -\frac{1}{24} \epsilon^{\mu\nu\rho\sigma} A_\mu^I \left[ F_{\nu\rho}^J F_{\sigma\tau}^K + d_{LM}^J A_\nu^L A_\rho^M \left( -\frac{1}{2} F_{\sigma\tau}^K + \frac{1}{10} f_{NP}^K A_\sigma^N A_\tau^P \right) \right] \right] \\ \left. -\frac{1}{8} i \bar{\lambda}^I \gamma^{\mu\nu} F_{\mu\nu}^J \lambda^K - \frac{1}{2} i \bar{\lambda}^I \gamma_5 \not{D} \lambda^J + \frac{1}{4} i \sigma^I \sigma^J \lambda^L \lambda^M f_{LM}^K \right]$$

Let us expand the CS-term:

$$A^{(1)} \wedge F^{(1)} \wedge F^{(1)}$$

$$C_{IJK} A_m^I F_{\nu\sigma}^J F_{\rho\lambda}^K \epsilon^{\mu\nu\sigma\rho\lambda}$$

$$C_{IJK} (A_\mu^I F_{\nu\sigma}^J F_{\rho\lambda}^K \epsilon^{\mu\nu\sigma\rho\lambda} + A_\mu^I F_{\nu\sigma}^J F_{\rho\lambda}^K \epsilon^{\mu\nu\sigma\rho\lambda})$$

$$C_{IJK} (\sigma^I F_{\nu\sigma}^J F_{\rho\lambda}^K \epsilon^{\mu\nu\sigma\rho\lambda} + A_\mu^I F_{\nu\sigma}^J F_{\rho\lambda}^K \epsilon^{\mu\nu\sigma\rho\lambda})$$

$$\begin{aligned}
 & -\triangleright (\partial_\mu A_\sigma^\tau - \partial_\sigma A_\mu^\tau) \epsilon^{\nu\sigma\rho\lambda} F_{\rho\lambda}^\tau \\
 & \tilde{w} : (-\partial_\sigma \sigma^\tau \epsilon^{\sigma\rho\lambda} \sigma^\tau F_{\rho\lambda}^\tau) + F_{\nu\sigma}^\tau \sigma^\tau F_{\rho\lambda}^\tau \epsilon^{\nu\sigma\rho\lambda} \\
 & = -\partial_\sigma \sigma^\tau \epsilon^{\sigma\rho\lambda} \sigma^\tau F_{\rho\lambda}^\tau + (\partial_\nu A_\sigma^\tau - \partial_\sigma A_\nu^\tau) \sigma^\tau F_{\rho\lambda}^\tau \epsilon^{\nu\sigma\rho\lambda} \\
 & = (-\partial_\sigma \sigma^\tau) \sigma^\tau (\epsilon^{\sigma\rho\lambda} F_{\rho\lambda}^\tau + \epsilon^{\sigma\rho\lambda} F_{\rho\lambda}^\tau) \\
 & + (\partial_\nu \sigma^\tau) \sigma^\tau F_{\rho\lambda}^\tau \epsilon^{\nu\sigma\rho\lambda} + F_{\nu\sigma}^\tau \sigma^\tau F_{\rho\lambda}^\tau \epsilon^{\nu\sigma\rho\lambda}
 \end{aligned}$$

$$\begin{aligned}
 & = [-\partial_\sigma \sigma^\tau] \sigma^\tau \epsilon^{\sigma\rho\lambda} (-\partial_\lambda \sigma^\mu) + (-\partial_\sigma \sigma^\tau) \sigma^\tau \epsilon^{\sigma\rho\lambda} F_{\rho\lambda}^\tau \\
 & + \partial_\nu \sigma^\tau \sigma^\tau F_{\rho\lambda}^\tau \epsilon^{\nu\sigma\rho\lambda} + F_{\nu\sigma}^\tau \sigma^\tau F_{\rho\lambda}^\tau \epsilon^{\nu\sigma\rho\lambda}
 \end{aligned}$$

$$\begin{aligned}
 & = \cancel{\partial_\sigma \sigma^\tau \sigma^\tau \epsilon^{\sigma\rho\lambda}} \partial_\lambda \sigma^\mu + (-\cancel{\partial_\sigma \sigma^\tau} \sigma^\tau \epsilon^{\sigma\rho\lambda}) (\partial_\rho \sigma^\mu) \\
 & + \partial_\sigma \sigma^\tau \sigma^\tau \epsilon^{\sigma\rho\lambda} F_{\rho\lambda}^\tau + \partial_\nu \sigma^\tau \sigma^\tau F_{\rho\lambda}^\tau \epsilon^{\nu\sigma\rho\lambda} \\
 & + F_{\nu\sigma}^\tau \sigma^\tau F_{\rho\lambda}^\tau \epsilon^{\nu\sigma\rho\lambda}
 \end{aligned}$$

$$\begin{aligned}
 & = \partial_\sigma \sigma^\tau \sigma^\tau \epsilon^{\sigma\rho\lambda} F_{\rho\lambda}^\tau + \partial_\nu \sigma^\tau \sigma^\tau (-\cancel{\partial_\lambda \sigma^\mu}) \epsilon^{\nu\sigma\rho\lambda} \\
 & + \partial_\nu \sigma^\tau \sigma^\tau F_{\rho\lambda}^\tau \epsilon^{\nu\sigma\rho\lambda} + F_{\nu\sigma}^\tau \sigma^\tau F_{\rho\lambda}^\tau \epsilon^{\nu\sigma\rho\lambda}
 \end{aligned}$$

$$\begin{aligned}
 & = \partial_\sigma \sigma^\tau \sigma^\tau \epsilon^{\sigma\rho\lambda} F_{\rho\lambda}^\tau + \partial_\nu \sigma^\tau \sigma^\tau \epsilon^{\nu\sigma\rho\lambda} (F_{\rho\lambda}^\tau + F_{\rho\lambda}^\tau) \\
 & + F_{\nu\sigma}^\tau \sigma^\tau (F_{\rho\lambda}^\tau \epsilon^{\nu\sigma\rho\lambda} + F_{\rho\lambda}^\tau \epsilon^{\nu\sigma\rho\lambda})
 \end{aligned}$$

$$\begin{aligned}
 & = \partial_\nu \sigma^\tau \sigma^\tau \epsilon^{\sigma\rho\lambda} F_{\rho\lambda}^\tau + \cancel{\partial_\nu \sigma^\tau \sigma^\tau} + \partial_\nu \sigma^\tau \sigma^\tau \epsilon^{\nu\sigma\rho\lambda} F_{\rho\lambda}^\tau \\
 & + F_{\nu\sigma}^\tau \sigma^\tau (-\partial_\lambda \sigma^\mu) \epsilon^{\nu\sigma\rho\lambda} + F_{\nu\sigma}^\tau \sigma^\tau F_{\rho\lambda}^\tau \epsilon^{\nu\sigma\rho\lambda}
 \end{aligned}$$

$$\begin{aligned}
 & = 2 \partial_\sigma \sigma^\tau \sigma^\tau \epsilon^{\sigma\rho\lambda} F_{\rho\lambda}^\tau - F_{\nu\sigma}^\tau \sigma^\tau \partial_\lambda \sigma^\mu \epsilon^{\nu\sigma\rho\lambda} \\
 & + F_{\nu\sigma}^\tau \sigma^\tau \partial_\rho \sigma^\mu \epsilon^{\nu\sigma\rho\lambda} + F_{\nu\sigma}^\tau \sigma^\tau F_{\rho\lambda}^\tau \epsilon^{\nu\sigma\rho\lambda} \\
 & = 2 \partial_\sigma \sigma^\tau \sigma^\tau \epsilon^{\sigma\rho\lambda} F_{\rho\lambda}^\tau - 2 F_{\nu\sigma}^\tau \sigma^\tau \partial_\rho \sigma^\mu \epsilon^{\nu\sigma\rho\lambda} \\
 & + F_{\nu\sigma}^\tau \sigma^\tau F_{\rho\lambda}^\tau \epsilon^{\nu\sigma\rho\lambda}
 \end{aligned}$$

Multiplying this by  $C_{\tau\mu\kappa}$  the first two terms cancel, leaving

$$C_{\tau\mu\kappa} F_{\nu\sigma}^\tau \sigma^\tau F_{\rho\lambda}^\tau \epsilon^{\nu\sigma\rho\lambda}$$

Err... probably not relevant.....

\* The above Lagrangian is superconf. invariant.

$A_\mu$  have zero conf. weight and  $\sigma^I$  has conf weight  $\frac{1}{2}$  gaug

$$A_\mu: \quad h=0$$

$$\sigma^I: \quad h=1$$

$$\lambda^{iI}: \quad h=3/2$$

$$\vec{\psi}^I: \quad h=2$$

$$S\text{-susy: } \epsilon^i \mapsto \epsilon^i + i\chi^\mu \gamma_\mu \eta^i$$

$$\delta_{\text{extra}} \lambda^{iI} = \sigma^I \eta^i \quad | \quad \delta_{\text{extra}} \vec{\psi}^I = \frac{1}{4} i \vec{\epsilon}_{ij} \eta^j \lambda^{iI}$$

• Also an extra  $SU(2)_R$  symm:  $\lambda^i$  and  $\vec{\psi}^I$  in adjoint.

• To see the CS-term is Gauge invariant:

$$A^{(1)} \rightarrow A^{(1)} + d\theta$$

$$dA^{(1)} \rightarrow d(A^{(1)} + d\theta) = dA^{(1)} - F^{(2)}$$

$\therefore$

$$\begin{aligned} S(A \wedge F \wedge F) &= \delta A \wedge F \wedge F - A \wedge (\delta F) \wedge F \\ &\quad - A \wedge F \wedge \delta F \end{aligned}$$

$$\begin{aligned} &= \int \delta A \wedge F \wedge F = \int d\theta \wedge F \wedge F \\ &= \int d(\theta F \wedge F) + 0. \end{aligned}$$

$$\begin{aligned} \bullet \quad -\frac{1}{2} \lambda^{iI} \gamma^\mu D_\mu \lambda_i^J &= -\frac{1}{2} \lambda^{iI} i D_0 \lambda_i^J + \frac{1}{2} \lambda^{iI} \vec{p} \lambda_i^J \\ &\quad + \frac{1}{2} \lambda^{iI} \sigma^K F_{KL} \lambda_i^L - \frac{1}{2} \lambda^{iI} \not{D} \lambda_i^J \end{aligned}$$

Reduction:  $D=5 \rightarrow D=4$ :



$$\delta A_{(u)}^I = \frac{1}{2} \bar{E}^i \gamma_* (-\Omega^{ij} \xi_{ji} + \Omega_i^I) = \frac{1}{2} \bar{E}^i (\Omega^i \xi_{ji} + \Omega_i^I)$$

$$\delta \sigma^I = \frac{1}{2} i \bar{E}^i (-\Omega^{ij} \xi_{ji} + \Omega_i^I)$$

$$\rightarrow \delta \sigma^I = \frac{1}{2} \bar{E}^i (-\Omega^{ij} \xi_{ji} + \Omega_i^I)$$

$$\therefore \frac{1}{2} (\delta A_{(u)}^I - i \delta \sigma^I) = \frac{1}{2} \bar{E}^i \Omega_i^I$$

$$\text{Since } \chi^I = \frac{1}{2} (A_{(u)}^I - i \sigma^I)$$

$$\boxed{\delta \chi^I = \frac{1}{2} \bar{E}^i \Omega_i^I}$$

(ii) The remaining terms for  $A_m^I$ :

$$\delta A_m^I = \frac{1}{2} \bar{E}^i \gamma_m (-\Omega^{ij} \xi_{ji} + \Omega_i^I)$$

$$= -\frac{1}{2} \bar{E}^i \gamma_m \Omega^{ij} \xi_{ji} + \frac{1}{2} \bar{E}^i \gamma_m \Omega_i^I$$

$$\text{should be } \frac{1}{2} \xi^{ij} \bar{E}_i \gamma_m \Omega_j^I + \text{h.c.}$$

not sure....

$$(iii) \delta \lambda^{iI} = -\frac{1}{4} \gamma^{\mu\nu} F_{\mu\nu}^I \bar{E}^i - \frac{i}{2} \not{D} \sigma^I \bar{E}^i - \gamma^{iiI} \bar{E}_i$$

$$\delta (-\Omega^{ij} + \xi^{ij} \Omega_j^I)$$

$$\text{RHS: } -\frac{1}{4} \gamma^{\mu\nu} F_{\mu\nu}^I \bar{E}^i - \frac{1}{4} \gamma^{\mu\nu} F_{\mu\nu}^I \bar{E}^i$$

$$\rightarrow -\frac{1}{4} (\gamma_* \gamma^{\mu\nu} - \gamma^{\mu\nu} \gamma_*) (\partial_\mu A_\nu^I - \partial_\nu A_\mu^I) \bar{E}^i - \frac{1}{4} \gamma^{\mu\nu} F_{\mu\nu}^I \bar{E}^i$$

$$= \frac{1}{2} \gamma^\mu (-\partial_\mu A_{(u)}^I) \bar{E}^i - \frac{1}{4} \gamma^{\mu\nu} F_{\mu\nu}^I \bar{E}^i$$

$$\rightarrow -\frac{i}{2} \gamma^\mu (\not{D}_\mu \sigma^I - f_{KL}^I A_\mu^K \sigma^L) \bar{E}^i$$

$$= -\frac{1}{2} [-f_{KL}^I A_{(u)}^K \sigma^L] \bar{E}^i - \frac{i}{2} \not{D} \sigma^I \bar{E}^i$$

$$= \frac{i}{2} f_{KL}^I A_{(u)}^K \sigma^L \bar{E}^i - \frac{i}{2} \not{D} \sigma^I \bar{E}^i$$

$$\bullet + \bullet = \left( +\frac{1}{2} \not{D} A_{(u)}^I - \frac{i}{2} \not{D} \sigma^I \bar{E}^i - \frac{1}{4} \gamma^{\mu\nu} F_{\mu\nu}^I \bar{E}^i \right. \\ \left. + \frac{i}{2} f_{KL}^I A_{(u)}^K \sigma^L \bar{E}^i \right)$$

$$= \frac{1}{2} (\not{D} (A_{(u)}^I - i \sigma^I) \bar{E}^i) + \frac{i}{2} f_{KL}^I A_{(u)}^K \sigma^L \bar{E}^i \\ + \frac{i}{2} f_{KL}^I A_{(u)}^K \sigma^L \bar{E}^i - \frac{1}{4} \gamma^{\mu\nu} F_{\mu\nu}^I \bar{E}^i$$

$$\begin{aligned}
&= \cancel{D} X^I \epsilon^i + f_{KL}^I \left( \frac{1}{2} A_{(M)}^K - i \sigma^M \right) A_M^K + \frac{i}{2} f_{KL}^I A_M^K \sigma^L \epsilon^i \\
&\quad + \frac{i}{2} f_{KL}^I A_{(M)}^K \sigma^L \epsilon^i \\
&= \cancel{D} X^I \epsilon^i + f_{KL}^I X^L \cancel{A}^K + \frac{i}{2} f_{KL}^I \cancel{A}^K \frac{(X^L - \bar{X}^L)}{i} (L-1) \epsilon^i \\
&\quad + \frac{i}{2} f_{KL}^I (X^K + \bar{X}^K) (X^L - \bar{X}^L) / i \epsilon^i \\
&= \cancel{D} X^I \epsilon^i + f_{KL}^I X^L \cancel{A}^K + \frac{1}{2} f_{KL}^I \cancel{A}^K (X^L - \bar{X}^L) \epsilon^i \\
&\quad + \frac{1}{2} f_{KL}^I (X^K + \bar{X}^K) (X^L - \bar{X}^L) \epsilon^i
\end{aligned}$$

Not sure about the decomposition of the LHS:  
 $\delta(-\Omega^i + \epsilon^{ij} \Omega_j)$  However we take  $\delta \Omega_i$   
 multiply by  $\epsilon_{ik}$  and Ignore  $\cancel{A}$  terms  
 as well as  $X^K X^L$  terms.

• Perhaps we project with  $P_L$  on LHS  
 $P_L \{ \delta(-\Omega^i + \epsilon^{ij} \Omega_j) \} = \delta \epsilon^{ij} \Omega_j$

RHS: Taking account what we have just ignored:

$$\begin{aligned}
P_L \epsilon_{ki} \left( \cancel{D} X^I \epsilon^i - \frac{1}{4} \gamma^{\mu\nu} F_{\mu\nu} \epsilon^i - Y_{ij}^I \epsilon_j + X^S \bar{X}^U f_{\mu\nu}^S \epsilon^i \right) \\
\epsilon_{\mu\nu} \epsilon^{\mu\nu} = -\delta_{ij}
\end{aligned}$$

• Putting this all together we have:

5D-Susy:

$$\delta X^I = \frac{1}{2} \bar{\epsilon}^i \Omega_i$$

$$\delta \Omega_i^I = \cancel{D} X^I \epsilon_i + \frac{1}{4} \gamma^{\mu\nu} F_{\mu\nu} \epsilon_{ij} \epsilon^j + Y_{ij}^I \epsilon^j + X^S \bar{X}^U f_{\mu\nu}^S \epsilon_{ij} \epsilon^j$$

$$\delta A_\mu^I = \frac{1}{2} \epsilon^{ij} \bar{\epsilon}_j \gamma_\mu \Omega_i^I + \text{h.c.}$$

$$\delta \bar{\psi}^I = \frac{1}{4} \bar{\epsilon}^{ij} \bar{\epsilon}_j \cancel{D} \Omega_i^I - \frac{1}{2} f_{\mu\nu}^I \bar{\epsilon}^{ij} \bar{\epsilon}_j X^S \Omega_i^{\mu} + \text{h.c.}$$

The algebra:

$$[\delta \epsilon_1, \delta \epsilon_2] = \delta_{\text{eq}+}(\bar{\epsilon}_1 \epsilon_2) + \delta_0(\theta^T)$$

$$\xi^m = \frac{1}{2} \bar{E}_2^i \gamma^m E_{2i} + \text{h.c.} \quad | \mathcal{Q}^F = X^I \xi^{ij} \bar{E}_{2i} E_{1j} + \bar{X}^I \xi_{ij} \bar{E}_2^i E_1^j$$

## Proof of Central Charges:

$$(i) [\mathcal{S}_1, \mathcal{S}_2] X^I = \mathcal{S}_2 \left( \frac{1}{2} \bar{E}_2^i \Omega_{ij}^I \right) - \mathcal{S}_1 \left( \frac{1}{2} \bar{E}_1^j \Omega_{ij}^I \right)$$

$$\begin{aligned} & \frac{1}{2} \bar{E}_2^i \left( \gamma^I X^J E_{ij} + \frac{1}{4} \gamma^{\mu\nu} F_{\mu\nu}^I \xi_{ij} E_{(i}^j + \gamma_{ij}^J E_{(i}^j + X^J \bar{X}^K f_{JK}^I E_{ij} E_{(i}^j \right) \\ &= \frac{1}{2} \bar{E}_2^i \gamma^I X^J E_{ij} + \frac{1}{4} \bar{E}_{(i}^j \gamma^{\mu\nu} F_{\mu\nu}^I \xi_{j) i} E_{(i}^j + X^J \bar{X}^K f_{JK}^I \bar{E}_2^i \xi_{ij} E_{(i}^j \\ & \quad - (2 \leftrightarrow 1) \end{aligned}$$

$$= \frac{1}{4} \left( - \bar{E}_{(i}^j \gamma^{\mu\nu} F_{\mu\nu}^I E_{j) i} \xi_{(i}^j - \bar{E}_{(i}^j \gamma^{\mu\nu} F_{\mu\nu}^I E_{(i}^j \xi_{j) i} \right) \\ = 0 \quad \text{since } \bar{E}_{(i}^j \xi_{j) i} \gamma^{\mu\nu} F_{\mu\nu}^I E_{(i}^j = + \text{ and } -$$

$$\mathcal{S}_0 X^I = X^J \bar{X}^K f_{JK}^I \bar{E}_{(2)}^i \xi_{ij} E_{(1)}^j \quad - \leftrightarrow$$

✓ so this checks out.

(ii) Similar for  $\mathcal{S} A_m^I$  except  $\bar{E}_{(2)}^i \rightarrow \bar{E}_{(2)}^i \gamma_m$

\* The susy of  $X^I$  is chiral:  $E_i = \text{Pr} E_i$  does not appear,

→ For  $N=2$ , a chiral Mult. is reducible i.e.

the susy algebra closes on a subset of the fields of the Chiral Mult.

$N=2$ : full chiral (16 + 26)

↓ irrep  
(8 + 8)

\* The Gauged WZ vect. mult. is an irrep of the susy alg. While the Vector mult. is not in an irrep of  $Sp(p, q | N) / \sim$

## Construction of $N=2, D=4$ Action:

\* We first work w/  $N=2$  rigid susy:

The  $N=2$  vector multiplet

$$V = (\lambda_\alpha, A_\mu, D) \oplus \mathbb{E}(\phi, \psi_\alpha, F)$$

$\downarrow$  ( $\underline{N=2}$  WZ) ( $\underline{N=2}$  Chiral)  
 $W_\alpha$

This is a SYM theory w/ gauge group  $G$ , where we impose constraints on.

(i)  $\mathbb{E}$  should transform in the adjoint

$(\lambda_\alpha, \psi_\alpha)$ : transform in  $\underline{2}$  of  $SU(2)_R$

(ii)  $(Q^1_\alpha, Q^2_\beta)$  transform in the fundamental of  $SU(2)_L$

The  $N=2$  rigid case:

$$\begin{aligned} \mathcal{L}_{SYM}^{N=2} &= \frac{1}{32\pi} \int d^2\theta \text{Tr}(W^\alpha W_\alpha) + \int d^2\theta d^2\bar{\theta} \text{Tr}(\bar{\mathbb{E}} e^{gV} \mathbb{E}) \\ &= \text{Tr} \left( -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} - i \lambda^\mu D_\mu \bar{\lambda} - i \psi \sigma^\mu D_\mu \bar{\psi} + \bar{D}_\mu \mathbb{E} D^\mu \mathbb{E} \right. \\ &\quad \left. + \frac{g^2}{32\pi^2} F_{\mu\nu} \tilde{F}^{\mu\nu} + \frac{1}{2} D^2 + \bar{F} F + i\sqrt{2} g \bar{\phi} \{ \lambda, \psi \} \right. \\ &\quad \left. - i\sqrt{2} g \{ \bar{\phi}, \bar{\lambda} \} \psi + g D[\phi, \bar{\phi}] \right) \end{aligned}$$

$$\phi = \phi^A T_A, \quad \psi_\alpha = \psi_\alpha^A T_A \quad \text{etc.} \dots A=1, \dots, \dim(G)$$

$\Rightarrow$  All fields are in the adjoint rep.  $\square$

\* For order for there to be an  $SU(2)_R$  symm there must not be a superpotential  $W(\mathbb{E})$  as this would give  $\phi$  interactions/mass terms which are absent for  $\lambda$ . Instead our well known fermionic shift potential occurs for D-terms  $V$

$N=2$  Nonlinear Sigma models:

\* Relaxing renormalizability, start w/ pure SYM

as above:

(i) Take  $N=1$   $\sigma$ -model Lagrangian

(ii) Set superpotential  $W(\Phi)$  to zero.

(iii) Take into Account  $\Phi$  transforms in the Adj. of  $G$ .

→ In this way, we would expect the Kähler potential to be related to the chiral superfield  $F_{\pm 5}(\Phi)$  Adj.  $\otimes$  Adj. The  $N=2$   $\sigma$ -model, w/  $W=0$

$$\mathcal{L}^{N=1} = \left( \frac{1}{32\pi} \right) \text{Im} \left( \int d^2\theta F_{\pm 5}(\Phi) W^{\pm\alpha} W_{\alpha} \right) + \int d^4\theta K(\Phi^i, (\Phi^{\dagger})^{\dagger j})$$

Now, we would like to relate  $K_{\pm 5}$  to  $F_{\pm 5}$ . This is done by introducing a prepotential which is a holomorphic function  $F(\Phi)$

$$F_{\pm 5} = \partial^2 F(\Phi) / (\partial \Phi^{\pm} \partial \Phi^{\mp})$$

$$K(\Phi, \bar{\Phi}) = -\frac{i}{32\pi} \bar{\Phi}_A \frac{\partial^2}{\partial \Phi_A} + \frac{i}{32\pi} \overline{F_A(\Phi)} \Phi^A$$

This is a nontrivial statement! (Must look up Seiberg-Witten theory for full explanation).

→  $N=2$   $\sigma$ -model is uniquely determined by a single holomorphic function  $F$ .

Plugging this back into the above we find:

$$\mathcal{L}^{N=1} = \frac{1}{32\pi} \text{Im} \left( \int d^2\theta F_{\pm 5}(\Phi) W^{\pm\alpha} W_{\alpha} \right)$$

$$= \frac{1}{32\pi} \int d^4\theta (\bar{\Phi} e^{2gV})^T F_I + \frac{1}{32\pi} \int d^4\theta \bar{\Phi}^T (\bar{F}_I(\Phi))$$

Using the relationship between  $F$  and  $K$  above

-D Lower index: Anti-chiral:  $\bar{\Phi}^T$

Upper: Chiral  $\bar{\Phi}$

$$\frac{\partial}{\partial \Phi^I} K = -\frac{i}{32\pi} \bar{\Phi}_K F^K(\Phi) + \frac{i}{32\pi} \bar{F}_K(\Phi) \bar{\Phi}^K$$

$$\frac{\partial^2 K}{\partial \Phi^I \partial \bar{\Phi}^J} = \left( \frac{i}{32\pi} \right) \left( -\bar{\Phi}_K F_I^K(\Phi) + \bar{F}_I(\Phi) \bar{\Phi}^K \right)$$

$$\frac{\partial^2 K}{\partial \bar{\Phi}^J \partial \Phi^I} = \left( \frac{i}{32\pi} \right) \left( -F_I^J(\Phi) + F_I^J(\Phi) \right)$$

$$= \left( \frac{-i}{32\pi} \right) \left( \frac{\partial^2 F(\Phi)}{\partial \Phi^I \partial \bar{\Phi}^J} - \frac{\partial^2 \bar{F}}{\partial \bar{\Phi}^I \partial \Phi^J} \right)$$

Lowering  $K_I^J$  to  $K_{IJ}$  we find LHS

$$\text{LHS} = \frac{1}{16\pi} \text{Im}(F_{AB}(\Phi))$$

Back to Super: Here the pre potential  $F$  is not constrained.  $F(\Phi)$  has  $(16+16)$  components

$$\begin{aligned} \mathcal{L}_{\text{un}} = & i F_{IJ} D_\mu X^I D^\mu \bar{X}^J + \frac{1}{4} i K_{IJ} F_{\mu\nu}^{-I} (F^{-J})^{\mu\nu} + \frac{1}{2} i F_{IJ} \bar{\Omega}_i^I \not{D} \Omega^J \\ & - i K_{IJ} \bar{\Psi}^I \not{D} \Psi^J + \frac{1}{4} i K_{IJ} R \bar{\Psi}^I \not{D} \Psi^J + \frac{1}{2} i \bar{\Omega}_i^I \not{D} \Omega^J \\ & - \frac{1}{16} i F_{IJK} \epsilon^{ij} \bar{\Omega}_i^I \gamma^{\mu\nu} F_{\mu\nu}^{-J} \Omega_j^K + \frac{i}{48} F_{IJK} \bar{\Omega}_i^I \not{D}_\mu \bar{\Omega}_j^J \Omega_k^K \epsilon^{ij\mu\kappa} \\ & + \frac{1}{2} i F_{IJK} \bar{\Omega}^{iJ} \Omega^{iK} \epsilon_{ij} - \frac{1}{2} i F_{IJK} \bar{X}^K \bar{\Omega}_i^J \Omega_j^I \epsilon^{ij} \\ & - i F_I F_{JK} f_{\mu\nu} \bar{X}^K \bar{X}^L X^\mu + \text{h.c.} \end{aligned}$$

$$F_{\mu\nu}^{-I} := \frac{1}{2} (F_{\mu\nu}^I + \tilde{F}_{\mu\nu}^I) \quad \tilde{F}_{\mu\nu} = -i(F)_{\mu\nu} = -\frac{i}{2} \epsilon_{\mu\nu\alpha\beta} F^{\alpha\beta}$$

\* The scalar terms form a Kähler manifold as we discussed above however now the Relation is based on scalars, and not the Superfields

$$K(X, \bar{X}) = i X^I \bar{F}_I - i \bar{X}^I F_I$$

$$\square K \sim -i F_{IJ} \partial_m X^I \partial^m \bar{X}^J$$

$$D_m X^I = \partial_m X^I - A_m^K X^J f_{JK}^I$$

$$D_m \Omega_i^I = \partial_m \Omega_i^I - A_m^K \Omega_i^J f_{JK}^I$$

$$F_{mn}^{IJ} = 2 \partial_{[m} A_{n]}^{IJ} + A_m^K A_n^J f_{JK}^{IL}$$

A:  $Sp(2n, \mathbb{R})$ : fundamental rep.

$$S_G(\theta) q^K = \theta^T K_X^K(q)$$

This Kähler potential is invariant under the Hodge  
Kähler transformations (antisymm/real)

$$F = A_{IJ} X^I X^J$$

$$K \rightarrow K + F + \bar{F}$$

These terms vanish since there's a derivative  
in  $X^I$  and  $\bar{X}^J$ .

→ Need a better way to keep track of  $I, J$ , why  
does the text not include  $\bar{J}$  etc...

The Scalar Manifold is called a Rigid Special  
Kähler Manifold

• The Motion of special comes from the fact

that the Potential is not arbitrary but depends on the prepotential.

(i) Simplest Case:  $F(X) = \frac{1}{2} i \delta_{I5} X^I X^5$

$$\Rightarrow i \bar{F}_{I5} = -\delta_{I5}$$

$$K = X^I \delta_{I5} \bar{X}^5$$

In fact, if the Algebra is abelian  $f_{I5}^k = 0$  which leaves

$$\begin{aligned} \mathcal{L}_{uv} = & -\delta_{I5} D_\mu X^I D^\mu \bar{X}^5 - \frac{1}{4} \delta_{I5} F_{\mu\nu}^{-I} F^{-\mu\nu 5} \\ & - \frac{1}{2} \delta_{I5} \bar{\Omega}^I \not{D} \Omega^5 + \delta_{I5} \bar{\Psi}^I \not{D} \Psi^5 \end{aligned}$$

(ii) Dim Reduction of  $D=5 \rightarrow D=4$ :

We note that dim reduction of the 5D theory only gives us a specific type of theory. The one we have constructed is the most general one for some  $F$  term. Here, if we wish to obtain the theory obtained by reduction of the 5D theory we simply choose

$$F(X) = -\frac{1}{6} C_{I5K} X^I X^5 X^K$$

$$K = \frac{i}{2} \bar{X}^I C_{I5K} X^5 X^K$$

\* We assumed  $F$  is a Gauge invariant  $f_{\dots}^u$  under the central terms of the susy alg.

$$\mathcal{S}_G(\theta) = f_{KL}^I (\theta^{\pm 2i} \bar{e}_{\pm i} e_{\pm i} X^K + \theta^{\mp 2i} \bar{e}_{\mp i} e_{\mp i} \bar{X}^K)$$

$$\mathcal{S}_G(\theta) F = F_I \theta^K X^5 f_{5K}^I = 0$$

However, more generally, we can allow for.  $q$ uarter. variation of  $F$

$$\delta_G F = -\theta^k \tilde{C}_k \mp X^\mp X^\delta$$

$$\tilde{C}_k \mp X^\delta = \mp f_k^L(\mp F^\delta) \mp \frac{1}{2} F_{\delta L} X^m f_{mk}^L = \text{c.c.}$$

\* These terms enter in an additional CS term which is added to the action.

## Hypermultiplets:

$$(N=2 \text{ Hypermult}) \oplus H_2 = (\phi_{cl}^x, \zeta_{cl}^{\pm}, f_{cl}) \oplus \bar{H}_{(2)} = ( \quad )^{\Leftrightarrow 2}$$

Each  $H_{(i)}$  are  $N=1$  chiral multiplets

$\rightarrow H_1$  and  $\bar{H}_2$  transform in the same reps of the internal symmetries.

- They are the Analog of chiral mulTs in  $N=2$   
 $U(1)$  scalars, two spin  $1/2$ -fields. +  $Aux$ .  
 They are known as "matter multiplets".

\* Consider  $n_H$  hypermulTs:

Real scalars:  $q^x$ ,  $x=1, \dots, 4n_H$

$\rightarrow$  coords of a hypercomplex structure, Hyper-Kähler manifold

$\rightarrow$  For Sugra, these  $q^x$  span a Quaternionic manifold

Fermions:  $\psi^{\pm x}$ ,  $x=1, \dots, 2n_H$

- These are related by a frame field  
 $f_{iA}^X \in M_{4n_H \times 4n_H}(\mathbb{R})$ ,  $i: SU(2)_R$

$$f^{iA} \wedge f_{iA}^X = \delta_X^A$$

$$f^{iA} \wedge f_{jB}^X = \delta_j^i \delta_B^A$$

Dim. reg of Hypermult. Fermionic fields:  
 $\rho_{AB}: Sp(2)$  invariant.  $(S \rightarrow Y)$

$$\begin{aligned} \bar{\psi}^A &\rightarrow \bar{\psi}^A - \bar{\psi}^B \rho_{BA}^{\bar{A}} & \bar{\psi}^A &\rightarrow \bar{\psi}^A + \bar{\psi}^B \rho_{BA}^{\bar{A}} \\ \bar{\psi}_A &\rightarrow \bar{\psi}_B C_{BA} + \bar{\psi}^B d_{AB}^{\bar{B}} & \bar{\psi}_A &\rightarrow \bar{\psi}_B C_{BA} - \bar{\psi}^B d_{AB}^{\bar{B}} \end{aligned}$$

$C_{AB}$  is the charge conj. matrix.

The frame field satisfies the following reality condition:

$$(f_{iA}^X)^* = f_{iA}^{XB} \epsilon_{ij} \rho_{BA}^{\bar{A}} \quad | \quad (f_{iA}^X)^* = \epsilon^{ij} \rho_{AB}^{\bar{B}} f_{jB}^X$$

where

$$\rho_{AB}^{\bar{A}} \rho_{BC}^{\bar{B}} = -\delta_{AC}^{\bar{C}} \quad | \quad \rho_{AB}^{\bar{A}} = (\rho_{AB}^{\bar{B}})^*$$

- To see how we arrive at a Quaternionic Space note that the Quaternionic Algebra is given, for vectors  $\vec{A}$  and  $\vec{B}$ :

$$\vec{A} \cdot \vec{J}_x^{\bar{A}} \vec{B} \cdot \vec{J}_x^{\bar{B}} = -\delta_{AB}^{\bar{A}\bar{B}} + (\vec{A} \times \vec{B}) \cdot \vec{J}_x^{\bar{A}\bar{B}}$$

where  $\vec{J}_x^{\bar{A}\bar{B}}$ : Almost Quaternionic structure

We combine the above conditions to obtain

$$2 f_{\alpha}^{i\alpha} f_{j\alpha}^{\gamma} = \delta_{\alpha}^i \delta_j^{\gamma} + \vec{e}_j^{\gamma} \cdot \vec{J}_{\alpha}^{\gamma}$$

where

$$\vec{J}_{\alpha}^{\gamma} = (\vec{J}_{\alpha}^{\gamma})^* = -f_{\alpha}^{i\alpha} f_{j\alpha}^{\gamma} \vec{e}_i^j$$

Proof:  $\vec{A} \cdot \vec{J}_{\alpha}^{\gamma} \vec{B} \cdot \vec{J}_{\alpha}^{\gamma}$

$$\{\sigma_a, \sigma_b\} \neq [\sigma_a, \sigma_b] = 2 \sigma_a \sigma_b$$

$$\epsilon_{abc} \sigma_c + \sigma_{ab} = (\sigma_a \sigma_b)$$

$$\begin{aligned} & \vec{A} (f_{\alpha}^{i\alpha} f_{j\alpha}^{\gamma} \vec{e}_i^j) \vec{B} (f_{\alpha}^{k\alpha} f_{\beta}^{\gamma} \vec{e}_k^{\gamma}) \\ &= A^a \vec{e}_a^i B^b \vec{e}_b^j f_{\alpha}^{i\alpha} f_{j\alpha}^{\gamma} \\ &= A^a B^b (\sigma_{ab})^i_j + \epsilon_{abc} \sigma_c f_{\alpha}^{i\alpha} f_{j\alpha}^{\gamma} f_{\alpha}^{k\alpha} f_{\beta}^{\gamma} \\ &= A^a B^b (\sigma_{ab} + \epsilon_{abc} \sigma_c) f_{\alpha}^{i\alpha} f_{\beta}^{\gamma} \delta_j^k \delta_{\alpha}^{\beta} \\ &= A^a B^b (\sigma_{ab} \delta_{\alpha}^{\gamma} + \epsilon_{abc} \sigma_c) \\ &= \vec{A} \cdot \vec{B} \delta_{\alpha}^{\gamma} + (\vec{A} \times \vec{B}) \cdot \vec{J}_{\alpha}^{\gamma} \quad \checkmark \end{aligned}$$

The frame field is covariantly constant using  $\nabla_{\alpha}$  and torsionless wrt  $X$ :

$$\begin{aligned} \nabla_{\gamma} f_{i\alpha}^{\alpha} &\equiv \partial_{\gamma} f_{i\alpha}^{\alpha} - \omega_{\gamma\alpha}^{\beta}(\gamma) f_{i\beta}^{\alpha} + \Gamma_{\gamma\alpha}^{\alpha}(\gamma) f_{i\alpha}^{\alpha} = 0 \\ \nabla_{\gamma} f_{i\alpha}^{\alpha} &\equiv \partial_{\gamma} f_{i\alpha}^{\alpha} + f_{i\beta}^{\alpha} \omega_{\gamma\alpha}^{\beta}(\gamma) - \Gamma_{\gamma\alpha}^{\alpha}(\gamma) f_{i\alpha}^{\alpha} = 0 \end{aligned}$$

The second eq. follows from using  $(f_{i\alpha}^{\alpha})^*$  on the first. This is the Quaternionic vielbein postulate. Solving for  $\omega_{\gamma\alpha}^{\beta}(\gamma)$ :

$$\begin{aligned} f_{i\alpha}^{\alpha} \partial_{\gamma} f_{i\alpha}^{\alpha} - \omega_{\gamma\alpha}^{\beta} f_{i\alpha}^{\alpha} f_{i\beta}^{\alpha} + f_{i\alpha}^{\alpha} \Gamma_{\gamma\alpha}^{\alpha} f_{i\alpha}^{\alpha} &= 0 \\ f_{i\alpha}^{\alpha} \partial_{\gamma} f_{i\alpha}^{\alpha} - \omega_{\gamma\alpha}^{\beta} 2\delta_{\alpha}^{\beta} + f_{i\alpha}^{\alpha} \Gamma_{\gamma\alpha}^{\alpha} f_{i\alpha}^{\alpha} &= 0 \end{aligned}$$

$$\omega_{XA}{}^B = \frac{1}{2} f_Y^{iB} \left( \partial_X f_{iA}^Y + \Gamma_{XZ}^Y f_{iA}^Z \right)$$

$\Gamma_{YZ}^X$  is the unique connection on the scalar manifold  $\mathcal{M}^X$  which "preserves" the complex structure.

$$\nabla_Z \vec{J}_X^Y \equiv \partial_Z \vec{J}_X^Y - \Gamma_{ZX}^U \vec{J}_U^Y + \Gamma_{ZU}^Y \vec{J}_X^U = 0$$

A connection of this type is called an Onsager connection.

- The above condition promotes the almost quaternionic structure to a quaternionic structure.

$$(W_{XA}{}^B)^* \equiv \bar{W}_X{}^{\bar{A}}{}_{\bar{B}} = -\rho_{\bar{A}C} W_{XC}{}^D \rho_{D\bar{B}}$$

Here  $\rho_{AB}$  are also covariantly constant. enters also in the def. of symplectic Maj. Spinors of Hypermultiplets

D=6: The Hypermult. fermions are righthanded chiral

D=4: Charge conjugation raises and lowers the indices on the fermions and changes the chirality

Thus,

$$\underline{D=6}: (\zeta^A)^c = \zeta^B \rho_{BA}{}^{\bar{A}} \quad | \quad \zeta^A = P_R \zeta^A$$

$$\underline{D=5}: (\zeta^A)^c = \zeta^B \rho_{BA}{}^{\bar{A}}$$

$$\underline{D=4}: (\zeta^A)^c = \zeta^{\bar{A}} = P_R \zeta^{\bar{A}} \quad | \quad (\zeta^{\bar{A}})^c = \zeta^A = P_L \zeta^A$$

In D=6 the supersymmetry is:

D=6:  $\delta_Q \psi^X = \bar{\epsilon} i \gamma^\mu f_{i\mu}^X$   
 $\delta_Q \tilde{\gamma}^\mu = \frac{1}{2} f_{i\mu}^X \times \not{D} \psi^X G_i - \gamma^\mu \omega_{XB} \epsilon^B \delta_Q \psi^X$

Recall for chiral mults  $z \rightarrow z'(z)$

$$\chi^{\alpha'} = \frac{\partial z'^\alpha}{\partial z^\beta} P_L \chi^\beta$$

$$\therefore \delta \chi^{\alpha'} = \frac{\partial z'^\alpha}{\partial z^\beta \partial z^\gamma} \delta z^\alpha P_L \chi^\beta + \frac{\partial z'^\alpha}{\partial z^\beta} \delta P_L \chi^\beta$$

Which is not covariant. Instead, we define a covariant susy transformation wrt the  $z^\alpha$  manifold

$$\hat{\delta} P_L \chi^\alpha = \delta P_L \chi^\alpha + \Gamma_{\beta\gamma}^\alpha P_L \chi^\gamma \delta z^\beta$$

which yields

$$\hat{\delta} P_L \chi^{\alpha'} = \frac{\partial z'^\alpha}{\partial z^\beta} \hat{\delta} P_L \chi^\beta$$

Now, in order to obtain a covariant form of

$\delta_Q \tilde{\gamma}^\mu$  we must have  $\hat{\delta}$

$$\hat{\delta}_Q \tilde{\gamma}^\mu = \frac{1}{2} f_{i\mu}^X \times \not{D} \psi^X G_i = \gamma^\mu \omega_{XB} \epsilon^B \delta_Q \psi^X$$

So, the "normal" susy on  $\tilde{\gamma}^\mu$  would be

$$\delta_{old} \tilde{\gamma}^\mu = \frac{1}{2} f_{i\mu}^X \times \not{D} \psi^X G_i$$

where we have simply not written the Aux field. "F".

\* The Hypermultiplet also transforms under the Gauge group  $G: (I \text{ index})$   $\Theta^I K_I^X$

$$D_\mu \psi^X = \partial_\mu \psi^X - A_\mu^I K_I^X$$

The above susy must satisfy  $[\delta(\epsilon_1), \delta(\epsilon_2)] = \delta_{\text{gauge}}(\xi^\mu)$

which it does for  $\psi^X$ :

$$\delta_{\epsilon_2} (\delta_{\epsilon_1} \psi^X) = \delta_{\epsilon_2} (\bar{\epsilon} i \gamma^\mu f_{i\mu}^X)$$

$$= \bar{\epsilon}_{(1)} (\delta_{\epsilon_2} \tilde{\gamma}^\mu f_{i\mu}^X + \tilde{\gamma}^\mu \delta_{\epsilon_2} f_{i\mu}^X)$$

$$= \bar{\epsilon}^{(1)} \left[ \frac{1}{2} f_{i\mu}^X \times \not{D} \psi^X f_{i\mu}^{(2)} - \gamma^\mu \omega_{XB} \epsilon^B \delta_Q \psi^X \right] f_{i\mu}^X$$

$$+ \bar{E}_{(1)}^i \gamma^A \partial_z f_{iA}^X \delta q^z - 2 \leftrightarrow 1$$

$$= E_{(1)}^i \left( \frac{1}{4} \right) (\delta_Y^X \delta_i^j + \vec{E}_{(1)}^j \cdot \vec{\delta}_X^Y) \not{D} q^Y E_j^{(2)} + \dots$$

$$= \frac{1}{4} E_{(1)}^i \delta_Y^X \delta_i^j \not{D} q^Y E_j^{(2)} + \dots$$

$$= \frac{1}{4} \bar{E}_{(1)}^i \gamma^m \partial_m q^Y E_i^{(2)} = \frac{1}{4} \bar{E}_{(1)}^i \gamma^m E_i^{(2)} A_m^I K_I^X$$

$$+ \dots$$

extra terms, from expansion  $\frac{1}{4} \rightarrow \frac{1}{2}$   
 $= \gamma^m \partial_m q^Y + \text{Gauge as expected.}$

\* For  $\gamma^A$  the supersym goes as

$$[\delta_1, \delta_2] \gamma^A = \delta_{\text{Gauge}} \gamma^A + \delta_G(\Theta) \gamma^A + (\text{E.O.M})$$

D=6  $\rightarrow$  D=5: We must include factors of i so that the charge conj. matrices keep the required reality conditions.

- The scalars  $\sigma^I$  of the Gauge Multiplet enter (as a sixth component of the Gauge Vector of D=6) s.t. The 5D susy laws become:

$$\underline{D=5}: \delta q^X = -i \bar{E}^i \gamma^A f_{iA}^X$$

$$\delta \gamma^A = \frac{1}{2} i f_{iA}^X \not{D} q^X E_i - \gamma^B \omega_{XB}^A \delta q^X + \frac{1}{2} \sigma^I q_I^X f_{iA}^X E_i$$

$$(E^i) \rightarrow i E^i \Leftrightarrow (\bar{E}^i) \Rightarrow -i \bar{E}^i$$

$$A_m^I = \begin{pmatrix} A_m^I \\ \sigma^I \end{pmatrix} \quad \text{in } D=6$$

$$\sigma^I \equiv A_{(6)}^I$$

$$= \frac{1}{2} f_{iA}^X \gamma^{(6)} A_{(6)}^I K_I^X (E^i) \quad (\text{extra term for } \gamma^A)$$

$$= -\frac{1}{2} f_{\alpha}^{iA} \sigma^{\alpha} \kappa_{\alpha}^X (E^i) = \frac{1}{2} \sigma^{\alpha} \kappa_{\alpha}^X f_{\alpha}^{iA} E_i \quad \text{as above.}$$

$$\boxed{D=4: \delta q^X = -i \bar{E}^i \gamma^{\alpha} f_{\alpha}^X + i \bar{E}^i \gamma^{\alpha} \bar{p}^{\beta} \bar{E}_i \gamma_{\alpha} f_{\beta}^X}$$

$$\delta \bar{q}^{\alpha} = \frac{1}{2} i f_{\alpha}^{iA} \delta q^X E_i - \int^B \omega_{X B}^A \delta q^X + i \bar{X}^{\alpha} \kappa_{\alpha}^X f_{\alpha}^{iA} E_i E_i$$

Note that  $D=5 \rightarrow D=4$   $\bar{q}^{\alpha} \rightarrow \bar{q}^{\alpha} - \bar{q}^{\beta} p^{\beta} \bar{p}^{\alpha}$

which is where the extra terms come from

\* To construct an action we need a metric.

Define a Hermitian metric on the tangent space of the  $Sp(2n_H)$  space:

$$\boxed{d\bar{A}_B = (d\bar{B}_A)^{\dagger}}$$

• For Global susy:  $d\bar{A}_B = \delta\bar{A}_B$

• In general,  $d\bar{A}_B$  will respect the Quaternionic structure which can be expressed as the Antisym of a tensor  $C_{AB}$ :

$$\boxed{C_{AB} = -C_{BA}}$$

$$\boxed{C_{AB} \equiv p_A \bar{C} d\bar{B}}$$

at any point on the manifold we can write  $p$  as

$$\boxed{p_{AB} = \begin{pmatrix} 0 & \mathbb{1}_{n_H} \\ -\mathbb{1}_{n_H} & 0 \end{pmatrix}}$$

and at the same time

$$\boxed{d\bar{A}_B = \begin{pmatrix} \mathbb{1}_p & & & \\ & -\mathbb{1}_q & & \\ & & \mathbb{1}_p & \\ & & & -\mathbb{1}_q \end{pmatrix}} \quad \underline{p+q = n_H}$$

→ The metric for the scalar Manifold is defined by from the coset metric  $d\bar{A}_A$  @ frame fields:

$$g_{XY} = (f^i{}_{\bar{A}})_X^* d\bar{E}_B f^{iB}{}_Y = f^{iB}{}_X \epsilon_{ij} \rho^j{}_{\bar{A}} d\bar{E}_B f^{iB}{}_Y \\ = f^{iA}{}_X C_{AB} \epsilon_{ij} f^{jB}{}_Y$$

$$\therefore g_{XY} = f^{iA}{}_X \epsilon_{ij} C_{AB} f^{jB}{}_Y$$

- $C_{AB} \epsilon_{ij} \sim \eta_{mn}$  and  $f^{iA}{}_X$  are frames so that  $f^{iA}{}_X \sim e^a_\mu$

Require  $C_{AB}$  to be covariantly constant:

$$\nabla C = \nabla(\rho d) = \nabla \rho d + \rho \nabla d = 0, \text{ or}$$

$$\nabla_X C_{AB} := \partial_X C_{AB} + 2\omega_X{}^C{}_{[A} C_{B]C} = 0$$

and since  $\nabla \theta = 0$  and so  $\nabla g_{XY} = 0$  and so  $\Gamma_{XY}^Z(\varphi)$  are Christoffel connection of the metric.

\* Use  $C_{AB}$ :  $Sp(2n)$  metric so we use it to raise and lower  $\bar{A}$  and  $B$ :

$$\phi^{\bar{A}} = C^{\bar{A}B} \phi_B, \quad \phi_B = \phi^{\bar{A}} C_{\bar{A}B}, \quad C^{\bar{A}C} C_{BC} = \delta^{\bar{A}}_B$$

$$\text{Note } f_{Xi\bar{A}} = g_{XY} f^{jB}{}_Y = f^{jC}{}_X \epsilon_{jk} C_{CB} f^{kB}{}_Y f^{jB}{}_Y \\ = f^{jC}{}_X \epsilon_{jk} C_{CB} \delta^k{}_i \delta^B{}_A$$

$$f_{Xi\bar{A}} = f^{jC}{}_X \epsilon_{jk} C_{CB}$$

$$\text{Claim: } C^{\bar{A}\bar{B}} = (C^{\bar{A}D})^* = d\bar{E}_C d\bar{E}_D C^{CD} = \rho^{\bar{A}C} d\bar{E}_C$$

$$\text{Proof: } (\rho^{\bar{A}C} d\bar{E}_C)^* = (C^{\bar{A}B})^*$$

$$\text{LHS} = \rho^{\bar{A}C} (d\bar{E}_C)^* = \rho^{\bar{A}C} d\bar{E}_C = C^{\bar{A}\bar{B}}$$

$$\text{Now, since } C_{AB} = \rho_{\bar{A}\bar{C}} d\bar{E}_B$$

$$\text{then } \rho^{\bar{A}\bar{B}} C_{AB} = \rho^{\bar{A}\bar{B}} \rho_{\bar{C}\bar{D}} d\bar{E}_B d\bar{E}_C = \delta^{\bar{A}\bar{B}}_{\bar{C}\bar{D}} d\bar{E}_B d\bar{E}_C$$

using tws

$$\delta_D^C \rho^{\bar{A}D} d_C^{\bar{B}} = C^{CB} \rho_D^{\bar{A}} d_C^{\bar{B}} = +\delta_C^{\bar{A}} C^{CB} d_C^{\bar{B}} = \rho^{\bar{A}C} d_C^{\bar{B}}$$

$$\therefore C^{\bar{A}B} = (C_{AB})^* = d_C^{\bar{A}} d_B^{\bar{C}} C^{CB} = \rho^{\bar{A}C} d_C^{\bar{B}}$$

as claimed

Note by our conventions of raising and lowering

$$C^{AC} C_{BC} = C^{AC} \rho_{AC}^{\bar{D}} d_B^{\bar{C}} = \rho_{AC}^{\bar{D}} d_B^{\bar{C}} \rho_{\bar{D}}^C d_P^{\bar{C}} = \rho_{AC}^{\bar{D}} d_B^{\bar{C}} \rho_{\bar{D}}^C d_P^{\bar{C}}$$

\*Curvature of the spacetime: wrt.  $\omega_{XA}^B$

$$R_{XYD}^A \equiv 2\partial_X \omega_{YD}^A + 2\omega_X^C \hat{C}^A_C \omega_{YD}^B$$

- Integrability conditions of the covariantly constant  $C_{AB}$  and  $d^{\bar{A}}_B$  implies the following:

$$[\nabla_X, \nabla_Y] C_{AB} = 2R_{XYA}^C C_{BC} = -2R_{XY[AB]} = 0$$

$$[\nabla_X, \nabla_Y] d^{\bar{A}}_B = -(R_{XYA}^C)^* d_C^{\bar{A}} - R_{XYB}^C d_C^{\bar{A}} = 0$$

$$-2R_{XY[A B]} = 0$$

$$(R_{XYA}^C)^* d_C^{\bar{A}} + R_{XYB}^C d_C^{\bar{A}} = 0$$

These say that the curvature matrices preserve the anti-symmetric  $C_{AB}$  and  $d^{\bar{A}}_B$ : so they are in the algebra  $USp(2p, 2q)$ .

$$USp(2m, 2n) \equiv U(m, n: \mathbb{H})$$

(Structure Group)

$R$  can be cast in terms of  $\mathbb{H}/\mathbb{R}$  coordinates:

$$R_{XY}^W \equiv f_{iA}^W f_{jB}^C R_{XYD}^A$$

We can express this in terms of  $Sp(2n, \mathbb{H})$  coords:

$$W_{ABCD} = f_{iA}^X f_{jB}^Y f_{kC}^Z f_{lD}^W R_{XYZW}$$

These are related by:

$$R_{xy}{}^w{}_z = \frac{1}{2} f^A{}_i x f^B{}_j y f^K{}_c z f^W{}_{kD} W_{ABCD}$$

$$W_{ABCD} = \frac{1}{2} f^X{}_A f^Y{}_B f^Z{}_C f^W{}_{kD} R_{XYZW}$$

$W_{ABCD} = W_{APC}{}^E{}_{ED}$  is symmetric.

Claim: This manifold is Ricci-flat:

$$R_{yz} = R_{xy}{}^x{}_z = 0$$

Proof:

$$\begin{aligned} R_{xy}{}^x{}_z &= \frac{1}{2} f^A{}_i x f^B{}_j y f^K{}_c z f^W{}_{kD} W_{ABCD} \\ &= \frac{1}{2} \delta^i{}_k \delta^A{}_D f^B{}_j y f^K{}_c z W_{ABCD} \\ &= \frac{1}{2} f^B{}_j y f^K{}_c z W_{ABCD}{}^A{}^D = \frac{1}{2} ( ) W_{ABCD}{}^A{}^D \end{aligned}$$

Since  $W_{ABCD} = W_{BCD}{}^E{}_{AE} = 0$

$$R_{yz} = 0 \quad \square$$

The Lagrangians for different dim:

$$\begin{aligned} \mathcal{L}_6 = \mathcal{L}_5 &= -\frac{1}{2} g_{xy} \partial_\mu q^x \partial^\mu q^y + \bar{\psi}_A \not{\partial} \psi^A - \frac{1}{4} W_{ABCD} \bar{\psi}^A \gamma^B \bar{\psi}^C \gamma^D \\ \mathcal{L}_4 &= -\frac{1}{2} g_{xy} \partial_\mu q^x \partial^\mu q^y - d_{\bar{A}}^{\bar{B}} (\bar{\psi}_{\bar{A}} \not{\partial} \psi^{\bar{B}} + \bar{\psi}^{\bar{B}} \not{\partial} \psi_{\bar{A}}) \\ &\quad + \frac{1}{2} W_{ABCD} d_{\bar{C}}^{\bar{E}} d_{\bar{F}}^{\bar{D}} \bar{\psi}_{\bar{C}} \gamma^{\bar{E}} \bar{\psi}_{\bar{D}} \gamma^{\bar{F}} \psi^A \psi^B \end{aligned}$$

6/5  
4

$$\nabla_\mu \psi^A = \partial_\mu \psi^A + \partial_\mu q^x \omega_{xB}{}^A \psi^B$$

ex: Simplest case is  $n=1$ :  $\rho_{AB} = \epsilon_{AB}$

(i) The reality constraints allow the sol<sup>n</sup> where the

only nonvanishing elements of  $f^{\bar{A}}{}_x$  are

$$f^{\bar{1}2}{}_2 = f^{\bar{2}1}{}_1 = i \quad -f^{\bar{1}2}{}_2 = f^{\bar{2}1}{}_2 = 1$$

$$f^{\bar{1}1}{}_3 = -f^{\bar{2}2}{}_3 = i \quad f^{\bar{1}1}{}_4 = f^{\bar{2}2}{}_4 = 2$$

proof:  $(f^{\bar{1}A}{}_x)^* = f^{\bar{J}B}{}_x \epsilon_{\bar{J}\bar{A}} \epsilon_{BA}$

$$(f^{\bar{1}2}{}_x)^* = f^{\bar{1}B}{}_x \epsilon_{\bar{1}\bar{A}} \epsilon_{BA} = -f^{\bar{2}2}{}_x$$

$$\text{For } (f''_1)^* = f^{j\beta}_1 \epsilon_{j2} \epsilon_{\beta 2} = f^{22}_1$$

We see that  $f^{ac}_c$  is zero if  $a=1,2, c=1,2$

Since there is an  $\epsilon$  for (12).

$$\boxed{\begin{array}{l|l} f^{12}_1 = f^{21}_1 = i & -f^{12}_2 = f^{21}_2 = 1 \\ f^{11}_3 = -f^{22}_3 = i & f^{11}_4 = f^{22}_4 = 1 \end{array}}$$

do not get mixing for 3,4

→ Not sure how these vanish...

$$g_{xy} = f^{j\beta}_x \epsilon_{ji} \epsilon_{\beta A} d^{\bar{A}}_C f^{iC}_y$$

Choose  $d^{\bar{A}}_C$  s.t.  $g_{xy} = \delta_{xy}$

$$\begin{aligned} g_{xy} &= f^{j\beta}_x \epsilon_{ji} \epsilon_{\beta A} d^{\bar{A}}_C f^{iC}_y + f^{j\beta}_x \epsilon_{2i} \epsilon_{\beta A} d^{\bar{A}}_C f^{iC}_y \\ &= f^{10}_x \epsilon_{\beta A} d^{\bar{A}}_C f^{2C}_y - f^{20}_x \epsilon_{\beta A} d^{\bar{A}}_C f^{1C}_y \\ &= f^{11}_x \epsilon_{2A} d^{\bar{A}}_C f^{2C}_y + f^{12}_x \epsilon_{2A} d^{\bar{A}}_C f^{1C}_y - f^{21}_x \epsilon_{1A} d^{\bar{A}}_C f^{1C}_y \\ &\quad - f^{22}_x \epsilon_{2A} d^{\bar{A}}_C f^{2C}_y \\ &= f^{11}_x d^2_C f^{2C}_y - f^{12}_x d^2_C f^{2C}_y - f^{21}_x d^2_C f^{1C}_y \\ &\quad + f^{22}_x d^2_C f^{2C}_y \\ &= f^{11}_x (d^2_1 f^{21}_y + d^2_2 f^{21}_y) - f^{12}_x (d^1_1 f^{21}_y + d^1_2 f^{22}_y) \\ &\quad - f^{21}_x (d^2_1 f^{11}_y + d^2_2 f^{12}_y) + f^{22}_x (d^1_1 f^{11}_y + d^1_2 f^{12}_y) \end{aligned}$$

$$\boxed{g_{xy} = d^1_2 (f^{22}_x f^{11}_y - f^{12}_x f^{21}_y) + d^2_2 (f^{11}_x f^{21}_y - f^{21}_x f^{11}_y) + d^1_2 (f^{22}_x f^{12}_y - f^{12}_x f^{22}_y) + d^2_2 (f^{11}_x f^{21}_y - f^{21}_x f^{12}_y)}$$

Since these are all constants  $\int_{xy} = W_{xA}^B = W_{ABCP} = 0$

$$\vec{J}_x^{\alpha} = (J^2)_x^{\alpha}, (J^2)_x^{\alpha}, (J^3)_x^{\alpha} \quad \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$(J^{\alpha})_x^{\gamma} = -f^{i2}_x f^{\gamma}_{j2} \vec{e}_i^{\alpha} + f^{i2}_x f^{\gamma}_{j2} (\vec{e}^{\alpha})_j^i$$

... expanding

$$\boxed{(J^{\alpha})_{\beta}^{\gamma} = -\epsilon_{\alpha\beta\gamma}}$$

$$\boxed{(J^{\alpha})_{\beta}^{\gamma} = - (J^{\alpha})_{\gamma}^{\beta} = \delta^{\alpha}_{\beta}}$$

## Gauged Hypermultiplets:

\* We now gauge the sym. in the Action of the hyper-mult.  
leads to interactions of the Gauge mult. & Hyper. This  
will lead to massive Hypermults (Higgs?). This  
requires central charges. These were found  
above for nonvanishing scalars in the Gauge mult.  
 $\sigma^I$ . from V.

\* These symmetries are defined by Killing vectors  $K_I^X$   
the metric  $g_{XY}$

- For Kähler manifolds:  $K_I^X$  was holomorphic, this meant  
 $\nabla_X K_I^Y$  is a matrix which commutes w/  $J_X^Y$ .  
 $\mathcal{L}_K J_i^j = \nabla_i K^k J_k^j - \nabla_k K^j J_i^k = 0$

- For Hypermultiplets:  $K_I^X$ : triholomorphic

$$(\nabla_X K_I^Y) \vec{J}_Y^Z = \vec{J}_X^Y (\nabla_Y K_I^Z)$$

the triholomorphic condition is simply the statement  
that  $\mathcal{L}_K(J^a) = 0$  for all three complex structures.

→ Still have  $(\mathcal{L}_K g)_{XY} = 0$  as well. These  
two conditions:  $K^a \rightarrow \{K^a(z), K^a(\bar{z})\}$

$$\Omega = -J_{ij} d\phi^i \wedge d\phi^j \quad J_{ij} = J_i^k g_{kj}$$

$$d\Omega = 0$$

$$\mathcal{L}_K \Omega = (d i_K + i_K d) \Omega = 0$$

$$d(i_K \Omega) = 0 \quad \text{so } i_K \Omega \text{ is exact locally.}$$

$$\text{So } i_K \Omega =: -2dP$$

$$\text{Recall: } K^a = -i g^{a\bar{b}} \partial_{\bar{b}} P(z, \bar{z})$$

$$J_i^j = -i g^{j\bar{k}} \partial_{\bar{k}} P(z, \bar{z}) \quad J_{\bar{i}}^{\bar{j}} = J_i^j g_{j\bar{k}} g^{\bar{k}\bar{i}}$$

$$K_a \equiv g_{a\bar{b}} K^{\bar{b}} = i \partial_a P$$

$$\Rightarrow \nabla_a \partial_{\bar{a}} P = 0$$

\* We get a triplet of moment maps:

$$\partial_x \vec{P}_I = \vec{J}_x^Y K_{IY}$$

\* In Kähler  $\partial_x \vec{P}_I = \vec{J}_x^Y g_{Y\bar{Z}} K_I^{\bar{Z}} = \vec{J}_{x\bar{Z}} K_I^{\bar{Z}}$   
 $\partial_x \vec{P}_I = -i g_{x\bar{Z}} K_I^{\bar{Z}}$  which is what we have above  
 however the complex structure was not the important aspect of the theory.

• Susy implies, for a non-abelian alg.

$$K_I^x \vec{J}_{xy} K^y_J = f_{IJ}^K \vec{P}_K$$

Which is the analog of  $K_A^a g_{a\bar{b}} K_{\bar{b}}^{\bar{B}} - K_B^a g_{a\bar{b}} K_{\bar{b}}^{\bar{A}} = i f_{AB}^C P_C$

\* Consider the flat Mfd above w/  $SO(4)$  symm

$K_X = \lambda_{XY} \partial^Y$ ,  $\lambda_{XY} = -\lambda_{YX}$ . This is then an isometry:

$$\mathcal{L}_{K_X} = \nabla_Y K_X + \nabla_X K_Y = \lambda_{XY} + \lambda_{YX} = 0 \quad \checkmark$$

The complex structures are self dual

$$(J^\alpha)_\beta^\gamma = -\epsilon_{\alpha\beta\gamma} \quad (J^\alpha)_\beta^\gamma = -(J^\alpha)_\gamma^\beta = \delta_\beta^\gamma$$

$$\begin{aligned} (*) (J^\alpha)_\beta^\gamma &= \frac{1}{2} \epsilon_{\beta\delta\epsilon} (J^\alpha)_\delta^\epsilon = -\frac{1}{2} \epsilon_{\beta\delta\epsilon} \epsilon^{\alpha\delta\epsilon} = -\frac{1}{2} \epsilon_{\beta\delta\epsilon} \epsilon^{\alpha\delta\epsilon} \\ &= -\frac{1}{2} (-4) (\delta_{\beta\gamma}^\alpha) = \delta_{\beta\gamma}^\alpha = \delta_{\alpha\beta} \delta_\gamma^\alpha = \epsilon_{\alpha\beta\gamma} \\ &\quad \checkmark = -J_{\alpha\beta\gamma} \end{aligned}$$

$\rightarrow$  Any self dual  $\lambda_{XY}$  defines a triholomorphic symm

$$\nabla_X K_Y = \lambda_{XY} \Rightarrow \lambda_{XY} \delta^Y{}_{\bar{Z}} \vec{J}_{\bar{Z}}^W$$

Note

$$G^{+m} H^{\pm} = \mathfrak{sl}(2) \oplus \mathfrak{p}^{(m)} H^{\pm} N) = -\frac{1}{2} \text{rank } G^{\pm} \mathfrak{p} \text{ or } 1/2$$

$$G^+_{p,q} H^+_{p,q} = 0 \quad \textcircled{*}$$

Since  $\vec{J}_Z \omega$  is anti-self dual  $\vec{J}_Z \omega = \vec{J}_Z^* \omega$

Now, if we further require that  $\lambda_{xy}$  is self dual i.e.  $\lambda_{xy} = \lambda_{xy}^+$  and so we find, using  $\textcircled{*}$

$$\lambda_{xy}^+ \vec{J}_Z^* x \Leftrightarrow \lambda_{xy}^+ \vec{J}_Z^* x = \lambda_{xz}^+ \vec{J}_Y^* x$$

which is precisely the triholomorphic condition.

\* What do we learn from this?

→ From  $SU(4) \cong SU(2) \times \overline{SU(2)}$  Symm:  $\dagger$

$\vec{J}_X^+$ : sits in  $SU(2)$  (Complex structure)

$\lambda_{xy}^+$ : sits in  $SU(2)$  (Triholomorphic isometries)

Now, we choose  $\lambda_{xy}$  s.t.

$$k^1 = q^2, k^2 = -q^1, k^3 = q^4, k^4 = -q^3$$

$$\text{or } K_{xy} = \lambda_{xy} q^y = \begin{pmatrix} 0 & 2 & 0 & 0 \\ -2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{pmatrix} \begin{pmatrix} q^1 \\ q^2 \\ q^3 \\ q^4 \end{pmatrix}$$

We note here that  $\lambda_{xy}$  is simply  $-\mathcal{I} \leftarrow$  where

$\mathcal{I}$  is the first complex structure of the canonical triple  $(\mathcal{I}, \mathcal{J}, \mathcal{K})$  of the hyper Kähler mfd.

• If we take  $k^3 \rightarrow -k^3$  and  $k^4 \rightarrow -k^4$  then

$K$  no longer becomes a triholomorphic vec.

$$\text{Since } \partial_x \vec{P} = \vec{J}_X^+ K_y$$

$$\begin{aligned} \partial_x P^2 &= (\vec{J}_X^+)_x^y K_y = (\vec{J}_X^+)_x^1 K_1 + (\vec{J}_X^+)_x^2 K_2 + (\vec{J}_X^+)_x^3 K_3 + (\vec{J}_X^+)_x^4 K_4 \\ &= (\vec{J}_X^+)_x^1 q^2 + (\vec{J}_X^+)_x^2 (-q^1) + (\vec{J}_X^+)_x^3 (q^4) + (\vec{J}_X^+)_x^4 (-q^3) \end{aligned}$$

$$\begin{aligned} \partial_1 P^2 &= (\vec{J}_X^+)_1^1 q^2 + (\vec{J}_X^+)_1^2 (-q^1) + (\vec{J}_X^+)_1^3 q^4 + (\vec{J}_X^+)_1^4 (-q^3) \\ &= -q^3 \end{aligned}$$

$$\therefore P^1 = -q^3 q^1 + f(q)$$

$$\partial_2 P^1 = (\mathcal{J}_2^1)^0 + \mathcal{J}_2^1 + (\mathcal{J}^1)_2 q^4 + 0$$

$$= -q^4$$

$$\Rightarrow P_1 = -q^2 q^4 - q^3 q^1$$

$$\partial_x P^3 = (\mathcal{J}^3)_x q^2 + (\mathcal{J}^3)_x (-q^1) + (\mathcal{J}^3)_x (q^4) + (\mathcal{J}^3)_x (-q^3)$$

$$\partial_1 P^3 = -\varepsilon_{3,2} (-q^1) = q^1$$

$$\Rightarrow P^3 = \frac{1}{2} (q^1)^2 + f(q)$$

$$\partial_2 P^3 = -\varepsilon_{321} q^2 = q^2 \Rightarrow P^3 = \frac{1}{2} ((q^1)^2 + (q^2)^2) + f(q)$$

$$\partial_3 P^3 = -q^3 \Rightarrow P^3 = \frac{1}{2} ((q^1)^2 + (q^2)^2 - (q^3)^2) + f(q^4)$$

$$P^3 = \frac{1}{2} ((q^1)^2 + (q^2)^2 - (q^3)^2 - (q^4)^2)$$

$$P^2 = q^1 q^4 - q^2 q^3$$

\* The transformations for fermions  $\psi^A$  under the Killing symm  $K_I^X$ :

Since we covariantized the symm:

$$\tilde{\delta} P_L \chi^\alpha = \delta P_L \chi^\alpha + \Gamma_{\beta\gamma}^\alpha P_L \chi^\gamma \delta z^\beta$$

However  $\chi^\alpha = \chi^\alpha(z, x)$  so we must consider

$$\delta \chi = \delta \chi + \frac{\partial \chi}{\partial z} \delta z \quad \text{and putting}$$

both these two together we find

$$\tilde{\delta} P_L \chi^\alpha = \delta P_L \chi^\alpha + \nabla_\beta P_L \chi^\alpha \delta z^\beta$$

$$\therefore z'^\alpha(z) = z^\alpha + \theta^I K_I^\alpha(z), \quad \delta z^\alpha = \theta^I K_I^\alpha(z)$$

$$\delta(\theta) P_L \chi^\alpha = \theta^I \frac{\partial K_I^\alpha}{\partial z^\beta} P_L \chi^\beta$$

$$\text{Covariantly: } \tilde{\delta}(\theta) P_L \chi^\alpha = \theta^I \nabla_\beta K_I^\alpha P_L \chi^\beta$$

Thus, here we should have:

$$\tilde{\delta}_G \psi^A = \theta^I \psi^B t_{IB}^A$$

where

$$t_{IB}^A = \frac{1}{2} P_L^A \cdot \nabla_I \cdot P_L^B \cdot f^{AB}$$

$$* [t_I, t_J]^A = \int_{FS} t_{IJ}^A - K_I^X K_J^Y R_{XY}^A \quad (\star)$$

These satisfy the properties:

$$t_I^A{}^B \equiv C^{AC} t_{IC}{}^B = t_I^B{}^A$$

$$\bar{t}_I^{\bar{A}}{}_{\bar{B}} \equiv (t_I^A{}^B)^* = -\rho^{\bar{A}D} t_{ID}{}^C \rho_C{}_{\bar{B}} \quad t \in \text{usp}(2n_H - 2p, 2p)$$

$$\bar{t}_I^A{}^C d_{\bar{B}}^{\bar{C}} = -d_{\bar{C}}^{\bar{A}} t_{IC}{}^B$$

With this we can write the Gauged Hypermultiplets:

D=6:

$$\mathcal{L}_6 = -\frac{1}{2} g_{XY} D_\mu q^X D^\mu q^Y + \bar{Z}_A \widehat{D}^\mu Z^A - \frac{1}{6} W_{ABCD} \bar{Z}^A Z^B \bar{Z}^C Z^D \\ - 2 K_I^X f^{IA}{}_{\bar{X}} \bar{Z}_A \lambda_{\bar{I}}^{\bar{X}} + 2 \bar{P}_I \cdot \vec{\gamma}^I$$

D=5:

$$\mathcal{L}_5 = -\frac{1}{2} g_{XY} D_\mu q^X D^\mu q^Y - (\bar{Z}_A \widehat{D}^\mu Z^A + \text{h.c.}) + \frac{1}{2} W_{AB}{}^{\epsilon F} d_{\epsilon F}^{\bar{C}} d_{\bar{F}}^{\bar{D}} \bar{Z}_D Z^A Z^B \\ - 2i K_I^X f^{IA}{}_{\bar{X}} \bar{Z}_A \lambda_{\bar{I}}^{\bar{X}} + i \sigma^I \bar{Z}_B{}^A \Sigma_A Z^B + 2 \bar{P}_I \cdot \vec{\gamma}^I - \frac{1}{2} \sigma^I \sigma^J K_I^X K_J^X$$

D=4:

$$\mathcal{L}_4 = -\frac{1}{2} g_{XY} D_\mu q^X D^\mu q^Y - (\bar{Z}_A \widehat{D}^\mu Z^A + \text{h.c.}) + \frac{1}{2} W_{AB}{}^{\epsilon F} d_{\epsilon F}^{\bar{C}} d_{\bar{F}}^{\bar{D}} \bar{Z}_D Z^A Z^B \\ + (2 \chi^I t_{IA}{}^B \bar{Z}^A Z^B + 2i f_{\bar{X}}^{IA} K_I^X \bar{Z}_{\bar{B}} \Omega^{iI} \epsilon_{ij} d_{\bar{A}}^{\bar{B}} + \text{h.c.}) \\ + 2 \bar{P}_I \cdot \vec{\gamma}^I - 2 \bar{X}^I X^J K_I^X K_J^X$$

→ These terms are similar to the Global coupling terms

$$S_{\text{coup}} = \int d^4x \left[ -\sqrt{2} (\bar{\lambda}^A \bar{Z} t_A P_2 X - \bar{X} P_2 t_A Z \lambda^A) + i D^A \bar{Z} t_A Z \right]$$

instead of  $Z \mapsto K_I^X$  and  $t_A \mapsto f_{\bar{X}}^{IA}$

• The covariant derivatives are:

$$D_\mu q^X \equiv \partial_\mu q^X + A_\mu^I K_I^X$$

$$\widehat{D}^\mu Z^A = \partial^\mu Z^A + \partial^\mu X^I A_I{}^A{}_{\bar{B}} Z^{\bar{B}} = \partial^\mu Z^A + \partial^\mu X^I A_I{}^A{}_{\bar{B}} Z^{\bar{B}}$$

Note: The hypermultiplets are coupled to the  $N=2$  matter multiplets i.e.  $V_{WZ} \sim A_m^\pm$

\* For the gauging  $k_x = \lambda_{xy} q^y = -\gamma_{xy} q^y$

$$\Rightarrow t_{IA}^B = \frac{1}{2} f_{iA}^y \nabla_y k_x^x f_x^{iB}$$

$$\nabla_y k_x^x = \begin{pmatrix} \delta_y^2 \\ -\delta_y^1 \\ \delta_y^4 \\ -\delta_y^3 \end{pmatrix}$$

$$\begin{aligned} t_{IA}^B &= f_{iA}^1 \nabla_1 k_x^x f_x^{iB} + f_{iA}^2 \nabla_2 k_x^x f_x^{iB} + f_{iA}^3 \nabla_3 k_x^x f_x^{iB} + f_{iA}^4 \nabla_4 k_x^x f_x^{iB} \\ &= f_{iA}^2 (\nabla_2 k_x^x) f_x^{iB} + f_{iA}^3 (-\nabla_3 k_x^x) f_x^{iB} + f_{iA}^4 \nabla_4 k_x^x f_x^{iB} \\ &= -2 f_{iA}^2 f_1^{iB} - 2 f_{iA}^3 f_2^{iB} + \delta_i^1 \delta_A^B \\ &= 2 \left( \delta_A^B - (f_{iA}^1 f_1^{iB} + f_{iA}^2 f_2^{iB}) - (f_{iA}^3 f_3^{iB} + f_{iA}^4 f_4^{iB}) \right) \\ &= 2 \left( \delta_A^B - (f_{iA}^1 \delta_2^{iB} + f_{iA}^2 \delta_1^{iB}) - (f_{iA}^3 \delta_2^{iB} + f_{iA}^4 \delta_2^{iB}) \right) \end{aligned}$$

$$\therefore t_{IA}^B = 2(\delta_A^B) = 2 \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix}$$

- If we now choose  $x^I \equiv m$  and all other fields of the Gauge multiplet equal to zero ( $A_m^\pm = 0$  and  $\lambda^{iI} = 0$ )

$$\Rightarrow D_m q^x \rightarrow \partial_m q^x$$

$$\widehat{D}_m z^A \rightarrow \partial_m z^A$$

$$\text{Since } \omega_{xA}^B = 0 \Leftrightarrow \omega_{xA}^B = \frac{1}{2} (f_{iA}^{iB} \partial_x f_{iA}^i + 0)$$

- Recall,  $g_{xy} = \delta_{xy}$ :

$$\mathcal{L}_a = -\partial_m q^x \partial^m q^x - (\bar{z}_A \not{\partial} z^B d_B^{\bar{A}} + h.c.) + \dots$$

• Since  $K \times Y \times Z = 0 \Rightarrow W_{ABEF} = 0$  and so

$$W_{AB}{}^{EF} d\bar{z}_E d\bar{z}_F \bar{z}_C \bar{z}_D \bar{z}^A \bar{z}^B = 0$$

•  $\Omega^{i\bar{j}} = 0, \bar{\gamma}^{\bar{i}} = 0$

$$\mathcal{L}_4 = -\partial_\mu q^x \partial^\mu q^x - \bar{z}_A \not{\partial} z^A + 2(m)(\sigma_3)_A{}^B \epsilon_{BE} \bar{z}^A z^E - 2m^2 q^x q_x$$

$$\rightarrow 2m \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \bar{z}^A z^E = 2m \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix} \bar{z}^A z^E \\ = 2m (-1) (\bar{z}^1) \begin{pmatrix} z^2 \\ z^1 \end{pmatrix} = 2m (\bar{z}_1 z_2 - \bar{z}^2 z^2)$$

$\therefore$  In  $D=4$ , for a single Hypermultiplet Emb. into a flat Hyper-Kähler geometry coupled to a scalar matter field:

$$\mathcal{L}_4 = -\partial_\mu q^x \partial^\mu q^x - \bar{z}_A \not{\partial} z^A + 2im(\bar{z}_1 z_2 - \bar{z}^2 z^2) - 2m^2 q^x q_x$$

$\rightarrow$  We see that we get a simple massive theory for a scalar and fermion of the same mass. In this way, this is the simplest on-shell susy since the transformations of this Lagrangian reduce to

$$\delta q^x = -i \bar{\epsilon}^i z^A f_{iA}^x + i \bar{\epsilon}^{\dot{i}} \bar{\epsilon}_{\dot{j}} \epsilon^{AB} \bar{z}_{\bar{A}} f_{\bar{A}}^x \\ \delta z^A = \frac{1}{2} f_{iA}^x \not{\partial} q^x - i \bar{X}^{\dot{I}} \gamma_{\dot{I}}^x f_{iA}^x \epsilon_{\dot{I}} \epsilon^{\dot{i}}$$

## $N=2$ Superconformal Calculus:

$\rightarrow$  We have two compensating multiplets: one vector and one hypermultiplet. Only one field of these

compensating mult. is physical: the vector of the Gauge mult. This combines with the graviton & gravitino which are parts of the Weyl mult. to make pure  $N=2$  Poincaré sugra.

Weyl mult.  $\oplus$   $(n_v + 2)$ -vector mult.  $\oplus$   $(n_h + 2)$ -hyper mult.  
 $\Rightarrow$  embedding rigid special Kähler mld & hyper Kähler manifold w/ conformal symmetry

$\Downarrow$  Gauge fix  
 $\Downarrow$  + field Equations

Poincaré sugra  $\oplus$   $n_v$ -vector  $\oplus$   $n_h$  Hyper  
 $\Rightarrow$  Special Kähler and Quaternionic-Kähler mld for physical fields

The Superconformal Alg:

R-sym group  $D=6$ :  $USp(N_L) \times USp(N_R)$ . Here we are not distinguishing  $N_L$  and  $N_R$  ( $N_L, N_R$ ) instead we have  $N=2$  in general.

$$USp(2) \cong SU(2)_R \quad (D=6, D=5)$$

$Q_\alpha^i$  &  $\bar{Q}_{\dot{\alpha}}^i \Rightarrow$  8 fermions } 16 Real Supercharges  
 $S_\alpha^i$  &  $\bar{S}_{\dot{\alpha}}^i \Rightarrow$  8 fermi }

$$\text{In } D=4: U(N)_R: U(2)_R \cong SU(2) \times U(1)$$

The susy algebras which have this content are:

$$D=6: \underline{OSp(8^*(2))} \quad | \quad D=4: \underline{SU(2,2|2)}$$

$$D=5: \underline{F(4)}$$

which have very similar algebras. The superconf. alg. was previously noted: For the  $SU(2)$  algebra we use the triplet notation  $\vec{U}$  for the three generators  $\text{tr}(U) = 0$  and  $U_a^i$ . The  $SU(2)$  Alg:

$$[U_a^i, U_b^j] = \delta_a^j U_b^i - \delta_a^i U_b^j$$

$$\begin{aligned} U_a^i &\equiv i \vec{\sigma}_a^i \cdot \vec{U} \\ \vec{U} &= -\frac{1}{2} \vec{\sigma}_i^j U_j^i \end{aligned}$$

In triplet notation:

$$[U^1, U^2] = U^3$$

- For  $D=4$ , we denote the generator of the  $U(1)_R$  as  $T_R$  as in the  $N=1$  case  
 $[T_R, (\text{Bosonic symm})] = 0$

- The fermionic generators are symplectic Maj.-Weyl in  $D=6$  w/ chirality

$$Q^i \equiv P_R Q^i = -\gamma_* Q^i \quad | \quad S^i \equiv P_L S^i = \gamma_* S^i$$

$D=4$   $Q_i = P_L Q_i$  for susy. For the special susy we use the opposite:

$$S^i \equiv P_L S^i = \gamma_* S^i \quad | \quad Q_i \equiv P_R Q_i = -\gamma_* Q_i$$

Aside: Notation:  $\psi$ : Majorana  $|\chi = P_L \chi$  chiral  
 $\phi = P_L \chi + P_R \chi^c$

Note  $(\overline{P_R Q}) \equiv (P_R Q)^T C = Q^T P_R^T C = \overline{Q} P_R$

$$\{Q_\alpha, \overline{Q}^\beta\} = -\frac{1}{2} (\gamma_\mu)_\alpha^\beta P^\mu$$

$$P_L \{Q_\alpha, \overline{Q}^\beta\} = -\frac{1}{2} (P_L \gamma_\mu)_\alpha^\beta P^\mu$$

$$P_L \left[ (P_L Q_\alpha + P_R Q_\alpha) (\overline{Q}^\alpha P_R + \overline{Q}^\alpha P_L) \right] = ( ) ( )$$

$$P_L Q_\alpha Q^\alpha P_R + P_L (Q^\alpha P_R Q_\alpha + Q^\alpha P_L Q_\alpha) =$$

$$\therefore \{P_L Q_\alpha, (P_R Q)^\beta\} = -\frac{1}{2} (P_L \gamma_\mu)_\alpha^\beta P^\mu$$

N:  $i=1, \dots, N$   $Q_\alpha^i$ , we use  $i$  index to denote chirality.

$$Q_i = P_L Q_i \quad | \quad Q^i = P_R Q^i$$

i. Majorana spinor:  $\psi = Q^i + Q_i$

$$Q_i = (P_L Q_i) = \bar{Q}_i P_L \quad | \quad \bar{Q}^i = (P_R Q^i) = \bar{Q}^i P_R$$

$$[M_{ab}, Q_\alpha^i] = -\frac{1}{2} (\gamma_{ab} Q^i)_\alpha$$

$$[D, Q_\alpha^i] = \frac{1}{2} Q_\alpha^i$$

$$[U_i^i, Q_\alpha^k] = \delta_i^k Q_\alpha^i - \frac{1}{2} \delta_i^j Q_\alpha^k$$

$$[U_i^j, Q_{\alpha k}] = -\delta_k^j Q_{\alpha i} + \frac{1}{2} \delta_i^j Q_{\alpha k}$$

$$D=4: [T_R, Q_\alpha^i] = i \frac{1}{2} Q_\alpha^i$$

$$[K_a, Q_\alpha^i] = (\gamma_a S^i)_\alpha$$

$$D=5: [K_a, Q_\alpha^i] = i (\gamma_a S^i)_\alpha$$

$$D=6: [K_a, Q_\alpha^i] = -(\gamma_a S^i)_\alpha$$

$$[M_{ab}, S_\alpha^i] = -\frac{1}{2} (\gamma_{ab} S^i)_\alpha$$

$$[D, S_\alpha^i] = -\frac{1}{2} S_\alpha^i$$

$$[U_i^i, S_\alpha^k] = \delta_i^k S_\alpha^i - \frac{1}{2} \delta_i^j S_\alpha^k$$

$$[U_i^j, S_{\alpha k}] = \delta_k^j S_{\alpha i} - \frac{1}{2} \delta_i^j S_{\alpha k}$$

$$[T, S_\alpha^i] = \frac{i}{2} S_\alpha^i$$

$$[P_a, S_\alpha^i] = (\gamma_a Q^i)_\alpha$$

$$[P_c, S_\alpha^i] = -i (\gamma_c Q^i)_\alpha$$

$$[P_c, S_\alpha^i] = -(\gamma_a Q^i)_\alpha$$

Proof:  $e^{-\frac{i}{4} \lambda^{ab} M_{ab}} Q_\alpha e^{\frac{i}{4} \lambda^{ab} M_{ab}} = Q^\alpha = e^{\frac{1}{2} \gamma^{ab} \Sigma_{ab}} Q_\alpha$

$$(1 - \frac{i}{4} \lambda^{ab} M_{ab}) \hat{Q}_\alpha (1 + \frac{i}{4} \lambda^{ab} M_{ab})$$

$$(\hat{Q}_\alpha - \frac{i}{4} \lambda^{ab} M_{ab} \hat{Q}_\alpha) (1 + \frac{i}{4} \lambda^{ab} M_{ab}) = e^{\frac{1}{2} \gamma^{ab} \Sigma_{ab}} Q_\alpha$$

$$\hat{Q}_\alpha - \frac{i}{4} \lambda^{ab} M_{ab} \hat{Q}_\alpha + \frac{i}{4} \hat{Q}_\alpha \lambda^{ab} M_{ab} = Q_\alpha + \frac{i}{4} \gamma^{ab} \Sigma_{ab} Q_\alpha$$

$$\lambda^{ab} [\hat{Q}_\alpha, M_{ab}] = \lambda^{ab} \Sigma_{ab} Q_\alpha$$

$$[Q_\alpha, M_{ab}] = \frac{1}{2} \gamma_{ab} Q_\alpha$$

$$\therefore [M_{ab}, Q_\alpha^i] = -\frac{1}{2} (\gamma_{ab} Q^i)_\alpha$$

All others are found similarly found.

Int. - commutators are in the text.

Constr. of the Supercant. Alg.: Restrict to  $D=4$

$$S = E^A T_A = \frac{1}{2} \lambda^{ab} M_{ab} + \lambda_0 D + \lambda_a^i K_a + \lambda_i^j U_j + \lambda_T T_R \\ + \bar{E}^i Q_i + \bar{E}_i Q^i + \bar{\eta}^i S_i + \bar{\eta}_i S^i$$

• Spinors are split into left and right-handed chiral  
Impose constraints such that  $\psi_m^{ab}$ ,  $f_m^a$ ,  $\phi_m$  become composite fields.

~ In  $N=1$ : Multiplet w/  $8+8$  off-shell D.O.F. which was the Weyl Multiplet.

~ Counting does not work for  $N \geq 2$ : We do not have an equal number of D & F fields.

Thm: There are equal # of Bosonic & Fermionic D.O.F. in  
[ any realisation of sosy of the form  $\{Q, Q\} = P$   
Thus, we require further Auxiliary fields! The solution is not unique. We work with the "standard Weyl multiplet".

$$(\text{Standard Weyl Multiplet}) \cong \begin{matrix} 24 \\ D \end{matrix} + \begin{matrix} 24 \\ \bar{S} \end{matrix}$$

~ New Auxiliary fields:

$D$ : Real scalar

$T_{ab}$ : 2-form  $\rightarrow T_{[ab]}^+ + T_{[ab]}^-$

$\chi^i$ : Spinor

Parameters and Gauge fields for supercant.  $N=2$  Alg.

|               |                 |             |               |               |             |             |             |
|---------------|-----------------|-------------|---------------|---------------|-------------|-------------|-------------|
| $P_a$         | $M^{ab}$        | $D$         | $K_a$         | $U_i^j$       | $T_R$       | $Q^i$       | $S^i$       |
| $\tilde{g}^a$ | $\lambda^{ab}$  | $\lambda_D$ | $\lambda_k^a$ | $\lambda_i^j$ | $\lambda_T$ | $G_i$       | $N_i$       |
| $e_m^a$       | $\omega_m^{ab}$ | $b_m$       | $f_m^a$       | $V_{mij}$     | $A_m$       | $\psi_{im}$ | $\phi_{im}$ |

## Standard Weyl Multiplet

| field           | # D.O.F.  | $u$            | $c$           | L/R | Gauge-subtracted |
|-----------------|-----------|----------------|---------------|-----|------------------|
| $e_m^a$         | 5         | -1             | 0             |     | $P^a, M^{ab}, D$ |
| $b_m$           | 0         | 0              | 0             |     | $K^a$            |
| $\omega_m^{ab}$ | Composite | 0              | 0             |     |                  |
| $f_m^a$         | Composite | 1              | 0             |     |                  |
| $V_{ij}$        | 9         | 0              | 0             |     | $SU(2)_R$        |
| $A_m$           | 3         | 0              | 0             |     | $U(1)_R$         |
| $T_{ab}$        | 6         | 1              | 1             |     |                  |
| $D$             | 1         | 2              | 0             |     |                  |
| $\psi_a^i$      | 16        | $-\frac{1}{2}$ | $\frac{1}{2}$ | L   | $Q_i, S_i$       |
| $\phi_a^i$      | Composite | $+\frac{1}{2}$ | $\frac{1}{2}$ | R   |                  |
| $\chi^i$        | 8         | $+\frac{1}{2}$ | $\frac{1}{2}$ | L   |                  |

We find similar constraints such as

$$R_{mn}(P^a) = 0 \mid e_p^a \hat{R}_{mn}(M^{ab}) = 0 \mid \gamma^m R_{mn}(Q) = 0$$

which we used for superconformal gravity as well as for

$N=1$  sygn. Here we use:

$$\gamma^b \hat{R}_{ba}(Q^i) + \frac{1}{2} \gamma_a \chi^i = 0$$

$$\hat{R}_{ac}(M^{bc}) + i \tilde{R}_a^b(T) + \frac{1}{4} T_{ca} T + P_c + \frac{3}{2} \delta_a^b D = 0$$

Coefficients chosen for S-susy invariance.

The dilatation and  $U(1)$ -transformations of the fields follow from the weights mentioned above.  $\delta U(1)_R$  are fixed by the  $i$  position:

$$\delta_{SU(4)} \chi^i = \chi^i \lambda_j^i$$

$$\delta_{SO(6)} \chi_i = -\lambda_i^j \chi_j$$

The above is consistent w/ lowering/raising by canon:

$$\lambda_i = (\lambda^i)^c, \quad \lambda^i = -(\lambda_i)^c$$

$$\text{Recall } U U^\dagger = \mathbb{1}$$

$$U^{-1} = (U^\dagger)^T$$

$$U^* U^T = \mathbb{1}$$

$$\bullet (\delta_{SU(4)} \chi^i)^* = (\chi^j)^* (\lambda_j^i)^* = \chi_j (\lambda^{-1})^T = \chi_j \lambda_i^j$$

$$\rightarrow \delta_{SU(4)} \chi_i = \chi_j \lambda_i^j \text{ as above}$$

$$\bullet (\delta_{SO(6)} \chi_i)^* =$$

$\rightarrow \bar{E}_i \chi^i$  is invariant:

$$\delta_{SU(4)} (\bar{E}_i \chi^i) = -\lambda_i^j \bar{E}_j \chi^i + \bar{E}_i \lambda_i^j \chi_j = 0$$

\* For SCTs:  $\delta b_m = 2 \lambda_{km}$ , thus we must specify the Q & S

SUSY:

$$\delta e_\mu^a = \frac{1}{2} \bar{E}^i \gamma^a \psi_{\mu i} + \text{h.c.}$$

$$\delta b_\mu = \frac{1}{2} \bar{E}^i \psi_{\mu i} - \frac{1}{2} \bar{\psi}_\mu \psi_{\mu i} - \frac{3}{8} \bar{E}^i \gamma_\mu \chi_i + \text{h.c.}$$

$$\delta A_\mu = -\frac{1}{2} i \bar{E}^i \psi_{\mu i} - \frac{1}{2} i \bar{\psi}_\mu \psi_{\mu i} - \frac{3}{8} i \bar{E}^i \gamma_\mu \chi_i + \text{h.c.}$$

$$\delta V_{\mu i} = -\bar{E}_i \psi_\mu^i - \bar{\psi}_\mu \psi_\mu^i + \frac{2}{4} \bar{E}_i \gamma_\mu \chi^i + \bar{E}^j \psi_{\mu j} + \bar{\psi}_\mu \psi_{\mu j} - \frac{3}{4} \bar{E}^j \gamma_\mu \chi_j$$

$$- \frac{1}{2} \delta_i^j \left( -\epsilon_{\mu\nu} \psi_\mu^\mu \bar{\psi}_\nu^\mu + \frac{2}{4} \bar{E}_\mu \gamma_\mu \chi^\mu + \bar{E}^\mu \psi_{\mu\mu} + \bar{\psi}_\mu \psi_{\mu\mu} - \frac{3}{4} \bar{E}^\mu \gamma_\mu \chi_\mu \right)$$

$$\delta \psi_\mu^i = \left( \partial_\mu + \frac{1}{2} b_\mu + \frac{1}{4} \gamma^{ab} U_{\mu ab} - \frac{1}{2} i A_\mu \right) \epsilon^i + V_{\mu j}^i \epsilon^j - \frac{1}{16} \gamma^{ab} \bar{\tau}_{ab} \epsilon^{ij} \gamma_\mu \epsilon_j - \gamma_\mu \bar{\psi}^i$$

$$\delta \bar{\tau}_{ab} = 2 \bar{E}^i \hat{R}_{ab}(Q^j) \epsilon_{ij}$$

$$\delta \chi^i = \frac{1}{2} D \epsilon^i + \frac{1}{6} \gamma^{ab} \left[ -\frac{1}{4} \bar{\mathcal{D}} \bar{\tau}_{ab} \epsilon^{ij} \epsilon_j - \hat{R}_{ab}(V_j^i) \epsilon^j + i \hat{R}_{ab}(T) \epsilon^i + \frac{1}{2} \bar{\tau}_{ab} \epsilon^{ij} \gamma_\mu \epsilon_j \right]$$

$$\delta D = \frac{1}{2} \bar{E}^i \not{D} \chi_i + \text{h.c.}$$

Most of these are non-gauge terms as they are "M" terms

$$\begin{aligned}\hat{R}_{mn}^A &= F_{mn}^A - 2\beta_{mn}^P M_{PQ}^A \\ \delta(G) B_n^A &= \partial_n \epsilon^A + \epsilon^C \beta_n^B f_{BC}^A + \epsilon^B M_{n0}^A \\ F_{mn}^A &= 2\partial_{[m} B_{n]}^A + \beta_n^C \beta_m^B f_{BC}^A \\ \delta_{\text{ghost}}(\xi) B_n^A &= \xi^\nu \hat{R}_{\nu n}^A\end{aligned}$$

Here we have  $\hat{R}_{ab}(Q^i)$  for  $SU(2)$  and for  $U(1): \hat{R}_{ab}(U^i)$  which can be split into pure gauge parts  $r_{mn}$

$$\begin{aligned}\hat{R}_{mn}(Q^i) &= \tilde{R}_{mn}^i(Q^i) - 2\gamma_{[m}\phi_{n]}^i \\ \hat{R}'_{mn}(Q^i) &= r_{mn}(Q^i) - \frac{1}{8}\gamma^{ab}T_{ab}^c \epsilon^{ij} \gamma_{[m}\phi_{n]}^j \\ \hat{R}_{mn}(U^i) &= r_{mn}(U^i) + \frac{1}{2}\bar{\psi}_{[m}\hat{\gamma}_{n]}\chi^0 - \frac{3}{2}\bar{\psi}_{[m}\gamma_{n]}\chi_i - \frac{3}{4}\delta_i^j \bar{\psi}_{[m}\hat{\gamma}_{n]}\chi^k \\ &\quad + \frac{3}{4}\delta_i^j \bar{\psi}_{[m}\gamma_{n]}\chi_k \\ \hat{R}_{mn}(T) &= r_{mn}(T) - \frac{3}{4}i\bar{\psi}_{[m}\gamma_{n]}\chi_i + \frac{3}{4}i\bar{\psi}_{[m}\hat{\gamma}_{n]}\chi^i\end{aligned}$$

Here we ~~have~~ previously had:

$$\begin{aligned}\omega_\mu^{ab}(e, b, \psi) &= \omega_\mu^{ab}(e, b) + \frac{1}{2}\psi_\mu \gamma^{[a}\psi^{b]} + \frac{1}{4}\bar{\psi}^a \gamma_\mu \psi^b \\ \omega_\mu^{ab}(e, b) &= \omega_\mu^{ab}(e) + 2e_\mu^{[a} e^{b]} \wedge b_\nu\end{aligned}$$

$$\begin{aligned}\omega_\mu^{ab} &= \omega_\mu^{ab}(e, b) + \frac{1}{2}[\bar{\psi}_\mu^i \gamma^{[a}\psi_i^{b]} + \bar{\psi}_{\mu i} \gamma^{[a}\psi^{b]} + \bar{\psi}_i^a \gamma_\mu \psi^b] \\ f_\mu^a &= -\frac{1}{4}\hat{R}_\mu^a + \frac{1}{24}e_\mu^a \hat{R} + \frac{1}{4}i\hat{R}_\mu(T) - \frac{1}{16}T_{ab\mu} T^{ab} - \frac{1}{8}R_\mu^a D \\ \phi_\mu^i &= -\frac{1}{2}\gamma^\nu \tilde{R}_{\mu\nu}^i(Q^i) + \frac{1}{12}\gamma_\mu \gamma^{ab} \tilde{R}_{ab}(Q^i) + \frac{1}{4}\gamma_\mu \chi^i\end{aligned}$$

$U(1)$  covariant Lorentz curvature:

$$\hat{R}_{mn}^{ab} = r_{mn}^{ab} + [\bar{\psi}_{[m} \gamma_{n]} (\frac{3}{4}\gamma^{ab}\chi_i + \hat{R}^{ab}(Q^i)) + \frac{1}{4}\bar{\psi}_{[m} \gamma_{n]} \epsilon_{ij} T^{ab}]_{th.c}$$

where we have subtracted off terms used to solve for

the constraints.

$$F_{mn}^{ab} = 2\partial_m \omega_n^{ab} + 2\omega_m^a \omega_n^b - F_{ab} \gamma^{ab} \phi_{mn}$$

\* The relevant parts of the transformation of the Gauge fields are the non-gauge parts (M):

$$\delta \omega_m^{ab} = \dots - \frac{2}{8} \bar{E}^i \gamma_m \gamma^{ab} \chi_i - \frac{1}{2} \bar{E}^i \gamma_m \hat{R}^{ab}(Q_i) - \frac{1}{4} \bar{E}^i T_{ab}^{\pm} \phi_i + \text{h.c.}$$

$$\delta f_m^a = \dots - \frac{2}{16} e_m^a \bar{E}^i \not{D} \chi_i + \frac{1}{4} \bar{E}^i \gamma_m D_b \hat{R}^{ba}(Q_i) + \text{h.c.}$$

$$\delta \phi_m^i = \dots - \frac{1}{32} \not{D} T_{ab}^{\pm} \gamma^{ab} \gamma_m \bar{E}^i \epsilon_i - \frac{1}{8} \hat{R}_{ab}(U_i) \gamma^{ab} \gamma_m \bar{E}^i \epsilon_i - \frac{1}{8} \hat{R}_{ab}(U_i) \gamma^{ab} \not{D} \bar{E}^i \epsilon_i$$

\* The algebra, after imposing the constraints is a soft alg. in which the aux. fields appear in the structure functions

$$[\delta_Q(\xi_i), \delta_Q(\xi_j)] = \delta_{\text{rot}}(\xi_3^a) + \delta_m(\lambda_3^{ab}) + \delta_k(\Delta_3^a \kappa) + \delta_S(\eta_3^i)$$

$$\xi_3^a = \frac{1}{2} \bar{E}_L^i \gamma^a \epsilon_{ii} + \text{h.c.}$$

$$\lambda_3^{ab} = \frac{1}{4} \bar{E}_1^i \epsilon_2^j T^{ab} \epsilon_{ij} + \text{h.c.}$$

$$\Delta_3^a \kappa = \frac{1}{8} \bar{E}_1^i \epsilon_2^j D_b T^{ba} \epsilon_{ij} + \frac{2}{16} \bar{E}_2^i \gamma^a \epsilon_{ii} D + \text{h.c.}$$

$$\eta_3^i = \frac{3}{4} \bar{E}_1^i \epsilon_2^j \gamma_j$$

These give rise to central charge terms. When the Auxillary fields have non-zero VEV:

$$\langle 0 | D, T^{ab}, | 0 \rangle \neq 0$$

## Conformal Matter Mults

- Restrict to off-shell gauge mult. on on-shell Hyper
- The Global form of these mults were discussed above.

Extra Ingredients:

(i) Weights of the fields under dilatations and  $U(1)_R$

Supercurrent matter multiplets:

Field #  $u$   $c$   $L/R$

(Off-shell GAUGE Mult.)

$X^I$  2 1 1

$A_m^I$  3 0 0

$\tilde{\psi}^I$  3 2 0

$\Omega_i^I$  8  $\frac{1}{2}$   $\frac{1}{2}$  L

(On-shell Hypermult.)

$\phi^X$  4 1 0

$\chi^A$  4  $\frac{1}{2}$   $-\frac{1}{2}$

• Q & S - susy for these multiplets -

$$\delta X^I = \frac{1}{2} \bar{\epsilon}^i \Omega_i^I$$

$$\delta \Omega_i^I = \mathcal{D} X^I \epsilon_i + \frac{1}{4} f^{ab} F_{ab}^I \epsilon_{ij} \epsilon^j + \psi_{ij}^I \epsilon^j + X^J \bar{\chi}^K f_{JK}^I \epsilon_{ij} \epsilon^j + 2 \bar{\chi}^I \psi_i$$

$$\delta A_m^I = \frac{1}{2} \epsilon^{ij} \bar{\epsilon}_i \mathcal{D}_m \Omega_j^I + \epsilon^{ij} \bar{\epsilon}_i \psi_{mj} X^I + \text{h.c.}$$

$$\delta \tilde{\psi}^I = \frac{1}{4} \bar{\epsilon}^{ij} \epsilon_i \mathcal{D} \Omega_j^I - \frac{1}{2} f_{JK}^I \bar{\epsilon}_i \epsilon_j X^J \Omega^{IK} + \text{h.c.}$$

where

$$F_{ab}^{I*} := \hat{F}_{ab}^I - \frac{1}{2} \bar{X}^I T_{ab}$$

$$\hat{F}_{mn}^I \equiv F_{mn}^I + (-\epsilon_{ij} \psi_m^i \psi_n^j \Omega^{Ij} - \epsilon_{ij} \bar{\psi}_m^i \bar{\psi}_n^j \bar{X}^I + \text{h.c.})$$

$$F_{uv}^I = 2 \partial_m A_v^I - [A_m^I, A_v^I]$$

The covariant derivs are given by:

$$\mathcal{D}_m X^I = D_m X^I - \frac{1}{2} \phi_m^i \Omega_i^I$$

$$D_m X^I = (\partial_m - b_m - i A_m^I) X^I + 4 \bar{\chi}^J X^K f_{JK}^I$$

$$\mathcal{D}_m \Omega_i^I = D_m \Omega_i^I - \mathcal{D} X^I \psi_{mi} - \frac{1}{4} f^{ab} F_{ab}^I \epsilon_{ij} \psi_m^j - \psi_m^j \psi_n^j - X^J \bar{\chi}^K f_{JK}^I \epsilon_{ij} \psi_m^j - 2 \bar{\chi}^I \psi_{mi}$$

$$D_m \Omega_i^I = (\partial_m + \frac{1}{4} \omega_m^{ab} \gamma_{ab} - \frac{3}{2} b_m - \frac{1}{2} i A_m^I) \Omega_i^I + V_{mi}^j \Omega_j^I + A_m^J \Omega_i^K f_{JK}^I$$

\* For hypermultiplets, the conformal structure is determined from a closed homothetic Killing vector  $K_D^X$ , which generates the Dilatations

Previously, in the presence of a complex structure, the homothetic Killing vector gives rise to a Killing vector

$$(K_T)^i \equiv (K_D)^j J_j^i$$

For which we found  $\nabla_i (K_D^a J_{aj}) = J_{ij}$   $\leftarrow J_{ij} = -J_{ji}$

$$\nabla_i (K_T)_j + \nabla_j K_{Ti} = 0$$

→ For the present case we have

$$\vec{K}^X := \frac{1}{2} K_D^Y \vec{J}_Y^X$$

These Killing vectors determine  $SU(2)$  transformations of the superconformal alg.

Dilatation @ R-sym  $[SU(2) \times U(1)]_R$ : (Hypermult.)

$$\delta q^X = \lambda_D K_D^X - 2 \vec{X} \cdot \vec{K}^X$$

$$\delta \zeta^A = \left( \frac{3}{2} \lambda_D - \frac{1}{2} i \lambda_T \right) \zeta^A - \zeta^B \omega_{AB} \delta q^X$$

$$\delta \bar{\zeta}_{\bar{A}} = \left( \frac{3}{2} \lambda_D + \frac{1}{2} i \lambda_T \right) \bar{\zeta}_{\bar{A}} - \bar{\zeta}_{\bar{B}} \bar{\omega}^{\bar{B}A} \delta q^X$$

• Scalars are inert under  $U(1)_R$  but transform under  $SU(2)_R$  while the reverse is true for spinors when the covariant terms are ignored.

• The Q-susy formulae are superconf. covariantizations of the previous results

$$\delta q^X = -i \bar{\epsilon}^i \zeta^A f_{iA}^X + i \epsilon_{ij} \bar{\rho}^{\bar{A}B} \bar{\epsilon}_{\bar{A}} \bar{\zeta}_{\bar{B}} f_{ij}^X$$

$$\delta \zeta^A = \frac{1}{2} \epsilon f^{iA} \times \delta q^X \epsilon_i - \zeta^B \omega_{AB} \delta q^X + i \bar{X}^I K_I^X f_{iA}^X \epsilon_{\bar{A}}^i \epsilon^{\bar{A}}$$

By dimensional arguments,  $q$  must be Majorana  
 super-S transformations and  $\bar{z}^A$  is still affected.  
 $\delta_S q^x = 0$

$$\therefore [\delta_Q, \delta_S] q^x = -\delta_S (-i \bar{\epsilon}^i \gamma^A f_{iA}^x + i \epsilon^i \bar{\epsilon}_i \bar{z}_A f_{ip}^x)$$

$$\text{RHS} \sim (\delta_D + \dots - \delta_m - \delta_T + \delta_{SU(2)}) q^x$$

$$= \lambda_D K_D^x - \vec{\lambda} \cdot \vec{K}^x$$

$$\Rightarrow$$

$$\delta_S \bar{z}^A = i f_{iA}^x K_D^x \bar{z}_i$$

Claim:  $SU(2)$  and  $Q$ -susy implies

$$[\delta(\lambda), \delta(\bar{\epsilon})] = \delta(\bar{\epsilon}')$$

Proof: Since  $K_D^x$  is a homothetic Killing vec.

$$\nabla_x K_D^y = \delta_x^y \text{ and of course } \nabla_x \bar{z} = 0$$

$$\text{Lil Claim } \bar{z}_i^j \cdot \bar{z}_y^x f_{xA}^y = 2 \delta_{\mu}^i f_{jA}^x - \delta_i^j f_{\mu A}^x$$

Proof: Recall that

$$2 f_{\mu}^i f_{jA}^x = \delta_{\mu}^i \delta_j^x + \bar{z}_i^j \cdot \bar{z}_y^x$$

$$\therefore 2 f_{\mu B}^y f_{jA}^x f_{iA}^y = f_{\mu B}^y \delta_j^x \delta_i^y + \bar{z}_i^j \cdot \bar{z}_y^x f_{\mu B}^y$$

$$2 \delta_{\mu}^i \delta_B^A f_{jA}^x = \delta_j^i f_{\mu B}^x + \bar{z}_i^j \cdot \bar{z}_y^x f_{\mu B}^y$$

$$\Rightarrow \bar{z}_i^j \cdot \bar{z}_y^x f_{\mu B}^y = 2 \delta_{\mu}^i f_{jB}^x - \delta_i^j f_{\mu B}^x$$

We now that

$$[U_i^j, Q_a^k] = \delta_i^k Q_a^j - \frac{1}{2} \delta_i^j Q_a^k$$

$$[U_i^j, Q_{a\alpha}] = -\delta_{\alpha}^i Q_{a\alpha} + \frac{1}{2} \delta_i^j Q_{a\alpha}$$

$$\delta(\lambda) = \lambda_i^j U_j^i, \quad \delta(\bar{\epsilon}) = \bar{\epsilon}^i Q_i + \bar{\epsilon}_i Q^i$$

$$[\lambda_i^j U_j^i, \bar{\epsilon}^k Q_k] = \lambda_i^j \bar{\epsilon}^k \left( -\delta_{\alpha}^k Q_j^i + \frac{1}{2} \delta_i^j Q_{\alpha}^k \right)$$

$$= -\bar{\epsilon}^k \lambda_k^i Q_j^i + \frac{1}{2} \lambda_i^j \bar{\epsilon}^k Q_{\alpha}^k$$

But  $\lambda_i^i = 0$  since  $\lambda$  and  $\eta$  are traceless

$$(\lambda_i^a, \epsilon_k Q^a) = \epsilon_k \lambda_i^a Q^a$$

which is exactly as claimed.

\* On scalar fields:

$$\begin{aligned} [\delta(\lambda), \delta(\epsilon)] q^x &= \delta_\lambda(\delta_\epsilon q^x) - \delta_\epsilon(\delta_\lambda q^x) \\ &= \delta_\lambda(-\bar{\epsilon}^i \gamma^A f_{iA}^x + i \epsilon^{ij} \rho^{\bar{A}B} \bar{\epsilon}_i \gamma_{\bar{A}} f_{jB}^x) \\ &\quad - \delta_\epsilon(2 \vec{\lambda} \cdot \vec{k}^x) \\ &= -\bar{\epsilon}^i [\gamma^B \omega_{\gamma B}^A (\lambda_D k_D^x - 2 \vec{\lambda} \cdot \vec{k}^x)] f_{iA}^x \\ &\quad + i \epsilon^{ij} \rho^{\bar{A}B} \bar{\epsilon}_i [-\gamma_{\bar{B}} \bar{\omega}_{\gamma \bar{B}}^A (\lambda_D k_D^x - 2 \vec{\lambda} \cdot \vec{k}^x) f_{jB}^x] \\ &\quad - 2 \vec{\lambda} \cdot (\frac{1}{2} k_D^x \vec{\gamma}^x) \end{aligned}$$

$$\begin{aligned} &= -\bar{\epsilon}^i [\gamma^B \omega_{\gamma B}^A \lambda_D k_D^x - 2 \gamma^B \omega_{\gamma B}^A \vec{\lambda} \cdot \vec{k}^x] f_{iA}^x \\ &\quad - (i \epsilon^{ij} \rho^{\bar{A}B} \bar{\epsilon}_i \gamma_{\bar{B}} \bar{\omega}_{\gamma \bar{B}}^A \lambda_D k_D^x + 2 i \epsilon^{ij} \rho^{\bar{A}B} \bar{\epsilon}_i \vec{\lambda} \cdot \vec{k}^x) f_{jB}^x \\ &\quad - 2 (\lambda_i \vec{\gamma}_i) \cdot (\frac{1}{2} k_D^x \vec{\gamma}^x) \end{aligned}$$

$$\begin{aligned} &= -\bar{\epsilon}^i \gamma^B \omega_{\gamma B}^A \lambda_D k_D^x f_{iA}^x - 2 \bar{\epsilon}^i \gamma^B \omega_{\gamma B}^A (\frac{1}{2} \lambda_k \vec{\gamma}_k) \cdot (\frac{1}{2} k_D^x \vec{\gamma}^x) f_{iA}^x \\ &\quad + \dots \end{aligned}$$

$$\begin{aligned} &\rightarrow -\frac{1}{2} \bar{\epsilon}^i \gamma^B \omega_{\gamma B}^A \lambda_k \vec{\gamma}_k^A \cdot \delta_\omega k_D^x \vec{\gamma}^x f_{iA}^x \\ &= -\frac{1}{2} \bar{\epsilon}^i \gamma^B \omega_{\gamma B}^A (\lambda_k \vec{\gamma}_k^A) \cdot [f_{\omega}^{MD} f_{\gamma}^z] k_D^x \vec{\gamma}^x f_{iA}^x \\ &= -\frac{1}{2} \bar{\epsilon}^i \gamma^B \omega_{\gamma B}^A \lambda_k f_{\omega}^{MD} k_D^x (\vec{\gamma}_k^A f_{\gamma}^z \vec{\gamma}^x) f_{iA}^x \\ &= -\frac{1}{2} \bar{\epsilon}^i \gamma^B \omega_{\gamma B}^A \lambda_k f_{\omega}^{MD} k_D^x f_{iA}^x (2 \delta_m^k f_{\gamma}^z - \delta_{\gamma}^k f_{\gamma}^z) \\ &= -\bar{\epsilon}^i \gamma^B \omega_{\gamma B}^A \lambda_m f_{\omega}^{MD} k_D^x f_{iA}^x f_{\gamma}^z \end{aligned}$$

Note, we should still have the "long" expression for  $\delta_\epsilon q^x$  however we replace  $\epsilon$  with  $\epsilon'$ . In particular, from  $\nabla \gamma f$  we have an expression for  $\omega_{\gamma A}^B$  which we must substitute.

$$D_\mu q^X = \partial_\mu q^X + i \psi_\mu^i \bar{\lambda}^A f_{iA}^X - i \epsilon^{\mu\nu\rho\sigma} \psi_\mu^i \bar{\lambda}_\nu^A f_{iA}^X + i \bar{\lambda}^A f_{iA}^X$$

$$D_\mu q^X = \partial_\mu q^X - b_\mu K_D^X + 2 \vec{V}_\mu \cdot \vec{K}^X - \Delta_\mu^I K_I^X$$

$$\hat{D}_\mu \bar{\lambda}^A = \partial_\mu \bar{\lambda}^A - \frac{1}{2} i f_{iA}^X \not{D} q^X \psi_{\mu i} - i \bar{\lambda}^X K_I^X f_{iA}^X \epsilon_{ij} \psi_{\mu i} - i f_{iA}^X K_D^X \psi_{\mu i}$$

$$\hat{D}_\mu \bar{\lambda}^A = (\partial_\mu + \frac{1}{4} \omega_\mu^{ab} \gamma_{ab} - \frac{3}{2} b_\mu + \frac{1}{2} i A_\mu) \bar{\lambda}^A - \Delta_\mu^I \bar{\lambda}^A \bar{\lambda}^B + \partial_\mu q^X K_X^A \bar{\lambda}^B$$

$\hat{D}$  are also covariant wrt. target space transformations.

\* The alg. closes on scalars with the additional central charge terms. However the alg. does not close on the spinors. We find the extra terms are proportional to the field equations.

→ This addition fixes the action of the theory up to a proportionality  $f^u$ . The components of this  $f^u$  are  $\epsilon^{\mu\nu} \lambda_\nu$ !

## Supercentral Actions:

The actions are the covariantization of

$\mathcal{L}_{\text{gr}}$ : Gauge (from above)

$\mathcal{L}_{\text{hif}}$ : Hypermultiplet (from above)

→ Must include invariance under Dilations: all other symmetries follow

Since  $X^I: h_X = 1, \bar{\psi}^{\pm}, h_{\bar{\psi}} = 2$

then from the  $\mathcal{L}_{\text{gr}}$  action we have the term

$$F_{I5} \bar{\psi}^I \psi^5 \sim \frac{\partial F}{\partial (X^I X^5)} \bar{\psi}^I \psi^5 \Rightarrow (4-2) = h_F$$

$$\therefore h_F = 2$$

$$X^I \partial_I F(X) = 2 F(X)$$

This corresponds to zero weight for  $F_{IJ}$ .

We then define

$$N_{IJ} \equiv 2 \operatorname{Im}(F_{IJ}) = -i F_{IJ} + i \bar{F}_{IJ}$$
$$N \equiv N_{IJ} X^I X^J$$

$F_{IJ}$  also has zero chiral weight so  $N$  is consistent.

$N$ : weight  $h_w = 2$   
chiral  $h_c$ ??

For  $N=2$  there is no barred & unbarred  $I$ -index.

$$\Rightarrow \frac{\partial}{\partial X^I} \frac{\partial}{\partial \bar{X}^J} N = N_{IJ}$$

• From  $X^I \partial_I F = 2F$ :

$$\partial_\mu F + X^I \partial_\mu \partial_I F = 2 \partial_\mu F \Rightarrow X^I \partial_\mu \partial_I F = \partial_\mu F$$

$$\partial_\mu \partial_\nu F + X^I F_{I\mu\nu} = F_{\mu\nu} \Rightarrow X^I F_{I\mu\nu} = 0$$

written differently

$$F_{IJK} X^K = 0$$

\* This is very important to note: All previous expressions for  $N_{IJ}$  are valid however now  $N_{IJ} \rightarrow \hat{N}_{IJ}$   
The previous  $N_{IJ} \equiv \partial_I \partial_J N \rightarrow -i F_{IJK} \bar{X}^K$

We impose all of the same constraints i.e.

S-Gauge  $h_w = 0$ , N-Gauge  $N_I \Omega^I = 0$

etc... w/ added intricacies associated w/  $N=2$ .

Which we write below:

Lagrangian for Gauge Mult:

$$e\mathcal{L}_g^{-1} = -\frac{1}{6}NR - ND - N_{FS}D_m X^I D^m \bar{X}^J + N_{FS}\vec{\psi}^I \vec{\psi}^J + N_{FS}f_{KL}^I \bar{X}^K X^L f_{mn}^J \bar{X}^m X^n \\ + \left\{ -\frac{1}{4}i\hat{F}_{IS}\hat{F}_{n\bar{n}}^I \hat{F}^{+mn5} - \frac{1}{16}N_{FS}X^I X^J T_{ab}^+ T^{+ab} + \frac{1}{4}N_{FS}X^F \hat{F}_{a0}^+ T^{+ab} \right. \\ \left. - \frac{1}{4}N_{FS}\bar{\Omega}^I D\Omega_i^I + \frac{1}{6}N\psi_{in}\gamma^{mnp}D_n\psi_p^i + \dots \right\}$$

(see text).

Covariant der.:

$$D_m \psi_{vi} \equiv \left( \partial_m + \frac{1}{4}\omega_m^{ab}\gamma_{ab} + \frac{1}{2}b_m + \frac{1}{2}iA_m \right) \psi_{vi} + V_{mi}^j \psi_{vj}$$

(plus previous ders)

\* In coupling the hypermult to Gauge mult. in a superconf. setting we require that the Gauge trans. should commute w/ dilatations

$$0 = K_D^Y \partial_Y K_I^X - K_I^Y \partial_Y K_D^X = K_D^Y \nabla_Y K_I^X - K_I^X$$

Proof:

$$[\delta_D, \delta_G] q^X \stackrel{!}{=} 0 \\ \delta_G(\lambda_D K_D^X) = \delta_D(\theta^I K_I^X) \\ = \lambda_D \theta^I K_I^Y \partial_Y K_D^X - K_D^Y \partial_Y K_I^X \lambda_D \theta^I = 0 \\ \Rightarrow K_D^Y \partial_Y K_I^X - K_I^Y \partial_Y K_D^X = 0$$

Now, since these are related to the Killing Eq:

$$\partial_Y \rightarrow \nabla_Y \text{ and } \nabla_Y K_D^X = \delta_Y^X$$

$$K_D^Y \nabla_Y K_I^X - K_I^Y \nabla_Y K_D^X = 0 \Rightarrow K_D^Y \nabla_Y K_I^X - K_I^X = 0$$

\* In particular, we note that  $\vec{u}^X = \frac{1}{2} K_D^Y \vec{S}_Y^X$  generate the  $SU(2)_R$  symmetry of the theory since

$$\delta_{D,R} q^X = \lambda_D K_D^X - 2\vec{\lambda} \cdot \vec{R}^X$$

to see that they commute w/  $SU(2)$  we mult. by  $\vec{S}_Z^X$ :

$$K_D^Y \partial_Y K_I^X - J_X^Z K_I^X = 0$$

$$K_D^Y \vec{J}_Y^Z \nabla_X K_I^Z - \vec{J}_X^Z K_I^X = 0$$

$$\vec{K}^X \nabla_X K_I^Z - \frac{1}{2} \vec{J}_X^Z K_I^X = 0$$

$$\begin{aligned} \text{So, } [\delta_\eta, \delta_\theta] q^k &= \delta_\eta (\theta^I K_I^X) - 2\delta_\theta (\vec{\lambda} \cdot \vec{K}^X) \\ &= \theta^I \vec{\lambda} \cdot (\partial_Y K_I^X \vec{K}^Y) - \vec{\lambda} \cdot (\partial_Y \vec{K}^X) K_I^Y \theta^I = 0 \\ &= \vec{\lambda} \cdot (\partial_Y K_I^X \vec{K}^Y - 2 \partial_Y \vec{K}^X K_I^Y) \\ &= \vec{\lambda} \cdot (\partial_Y K_I^X \vec{K}^Y - 2 \nabla_Y \vec{K}^X K_I^Y) \\ &= \vec{\lambda} \cdot (\nabla_X K_I^Z \vec{K}^X - \nabla_Y K_D^X \vec{J}_X^Z K_I^Y) \\ &= \vec{\lambda} \cdot (\nabla_X K_I^Z \vec{K}^X - \vec{J}_X^Z K_I^X) \end{aligned}$$

off by factor of  $\frac{1}{2}$  However this may come from the replacement  $\partial_Y \rightarrow \nabla_Y$ .  $\square$

\* Thus  $SU(2)_R$  commutes w/ general Gauge transf.

• Weyl-weight of the moment map:

In  $D=4$  the Lagrangian contains the term  $2\vec{P}_I \cdot \vec{\psi}^I$

Since  $\vec{\psi}$  has weyl weight  $w_\psi = 2$  then  $\vec{P}_I$  must also have  $w_{\vec{P}} = 2$

$\rightarrow$  Recall the LHS is  $\bar{e}^i L_i$  where  $w_e = -4$

This fixes the solution of  $\partial_X \vec{P}_I = \vec{J}_X^Y K_{IY}$   
 $\vec{P}_I = \vec{K}_X K_I^X$

Proof: Recall for a closed homothetic Killing vector

$$\partial_X K_D^Y + \Gamma_{XZ}^Y K_D^Z = w \delta_X^Y$$

Since  $g^x$  has  $w=1$ , so too should  $K_D^x$

and so  $\nabla_x K_D^x = S_x^I$  as above

We will show this by simply plugging in this value:

→ Trivial: Since  $\vec{P}_I$  is a "scalar" then

$$\partial_x \vec{P}_I = \nabla_x \vec{P}_I.$$

Superconf. hypermult. action w/ gauged isometries:

$$\tilde{\mathcal{L}}_h = -\frac{1}{12} K_D^2 R + \frac{1}{4} K_D^2 D - \frac{1}{2} g_{\mu\nu} D^\mu g^\nu - 2 \bar{x}^I x^J K_I^x K_{Jx} + 2 \vec{P}_I \cdot \vec{q}^I$$

+ ... (see text).

→ Define:

$$A^i{}_A \equiv f^i{}_x K_D^x \quad | \quad P_{Eij} \equiv P_{Ei}{}^k g_{kj} \quad | \quad P_I{}^{ij} \equiv \epsilon^{ik} P_{Ik}{}^j$$

## Partial Gauge Fixing:

\* For  $N=1$  we had one compensating multiplet, for  $N=2$  we will need two compensating multiplets.

→ Different choices amounts to different Auxiliary fields

Compensating Multiplets:

|                 |                     |            |
|-----------------|---------------------|------------|
| [1] Gauge mult. | $V_{WZ}$            |            |
| [2] Hypermult.  | $H_{comp}$          | [on-shell] |
|                 | "curved" & "curved" |            |

The full Lagrangian is given by the sum of the previous Lagrangians:

$$\mathcal{L} = \mathcal{L}_g + \mathcal{L}_h$$

(i) K-gauge:  $b_\mu = 0$

The geometry for the scalars of the Matter mlt. in SUGRA is obtained as a projective mld from the Conformal invariant embedding mld.

Q: How can we make one projection for both gauge and hypermultiplets since there is only one dilatation generator in  $\text{SupConf}(4k)$ :

• The Aux. fields of the Weyl Mlt. (SUGRA mlt?) are of use:

The scalar terms of  $\mathcal{L}_g$  and  $\mathcal{L}_v$  tell us how to incorporate these Aux fields to define projective coords:

$$\begin{aligned} e^{-1}\mathcal{L} &= -\frac{1}{6}NR - ND - \frac{1}{12}K_0^2 R + \frac{1}{4}K_0^2 D + \dots \\ &= \left(-\frac{1}{6}N - \frac{1}{12}K_0^2\right)R + \left(-N + \frac{1}{4}K_0^2\right)D + \dots \end{aligned}$$

This is the only place where the aux field  $D$  occurs and hence its E.O.M yields

$$-N + \frac{1}{4}K_0^2 = 0 \Rightarrow \boxed{N = \frac{1}{4}K_0^2}$$

$\rightarrow$  D-gauge:  $-\frac{1}{6}N - \frac{1}{12}K_0^2 = \frac{1}{2}K^{-2}$

or

$$\boxed{N = -K^2 \quad \parallel \quad K_0^2 = -4K^2}$$

just as we had before w/  $a = K^{-2}$

• The two minus signs imply for both embedding mlds that there should be multiplets w/ negative kinetic terms. However, since the physical multiplets should have

all positive kinetic terms these embedding mlt's have signature  $(-t, \dots, +)$

R-symmetry Subgroup:  $SU(2)_R \times U(1)_R \cong U(2)$  ← near  $\mathcal{I}_{\text{CFT}}$

→  $U(1)_R$  can be used to fix the phase of the complex scalars of the compensating gauge mlt

→  $SU(2)_R$  used to fix gauge of the quaternions of the compensating hypermlt.

\* Fermionic Sector: For  $N=2$  we have an extra ferm. which is auxiliary  $\chi^i$ :

### Special Geometry:

\* The scalar sector of  $N=2$  sugra define

#### Special Geometry

→ R-symmetry plays an important role for the geometries of the scalars in sugra

$D=5/6$ : R-sym:  $SU(2)$

$D=4$ :  $SU(2)_R \times U(1)_R$

$SU(2)_R$ : Acts on the scalars of the Hypermult.

Which leads to three complex structures on the corresponding mlt's.

$U(1)_R$ : Acts on the complex scalars of the Gauge

mult. in  $D=4$ , whose mfd inherits are  $\mathbb{C}$ -struct.  
 $\rightarrow$  Kähler mfd.

### Family of Special mfd's:

- \* Three types of special mfd's, related to the real scalars of gauge mult. in  $D=5$ .  
 $\mathbb{C}$ -scalars at  $D=4$  of Gauge & quaternionic scalars of hyper.

$D=6$ : No scalars in the Gauge mult.: No geometry?

- For  $N=1$  (i) We had a projective mfd, which was defined by imposing conf. symm. on the global susy theory  
(ii) Included one mult. that contains 'compensating fields': Gauge DOF for the Superconf. theory.  
(Conf inv. theory)  $\xrightarrow{\text{Projection}}$  (Physical theory on CP mfd)

Def: Geometries related to global susy have been called Afine in mathematical literature and Rigid in physics

Def: A Hyper-Kähler Manifold is a special case of "Quaternionic-Kähler mfd's" w/ vanishing  $SU(2)$  curvature.

|                    | D=3 Gauge Mult.                   | D=4 Gauge Mult.             | Hypermult. |
|--------------------|-----------------------------------|-----------------------------|------------|
| Rigid (affine)     | affine<br>Very special real (VSR) | affine<br>special Kähler    | hyper-kä   |
| local (projective) | (projective) VSR                  | (projective) Special Kähler | Quatern-kä |



Def: s-map: Map between a very special real mld to a special Kähler

\* D=3: From D=4 a vector field ( $p=1$ )

$$\{ F^{(p+1)} = (-1)^{p-p} d b^{(D'-p-2)} \}$$

Can be transformed into a scalar. Thus, the dim reduction on the Bosonic part of Gauge mult. can be written in terms of scalars only.  $\rightarrow$  Becomes a Hyper.

C-map: Map between Vector mults & Hyper

IR

IR

$\Gamma$ -map:  $(\text{Geometry}) \rightarrow (\text{Geometry})$

• Fn terms of real dimensions

$$(n-2) \xrightarrow{\text{ramp}} 2n \xrightarrow{\text{C-map}} \mathbb{H}(n+1)$$

Supergrav. mult. in  $D=5$ :

$$(e_a^\mu, \text{gravitini}, \text{graviphoton}, \psi_{\alpha m})$$

$$\begin{array}{ccc} \downarrow & & \downarrow \\ (e_a^\mu \oplus \psi_{\alpha m} \oplus \phi) \oplus (\dots \oplus \phi) \oplus (\psi_{\alpha m} \oplus \chi_\alpha) & D=4 \end{array}$$

Thus, we end up w/ an extra WZ-term  
mult ✓

$$D=4 \rightarrow D=3$$

Same occurs except we have an extra Hyper mult.

### Very Special Real Geometry:

$I=1, \dots, n$ ,  $D=5$  Gauge mult. determined by a symmetric tensor  $C_{IJK}$

Scalars:  $\sigma^I \in \mathbb{R}$

$$\mathcal{L}_{5D} = -\frac{1}{2} C_{IJK} \sigma^K D_\mu \sigma^I D^\mu \sigma^J + \dots$$

So the metric is given by

$$g_{IJ} = C_{IJK} \sigma^K$$

Claim. The curvature of this metric is:

$$R_{FSKL} = -\frac{1}{2} g^{MN} C_N F [K C_L]_{SN}$$

Proof:

$$(i) \partial_N g_{FS} = \partial_N (C_{FSK} \sigma^K)$$

$$= \partial_N C_{FSK} \sigma^K + C_{FSN}$$

$$\Gamma_{SK}^F = \frac{1}{2} g^{IL} (\partial_S g_{LK} + \partial_K g_{LS} - \partial_L g_{SK})$$

$$= \frac{1}{2} g^{IL} ((\partial_S C_{LKM}) + \partial_K C_{LSM} - \partial_L C_{FSM}) \sigma^M$$

$$+ C_{FLK} + C_{KLS} - C_{LFS}$$

$$= \frac{1}{2} g^{IL} [(\partial_S C_{LKM} + \partial_K C_{LSM} - \partial_L C_{FSM}) \sigma^M + C_{FLK}]$$

$$R_{AB}^C = \partial_A \Gamma_{BD}^C - \partial_B \Gamma_{AD}^C + \Gamma_{AE}^C \Gamma_{BD}^E - \Gamma_{BE}^C \Gamma_{AD}^E$$

$$(a) \Gamma_{AE}^C \Gamma_{BD}^E$$

$$= \frac{1}{2} g^{CL} [(\partial_A C_{LEM} + \partial_E C_{LAM} - \partial_L C_{AEM}) \sigma^M + C_{AEL}]$$

$$\times \frac{1}{2} g^{EO} [\partial_B C_{ODU} + \partial_D C_{OBU} - \partial_O C_{BDU}) \sigma^U + C_{BDO}]$$

$$= \frac{1}{4} g^{CL} C_{AEL} g^{EO} C_{BDO} + \dots$$

$$R_{ABCD} = g_{CF} R_{AB}^F D /$$

$$= \frac{1}{4} \sum_C C_{AEL} g^{EO} C_{BDO} - \frac{1}{4} \sum_C C_{BEL} C_{ABO} g^{EO} + \dots$$

$$= \frac{1}{4} (C_{AEL} g^{EO} C_{BDO} - C_{BEL} C_{ABO} g^{EO})$$

$$= \frac{1}{4} (g^{MN}) (C_{AMC} C_{BDN} - C_{BMK} C_{ABN})$$

$$= \frac{1}{2}$$

$$\begin{aligned}
 & A \rightarrow I, B \rightarrow S, C \rightarrow K, D \rightarrow L \\
 & = \frac{1}{4} (g^{MN}) (C_{IMK} C_{S LN} - C_{SMK} C_{I LN}) \\
 & = -\frac{1}{2} g^{MN} \left( \frac{1}{2} \right) (C_N \\
 & \uparrow \text{ off by a } = \text{ sign} \quad \square
 \end{aligned}$$

\* This metric has a closed homothetic Killing vec.

$$\begin{aligned}
 \nabla_i K_j &= h g_{ij} \\
 \text{or } \partial_i K_j^i + \Gamma_{ik}^i K_j^k &= h \delta_j^i
 \end{aligned}$$

Claim: The Killing vector  $K_D^I = \sigma^I$  is a closed homothetic Killing vec. of  $g_{IJ}$ .

$$\begin{aligned}
 \partial_I^S + \sigma^K \frac{1}{2} g^{SL} (C_{LIK} + [\partial_I C_{LKM} + \partial_K C_{LIM} \\
 - \partial_L C_{IKM}] \sigma^M) \\
 \delta_I^S + \frac{1}{2} \delta_I^S + \frac{1}{2} g^{SL} (\sigma^K \sigma^M) [\partial
 \end{aligned}$$

$$\rightarrow \frac{1}{2} g^{SL} \sigma^K \sigma^M (\partial_I C_{LKM} + \partial_K C_{LIM} - \partial_L C_{IKM})$$

$$\begin{aligned}
 g^{IK} C_{SKM} \sigma^M &= \delta_S^I \\
 \sigma^S \partial_N g^{IK} C_{SKM} \sigma^M &= -\partial_N C_{SKM} \sigma^M g^{IK} \sigma^S \\
 &\quad - g^{IK} C_{SKN} \\
 \sigma^S \sigma^M (\partial_N g^{IK}) C_{SKM} &= -\partial_N C_{SKM} \sigma^M \sigma^S g^{IK} \\
 &\quad - g^{IK} C_{SKN} \sigma^S \\
 \sigma^S \sigma^M (\partial_N) C_{SKM} &= -\partial_N C_{SKM} \sigma^M \sigma^S g^{IK} \\
 &\quad - \delta_N^I
 \end{aligned}$$

$$\begin{aligned}
 & = \frac{1}{2} g^{SL} \sigma^K \sigma^M \partial_I C_{LKM} = -\frac{1}{2} \sigma^K \sigma^M (\partial_I g^{SL}) C_{LKM} \\
 & \quad - \frac{1}{2} \delta_I^S + \partial_I ( )
 \end{aligned}$$

Q: Not sure why  
 $\frac{1}{2} g^{\mu\nu} \sigma^{\mu} \sigma^{\nu} (\partial_I C_{\mu\nu} + \partial_{\mu} C_{\nu I} - \partial_{\nu} C_{\mu I})$   
vanishes (anti-sym)

$= \frac{1}{2} g^{\mu\nu} \sigma^{\mu} \sigma^{\nu} \partial_I C_{\mu\nu} ?$

\* With this, we can define a projective mfd by defining a 'radius':

$K_0^2 = -9\kappa^{-2}$      $C_{IJK} \sigma^I \sigma^J \sigma^K = -3\kappa^{-2}$

Similar to  $N = -\kappa^{-2}$  and  $K_0^2 = -4\kappa^{-2}$   
 for  $D=4$  partial Gauge fixing ( $b_m = 0$ )

\* Choose coordinates  $x^{\mu}$  on a  $(n-1)$  dim submfd of  $C_{IJK} \sigma^I \sigma^J \sigma^K$   
 $\rightarrow$  Rescale  $C$  and gauge mult. scalars

$h^I \equiv \kappa \sqrt{\frac{2}{3}} \sigma^I$  ,  $\tilde{C}_{IJK} \equiv -\frac{1}{2\kappa} \sqrt{\frac{3}{2}} C_{IJK}$

s.t.

$\tilde{C}_{IJK} h^I h^J h^K = 1$  

\* If we pull-back to the submfd using

$(\star g)_{xy} =$   
 $(\star g)(X, Y) = g(h_* X, h_* Y) =$   
 $(h_* Y)^i (f^j_{*}) = Y(f \circ h) = Y^{\mu} \frac{\partial}{\partial x^{\mu}} (f \circ h)$   
 $= Y^{\mu} \frac{\partial f^i}{\partial x^{\mu}} \frac{\partial x^i}{\partial x^{\mu}}$   
 $D = \frac{\partial \tilde{C}}{\partial x^{\mu}} \frac{\partial x^{\mu}}{\partial x^{\nu}} \frac{\partial x^{\nu}}{\partial x^{\rho}} g_{IJK}$

$$\begin{aligned}
&= \partial_x \sigma^I \partial_y \sigma^J g_{IJ} \\
&= \frac{1}{k^2} \left(\frac{3}{2}\right) \partial_x h^I \partial_y h^J (C_{IJK} \sigma^K) \\
&= \frac{1}{k^2} \left(\frac{3}{2}\right) \partial_x h^I \partial_y h^J (-2^K) \left(\sqrt{\frac{2}{3}}\right) \tilde{C}_{IJK} \left(\frac{1}{k} \sqrt{\frac{3}{2}} h^K\right) \\
&= -3 k^{-2} (\partial_x h^I \partial_y h^J) \tilde{C}_{IJK} h^K
\end{aligned}$$

Metric on the scalar Mfd

$$g_{xy} = -3 (\partial_x h^I) (\partial_y h^J) \tilde{C}_{IJK} h^K$$

(off by  $k^{-2}$ ?)

\* Very special Real Mfds are not homogeneous mfds. This mfd turns out to be homogeneous for certain choices of  $\tilde{C}$ .

$$\begin{aligned}
\hookrightarrow L(q, p), \quad q &= -1, 0, 1, 2, \dots \\
p &= 0, 1, 2, \dots
\end{aligned}$$

$\rightarrow q = 4m$ :  $\exists$  extra possibilities  
 $L(q, p, \dot{p}) = L(q, \dot{p}, p)$

$\rightarrow \tilde{C}$  is homogeneous for choices w/  $\mathbb{E}$  gamma matrices:

$$\tilde{C}_{IJK} h^I h^J h^K = 3 \{ h^2 (h^2)^2 - h^1 (h^m)^2 - h^2 (h^i)^2 + \gamma_m{}_{ij} h^m h^i h^j \}$$

$$\begin{array}{l}
I = 1, 2, m, i \\
m = 1, \dots, q+2
\end{array} \quad \left| \quad \gamma_m{}_{ij} : \mathbb{E}^{q+2}
\right.$$

\* A general irrep consists of the direct prod. of  $P$  minimal reps if  $q$  is not a multiple of 4

Special Kähler Geometries:

\* Gauge Mult. in  $D=4, N=2$  have complex scalars, the kinetic terms determine Kähler manifolds: Special Kähler mfd's.

→ For Global susy the Kähler potential is obtained by the Pre-potential  $F(X)$ .

$$K = i(X^I \bar{F}_I(\bar{X}) - \bar{X}^{\bar{I}} F_{\bar{I}}(X))$$

↳ This is for specific coordinates  $X^I$  and this relation is not preserved under a change of coordinates.

\* Recall, for Generalized Gauge theory w/  $F_{\mu\nu}^{(I)}$  and  $*F$  had symmetries between the E.O.M and Bianchi. These were symplectic symmetries.

→  $F$  and  $G$  Define a symplectic group.  $f_{AB}$  there depends on the scalars of the Gauge Mult.

$$f^I_J = (C + iDf)(A + iBf)^{-1}$$

## Rigid Special Kähler Geometry:

\* Special Kähler: Kähler & Connection

→ Connection: Connection which preserves the symplectic structure and has zero curvature

$$\underline{R = 0}$$

$$I = 1, \dots, n$$

$$\alpha = 1, \dots, n$$

Rigid Special Kähler: Kähler w/ holo. coords

$z^\alpha$  and 'holomorphic symplectic sections'.

→  $S: M \rightarrow BM \leftarrow$  Bundle over  $M$

$$S(x) \equiv (X^I(x), F_I(x))$$

For which the Group action  
 $G \times BM$  transforms  $X^I(x), F_I$  as  
 vectors under the symplectic trans.  
 if  $' = ''$  above.

$$C_n: \mathfrak{sp}(2n) \quad 0-0-\dots-0 \Rightarrow 0$$

$$M^T \Omega M = \Omega, \quad \Omega_{ab} = -\Omega_{ba}$$

### Mathematical Definitions: $\checkmark$ (RSKM)

Def: A rigid Special Kähler Mfd w/ Kähler form  $\Omega$  and complex structure  $J$  s.t.  $\exists$  a  
 real, flat, torsion-free, symplectic connection  
 $\nabla_s$ , whose connection comp. we denote  $A_{ij}^k$   
 $"(\nabla_s)_i = \partial_i + A_{ij}^k"$

$\rightarrow$  The differential  $d\sigma_s$ , built from  $\nabla_s$  annihilates  
 $J = J_i^j d\phi^i$

Let  $\phi^i$  denotes coordinates on  $M$

$$\phi: \Sigma \rightarrow M$$

Then, define the "1-form"?

$$J^i = d\phi^i J_i^j$$

$$\begin{aligned} d_{\nabla_s}(d\phi^i J_i^j) &= (d_{\nabla_s} J_i^j) \wedge d\phi^i \\ &= (\partial_k J_i^j + A_{kl}^j J_i^l) d\phi^k \wedge d\phi^i \end{aligned}$$

$$i, j = 1, \dots, 2n$$

Kähler form:  $\Omega = -J_{ij} d\phi^i \wedge d\phi^j$

$$J_{ij} = J_i^k g_{kj}$$

Recall  $J_{ij} = -J_{ji}$

\* Holomorphic 1-forms:  $\pi^i$

$$(\pi^i) \gamma_{ij} = i \pi^j$$

as such, we write

$$\pi^i = \pi^i_k d\phi^k$$

$$\pi^i_k d\phi^k \gamma_{ij} = i \pi^j_k d\phi^k$$

$$\pi^i_k \gamma_{ij} = i \pi^j_k = i \delta^j_i \pi^i_k$$

$$-i \gamma^j_i \dots \text{Not sure but}$$

$$\pi^i \equiv \frac{1}{2} (d\phi^j) (\delta^i_j - i \gamma^i_j)$$

\* This def. involves another connection  $\nabla_S$ , i.e. this holo-1-form is exact

(i) Torsion-free:

$$A^k_{ij} = A^k_{ji} \quad \text{i.e. } d\nabla_S(d\phi) = A^i_{jk} d\phi^j \wedge d\phi^k = 0$$

(ii) Symplectic:

$$\partial_i \gamma_{jk} + 2 A_{[i} \gamma^l_{j]k} = 0 \iff \nabla_S \Omega = 0$$

(iii) Flat:

$$\partial_i A^l_{jk} + A^l_{m[i} A^m_{j]k} = 0, \quad R = 0$$

$$(iv) \quad d\nabla_S(d\phi^i \gamma^j_i) = 0$$

$$\partial_i \gamma^j_k + A^k_{[i} \gamma^j_{k]} = 0$$

Proof: (i) trivial.

$$(ii) \quad \nabla_S \Omega = \nabla_S (-\gamma_{ij} d\phi^i \wedge d\phi^j)$$

$$\nabla_S \gamma_{jk} = \partial_i \gamma_{jk} - A^l_{[i} \gamma^l_{j]k} = 0$$

Somehow a factor of 2

$$(iii) \quad R^l_{ijk} = 0,$$

$$\partial_i A^l_{jk} + A^l_{m[i} A^m_{j]k} = 0$$

(iv) This is simply the above expression

$$d\nabla_S = \nabla_m \otimes d\phi^m$$

$$d\nabla_S(d\phi^i \gamma^j_i) = d\phi^a \wedge (\partial_a \gamma^i_j + A^j_{[a} \gamma^i_{b]}) [d\phi^b \gamma^j_i]$$

$$= \frac{1}{2} (\partial_b g_{ij}) \partial^i \partial^j d\varphi^i \wedge d\varphi^j$$

$$\partial_a \Gamma_{ij}^l + A_{ijk}^l \Gamma_{ij}^l = 0 \quad \square$$

example: Let  $(x, y) \in \mathbb{R}^2$  and write the metric

$$ds^2 = (1+y)(dx^2 + dy^2)$$

where

$$\Gamma_{xx}^x = \Gamma_{xy}^x = \Gamma_{yy}^x = \frac{1}{2}(1+y)^{-1}$$

proof:

$$\Gamma_{bc}^a = \frac{1}{2} g^{ad} (\partial_b g_{ac} + \partial_c g_{ab} - \partial_d g_{bc})$$

$$g_{xx} = (1+y), \quad g_{yy} = (1+y)$$

$$\begin{aligned} \Gamma_{bc}^x &= \frac{1}{2} g^{xx} (\partial_b g_{xc} + \partial_c g_{bx} - \partial_x g_{bc}) \\ &= \frac{1}{2} (1+y)^{-1} (\partial_b (1+y) + \partial_c (1+y) - \partial_x g_{bc}) \end{aligned}$$

$$(a) \Gamma_{xy}^x = \frac{1}{2} (1+y)^{-1} (1) = \frac{1}{2} (1+y)^{-1}$$

$$(b) \Gamma_{xx}^x = 0$$

$$(c) \Gamma_{yy}^x = \frac{1}{2} (1+y)^{-1} (2) = (1+y)^{-1}$$

~~~> Not sure why these are not there...~~

$$\begin{aligned} \Gamma_{yy}^x &= \frac{1}{2} g^{xx} (\partial_y g_{xy} + \partial_y g_{xy} - \partial_x g_{yy}) \\ &= 0 \end{aligned}$$

etc...

This manifold has a complex structure, i.e. the metric is Hermitian.

Recall:  $\nabla_k \Gamma_{ij}^l = 0$  is sufficient to ensure that  $N_{ij}^k = 0$

$$\rightarrow J_x^Y = -J_Y^X = 1$$

$$J_{ik}^X J_{\mu}^j = -J_i^j$$

$$\partial_\mu J_i^j + \Gamma_{\mu e}^j J_i^e - \Gamma_{\mu i}^e J_e^j$$

$$\partial_\mu J_i^j = 0, \text{ trivially:}$$

$$\Gamma_{\mu e}^X J_X^e + \Gamma_{\mu e}^Y J_Y^e - \Gamma_{\mu i}^X J_X^j - \Gamma_{\mu i}^Y J_Y^j =$$

$$\Gamma_{XY}^X J_X^Y + \Gamma_{XX}^Y J_Y^X - \Gamma_{Xj}^X J_X^j - \Gamma_{Xj}^Y J_Y^j$$

$$a(1) + (-a)(1) - (a) + \dots = 0.$$

✓

\* Now, the conditions (i)-(iv) are satisfied for a connection

$$A_{Yj}^Y = -A_{Xj}^Y = (1+\gamma)^{-1}$$

Q: How is  $A_{ij}^u$  determined? What existing data can be used to find  $A_{ij}^k$  and to what degree is it unique?

(i) Trivial

$$\text{ii) } J_i^X K_{jk} = J_i^X g_{Xj} + J_i^Y g_{jY}$$

$$= -\delta_{ij} g_{ik} + \delta_{ix} g_{iy}$$

$$= -\delta_{iy} \delta_{ix} (1+\gamma) + \delta_{ix} \delta_{iy} (1+\gamma)$$

$$= (1+\gamma) (\delta_{ix} \delta_{iy} - \delta_{iy} \delta_{ix}) \leftarrow \textcircled{\#}$$

$$J_{ij} = (1+\gamma) ( )$$

$$\partial_Y (J_{jk}) + 2 A_{Yj}^l J_{kl}$$

$$\textcircled{\#} + 2 A_{Yj}^X J_{Xk} + 2 A_{Yj}^Y J_{Yk}$$

$$\textcircled{\#} (\pm) (A_{Yj}^X J_{Xk} - A_{Yk}^X J_{jX})$$

$$J_{jk} = \delta_{jk} - \delta_{ij} \delta_{kr} = \delta_{jk}$$

$$(\star) (\pm) (\delta_{ux} - \delta_{jx})$$

etc....

Claim: Since  $\Gamma_{\mu\nu}^{\rho}(x') = \frac{\partial x'^{\rho}}{\partial x^{\sigma}} \left( \frac{\partial^2 x^{\sigma}}{\partial x'^{\mu} \partial x'^{\nu}} + \frac{\partial x^{\alpha}}{\partial x'^{\mu}} \frac{\partial x^{\beta}}{\partial x'^{\nu}} \Gamma_{\alpha\beta}^{\sigma} \right)$

then, with  $\frac{\partial^2 x^{\sigma}}{\partial x'^{\mu} \partial x'^{\nu}} = 0$  and  $y' = -y + \frac{1}{2}(x^2 - v^2)$

$$A_{\mu\nu}^{\rho}(x') = \frac{\partial x'^{\rho}}{\partial x^{\sigma}} \left( \frac{\partial^2 x^{\sigma}}{\partial x'^{\mu} \partial x'^{\nu}} + \frac{\partial x^{\alpha}}{\partial x'^{\mu}} \frac{\partial x^{\beta}}{\partial x'^{\nu}} A_{\alpha\beta}^{\sigma} \right)$$

$$A_{\mu\nu}^{\rho}(x') = \frac{\partial x'^{\rho}}{\partial y} \left( \frac{\partial^2 y}{\partial x'^{\mu} \partial x'^{\nu}} + \frac{\partial x^{\alpha}}{\partial x'^{\mu}} \frac{\partial x^{\beta}}{\partial x'^{\nu}} A_{\alpha\beta}^{\rho} \right)$$

$$A^x = 0$$

$$A_{\mu\nu}^y = -(1+y) \left( \frac{\partial^2 y}{\partial x'^{\mu} \partial x'^{\nu}} + \frac{\partial x^{\alpha}}{\partial x'^{\mu}} \frac{\partial x^{\beta}}{\partial x'^{\nu}} A_{\alpha\beta}^y \right)$$

$$\rightarrow D(x, x) : -(1+y) \left( \frac{\partial^2 y}{\partial x'^2} + \frac{\partial x^{\alpha}}{\partial x'} \frac{\partial x^{\beta}}{\partial x'} A_{\alpha\beta}^y \right)$$

$$-y' + \frac{1}{2}(x')^2 = y + \frac{1}{2}y^2$$

$$\frac{\partial y}{\partial x^2} x'' = \frac{\partial y}{\partial x'} + \frac{1}{2} \frac{\partial y}{\partial x'}$$

$$1 = \frac{\partial^2 y}{\partial x'^2} + \left( \frac{\partial y}{\partial x'} \right)^2 + \frac{1}{2} \frac{\partial^2 y}{\partial x'^2}$$

$$\left( 1 - \left( \frac{\partial y}{\partial x'} \right)^2 \right) = (1+y) \frac{\partial^2 y}{\partial x'^2}$$

$$\frac{\partial^2 y}{\partial x'^2} = (1 - \left( \frac{\partial y}{\partial x'} \right)^2) / (1+y)$$

(yy) etc  $\frac{\partial x'^{\mu}}{\partial x^{\nu}} \frac{\partial x'^{\nu}}{\partial x^{\mu}} = 1$

$$J_i^j = \frac{\partial x'^j}{\partial x^i} \frac{\partial x'^i}{\partial x^j} J_m^l$$

$$J_{ij} = J_i^k g_{kj} = g'_{ij} \frac{\partial x'^k}{\partial x^i} \frac{\partial x'^j}{\partial x^k} J_m^l$$

$$\left( \frac{\partial x^{\alpha}}{\partial x'^i} \right) \left( \frac{\partial x^{\beta}}{\partial x'^j} \right) \left( \frac{\partial x'^k}{\partial x^{\alpha}} \right) \left( \frac{\partial x'^l}{\partial x'^k} \right) J_m^l$$

$$= \delta_{\alpha}^{\beta} \left( \frac{\partial x^{\beta}}{\partial x'^i} \right) \left( \frac{\partial x'^l}{\partial x'^i} \right) J_m^l g_{\alpha\beta}$$

$$= g_{\alpha\beta} J_m^{\alpha} \left( \frac{\partial x^{\beta}}{\partial x'^i} \right) \left( \frac{\partial x'^l}{\partial x'^i} \right)$$

$$J_{x'y'} = (g_{\alpha\beta} J_m^{\alpha}) \left( \frac{\partial x^{\beta}}{\partial y'} \right) \left( \frac{\partial x'^m}{\partial x'^{x'}} \right)$$

$$\rightarrow J_{x'y'} = 1$$

\* Main properties of the Mfd can be seen from comparing two sets of coordinates:

(i) The  $\mathbb{C}$ -coords  $z^\alpha$  &  $\bar{z}^{\bar{\alpha}}$  for Kähler mfd

s.t. 
$$J = \begin{pmatrix} i\delta_\alpha^\beta & 0 \\ 0 & -i\delta_{\bar{\alpha}}^{\bar{\beta}} \end{pmatrix}$$

(ii) Flat real components. Owing to the flatness of the connection, one can choose a frame w/  $A_{ij}^\mu = 0$  &  $J_{ij}$  is the standard form [i].

$\rightarrow$  In  $z^\alpha$ :  $A_{\alpha\beta}^\gamma = \Gamma_{\alpha\beta}^\gamma$   $\leftarrow$  Kähler metric  
 $A_{\alpha\beta}^{\bar{\gamma}} = g^{\bar{\gamma}\delta} C_{\alpha\beta\delta}(z)$

$C_{\alpha\beta\gamma}(z)$ : Symmetric  $(\alpha\beta\gamma)$  with holomorphic components

$$\nabla_{[\delta} C_{\alpha]\beta\gamma} \equiv \partial_{[\delta} C_{\alpha]\beta\gamma} - \Gamma_{\beta[\delta}^\epsilon C_{\alpha]\epsilon\gamma} - \Gamma_{\gamma[\delta}^\epsilon C_{\alpha]\beta\epsilon} = 0$$

\* Recall

$$\Gamma_{\beta\gamma}^\alpha = g^{\alpha\delta} \partial_\beta g_{\gamma\delta}$$

Then the flatness condition can be split into contributions from  $\Gamma$  and from  $C$ :

$$R_{\beta\alpha\bar{\gamma}}^\delta(\Gamma) = C_{\alpha\beta\epsilon} g^{\epsilon\bar{\epsilon}} \bar{C}_{\bar{\epsilon}\bar{\beta}\bar{\delta}} g^{\bar{\delta}\delta}$$

LHS: metric curvature

\* For real components  $\{\phi_i\}$ :  $2n$

We split these into coordinates

$$\{\phi_i\} = \{x^I, f_I\}, \quad I=1, \dots, n$$

and so we have a symplectic Bundle

$x^I$ : coordinates on Base

$f_I$ : fibre coordinates

On  $T^*M$  we have the Poincaré 1-form

$$\lambda = -2 f_I dx^I$$

$$d\lambda = \Omega = 2 dx^I \wedge df_I$$

\*  $d\phi^j \otimes \delta_j^i$  is closed in a Frame w/ zero  
coeff.  $\Lambda_{ij}^k$ . Hence the 1-forms  $\{\pi^I, \pi_I\}$

are closed

$$\pi^i = \frac{1}{2} (d\phi^j) (\delta_j^i - i \delta_j^{\bar{i}}) \quad \leftarrow \quad \delta_j^{\bar{i}} = i \delta_j^{\bar{i}}$$

$$\pi^I = \left(\frac{1}{2}\right) (dx^I) (\delta_I^I - i \delta_I^{\bar{I}}) \\ = dx^I$$

$$\pi_I = df_I$$

$$\pi_{\bar{I}} = df_I$$

Purely Holomorphic

$\pi^i$ : Holomorphic  $f^u \in \mathbb{C}^n$

$$dx^I = dX^I(z) + d\bar{X}^I(\bar{z})$$

$$df_I = dF_I(z) + d\bar{F}_I(\bar{z})$$

$$X^I(z) \in \mathbb{C}$$

$$x^I \in \mathbb{R}$$

ex: In the flat frame  $(x^I, f_I)$

the latter are  $x$  and  $y$

$\Sigma$

$F$

$$\pi^x = \frac{1}{2} (d\phi^j) (\delta_j^x - i \delta_j^{\bar{x}})$$

$$= \frac{1}{2}(dx - i(-dy)) = \frac{1}{2}(dx + i dy)$$

$$= \frac{1}{2} dz$$

$$\pi^y = \frac{1}{2}(dy - i dx) = -\frac{i}{2} dz$$

$$z = x + i y$$

$$x' = x \Rightarrow \pi^x = \frac{1}{2} dz$$

$$f_1 = -y + \frac{1}{2}(x^2 - y^2)$$

$$df_1 = -dy + x dx - y dy$$

$$= -(1+y)dy + x dx$$

$$\pi_1 = \frac{\partial x}{\partial f_1} \pi_x + \frac{\partial y}{\partial f_1} \pi_y$$

$$\pi_1 = \frac{1}{2}(i + z) dz$$

$$\pi^1 = dx^1 = \frac{\partial x^1}{\partial z} dz = \frac{1}{2} dz$$

$$x^1 = \frac{1}{2} z + (\text{const.})$$

$$df_1 = \frac{1}{2}(i + z) dz$$

$$F_1 = \frac{1}{2}(i z + \frac{1}{2} z^2) + \text{const.}$$

$$F_1 = i(x^1) + (x^1)^2 + \text{const.}$$

Note that we can get rid of this const.

$$F_1 = \frac{\partial}{\partial x^1} \left[ \frac{1}{2} i (x^1)^2 + \frac{1}{3} (x^1)^3 \right]$$



Comparing  $\Omega$  w/  $\Omega = -2i g_{\alpha\bar{\beta}} dz^\alpha \wedge d\bar{z}^\beta$

Note:

$$(dx^1 + d\bar{x}^1) \wedge (dF_1 + d\bar{F}_1)$$

$$= dx^1 \wedge dF_1 + dx^1 \wedge d\bar{F}_1 + d\bar{x}^1 \wedge dF_1 + d\bar{x}^1 \wedge d\bar{F}_1$$

$$= -2i g_{\alpha\bar{\beta}} dz^\alpha \wedge d\bar{z}^\beta$$

→ This is Quadratic in holo. variables and should vanish.

$$dx^I \wedge dF_I = 0 = d\bar{x}^I \wedge d\bar{F}_I$$



$\therefore$

$$-2ig_{\alpha\bar{\beta}} dz^\alpha \wedge d\bar{z}^\beta = d\bar{x}^I \wedge dF_I + dx^I \wedge d\bar{F}_I$$

\* If  $g_{\alpha\bar{\beta}}$  is invertible, then  $X(z)$  can be inverted to obtain  $z(X)$ , and  $F_I(X) = F_I(z(X))$

In this case:  $dF_I(z) = \frac{\partial F_I(z)}{\partial x^J} dx^J(z)$

⊕ Becomes the integrability condition for the existence of a holo f<sup>ld</sup>  $F(X)$ , called the prepotential  $F_I(X) = \frac{\partial F(X)}{\partial x^I}$

### Back to NSUG:

\* Susy makes it so that Vect  $\leftrightarrow$  ferm.  
ferm  $\leftrightarrow$  Scalars

⊂ This is s.t. the symplectic transf. of the vectors are connected are connected to the Scalars sector by

$$V = \begin{pmatrix} X^I \\ \bar{F}_I \end{pmatrix} \Leftrightarrow \begin{pmatrix} F_{mn}^I \\ g_{mnI} \end{pmatrix}$$

We have a symplectic inner product  $\langle V, \bar{V} \rangle \equiv (X^I, \bar{F}_I) \Omega \begin{pmatrix} \bar{X}^I \\ \bar{\bar{F}}_I \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} \bar{X}^I \\ \bar{\bar{F}}_I \end{pmatrix}$

Here  $\Omega = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$

$$V \Omega \bar{V} = \begin{pmatrix} X^I & \bar{F}_I \end{pmatrix} \begin{pmatrix} \bar{\bar{F}}_I \\ -\bar{X}^I \end{pmatrix} = \langle V, \bar{V} \rangle_{\text{or}}$$

$$= X^I \bar{F}_I - F_I \bar{X}^I$$

$$\langle V, \bar{V} \rangle_S = V^T \Omega \bar{V} \equiv X^I \bar{F}_I - F_I \bar{X}^I$$

\* Special Geom. can be defined as a Kähler mfd. w/ metric  $g_{\alpha\bar{\beta}}$  based on  $\rightarrow$  Previously:

$$\begin{aligned} dX^I \wedge dF_I &= 0 \\ \frac{\partial X^I}{\partial z} dz \wedge dF_I &= \frac{\partial X^I}{\partial z} dz \wedge dz \frac{\partial F_I}{\partial z} = 0 \end{aligned}$$

which is similar to

$$\langle \partial_\alpha V, \partial_{\bar{\beta}} V \rangle = 0$$

\* Earlier we found:

$$\begin{aligned} -2i g_{\alpha\bar{\beta}} dz^\alpha \wedge d\bar{z}^\beta &= \frac{\partial X^I}{\partial z^\alpha} dz^\alpha \wedge d\bar{z}^\beta \frac{\partial F_I}{\partial \bar{z}^\beta} \\ &\quad + \frac{\partial \bar{X}^I}{\partial \bar{z}^\beta} d\bar{z}^\beta \wedge dz^\alpha \frac{\partial F_I}{\partial z^\alpha} \end{aligned}$$

$$-2i g_{\alpha\bar{\beta}} dz^\alpha \wedge d\bar{z}^\beta = \left( \frac{\partial X^I}{\partial z^\alpha} \frac{\partial F_I}{\partial \bar{z}^\beta} - \frac{\partial F_I}{\partial z^\alpha} \frac{\partial \bar{X}^I}{\partial \bar{z}^\beta} \right) dz^\alpha \wedge d\bar{z}^\beta$$

$$\therefore -i g_{\alpha\bar{\beta}} = \partial_\alpha \partial_{\bar{\beta}} \langle V, \bar{V} \rangle_S$$

or

$$g_{\alpha\bar{\beta}} = i \partial_\alpha \partial_{\bar{\beta}} \langle V, \bar{V} \rangle_S \quad (1)$$

and so, the Kähler potential is given by

$$\mathcal{K}(z, \bar{z}) = i \langle V, \bar{V} \rangle_S \quad (1')$$

is a symplectic invariant expression.

\* If  $X^I$  are independent variables i.e. if  $\partial_\alpha X^I$  is invertible, then the first condition is the integrability condition for the

local existence of a local prepotential  
 $F(x)$  s.t.  $F_I(x) \equiv \frac{\partial F(x)}{\partial x^I}$

or s.t.  
 $\frac{\partial}{\partial x} dF \neq 0 =$

• If  $g_{\alpha\bar{\beta}} > 0$ ,  $\partial_\alpha x^I$  is invertible and so a prepotential can be defined:

$$g_{\alpha\bar{\beta}} \equiv N_{I\bar{J}} \partial_\alpha x^I \partial_{\bar{\beta}} \bar{x}^{\bar{J}}$$

Since

$$g_{\alpha\bar{\beta}} = \partial_\alpha \partial_{\bar{\beta}} (x^I F_I - F_I \bar{x}^{\bar{I}}) i \\ = i \partial_\alpha x^I \partial_{\bar{\beta}} \bar{x}^{\bar{J}} F_{I\bar{J}} - \partial_\alpha x^{\bar{J}} F_{\bar{J}I} \partial_{\bar{\beta}} \bar{x}^{\bar{I}}$$

where

$$N_{I\bar{J}} \equiv 2 \operatorname{Im}(F_{I\bar{J}}) = -i F_{I\bar{J}} + i \bar{F}_{\bar{J}I}$$

\* In superspace:  
 $Sd^4 x d^2 \theta F + h.c.$

\* Previously, we had  $f_{AB}$ , here  $f_{AB} \rightarrow f_{I\bar{J}}$   
 and for extended susy

$$N_{I\bar{J}} \equiv -i f_{I\bar{J}}$$

Spin 1 Kinetic terms:

$$N = N_{I\bar{J}} x^I \bar{x}^{\bar{J}}$$

Recall

$$\mathcal{L}_1 = \frac{1}{4} e \operatorname{Im}(N_{I\bar{J}}) F_{\mu\nu}^I F^{\mu\nu \bar{J}} - \frac{1}{8} (\operatorname{Re}\{N_{I\bar{J}}\}) \epsilon^{\mu\nu\rho\sigma} F_{\mu\nu}^I F_{\rho\sigma}^{\bar{J}} \\ = \frac{1}{2} e \cdot \operatorname{Im}(N_{I\bar{J}} F^+ I F^+ + h.c.)$$

and so

$$N' = (C + DN)(A + BN)^{-2}, \quad N_{FS} \equiv -i \bar{F}_{FS}$$

The general form

$$\mathcal{L} = -\frac{1}{2} \text{Re} (f_{AB} F_{\mu\nu}^A F^{\mu\nu \cdot B}) = -\frac{1}{2} \text{Im} (F_{\mu\nu}^A G_A^{\mu\nu}) \\ = +\frac{1}{2} \text{Re} (i \bar{N}_{FS} F_{\mu\nu}^A F^{\mu\nu \cdot B})$$

and so  $N_{FS} = \bar{F}_{FS}$

\* A more symplectic invariant def. is

$$\bar{N}_{FS} \partial_\alpha x^5 = \partial_\alpha \bar{F}_I$$

## Projective Special Kähler Mds:

\* Require a Dilatation operator.

$X^I$ : Weyl weight 1  $= h_{X^I}$

$F$ :  $h_F = 2$

$\bar{F}_I$ :  $h_{\bar{F}_I} = 1$

ex: For any Kähler the condition for a closed homothetic Killing vec.

$$\partial_i K_j + \Gamma_{ik}^j K_D^k = h \delta_i^j$$

$$K_D^i = \{K_D^\alpha, \bar{K}_D^{\bar{\alpha}}\}$$

$h_{D0}$ , anti- $h_{D0}$ .

For  $h=1$ ,  $i=\alpha$ ,  $j=B$ :

$$\partial_\alpha K_D^B + \Gamma_{\alpha\gamma}^B K_D^\gamma = h \delta_\alpha^B$$

Since  $K$ :  $\Gamma_{\alpha\gamma}^B = g^{B\bar{\delta}} \partial_\alpha g_{\gamma\bar{\delta}}$

$$\partial_\alpha K_D^B + (g^{B\bar{\delta}} \partial_\alpha g_{\gamma\bar{\delta}}) K_D^\gamma = \delta_\alpha^B$$

$$g_{B\bar{E}} (\partial_\alpha K_D^B) + (\partial_\alpha g_{\gamma\bar{E}}) K_D^\gamma = g_{\alpha\bar{E}}$$

$$\partial_\alpha (g_{\gamma\bar{\delta}} K_D^\gamma) = g_{\alpha\bar{\delta}}$$

Using  $g_{\gamma\bar{\delta}} = i \langle \partial_\gamma V, \partial_{\bar{\delta}} \bar{V} \rangle_S$

$$\partial_\alpha (i \langle \partial_\gamma V, \partial_{\bar{\delta}} \bar{V} \rangle_S K_D^\gamma) = i \langle \partial_\alpha V, \partial_{\bar{\delta}} \bar{V} \rangle_S$$

$$\partial_\alpha \langle K^\gamma \partial_\gamma V, \partial_{\bar{\gamma}} \bar{V} \rangle_s = \langle \partial_\alpha V, \partial_{\bar{\gamma}} \bar{V} \rangle_s$$

$$\langle \partial_\alpha (K^\gamma \partial_\gamma V) - \partial_\alpha V, \partial_{\bar{\gamma}} \bar{V} \rangle_s = 0$$

$$\therefore \partial_\alpha (K^\gamma \partial_\gamma V - V) = 0$$

which can be solved if

$$K_D^\gamma \partial_\gamma V(z) = V(z)$$

Claim: Using  $F_I(x) = \frac{\partial F(x)}{\partial x^I}$  we find

$$K_D^\gamma \partial_\gamma \begin{pmatrix} x^I \\ \bar{F}_I \end{pmatrix} = \begin{pmatrix} x^I \\ \bar{F}_I \end{pmatrix}$$

$$\Rightarrow \begin{cases} K_D^\gamma \partial_\gamma x^I = x^I \\ K_D^\gamma \partial_\gamma \bar{F}_I = \bar{F}_I \end{cases}$$

The Dilatation op  $K_D^\gamma \partial_\gamma = \tilde{D}$

$$\begin{cases} \tilde{D} x^I = h_{x^I} x^I \\ \tilde{D} \bar{F}_I = h_{\bar{F}_I} \bar{F}_I \end{cases}$$

\*For a projective Kähler Geometry:

$$N = -K^{-2}, \quad K_D^2 = -4K^{-2}$$

• Recall for  $N=1$

$$N(x, \bar{x}) = -a, \quad 1/y = -a [z^I \partial_I \bar{z}^{\bar{J}} \bar{z}^{\bar{J}}(z)]^{-1}$$

• Here we have:  $a = -K^{-2}$

We now define projective coordinates:

$$V =: y N(z)$$

$$V = y \begin{pmatrix} z^I(z) \\ \bar{F}^{\bar{I}}(z) \end{pmatrix}$$

$N(z)$ : symplectic vector w/  
 $2(n_v+2)$  components

$$\dim(\mathbb{P}^n) = n_v$$

$$\dim(DS(KN)) = n_v + 1$$

Holomorphic coordinates:  $\{y, z\}$

As before,  $y$  is the direction of the holomorphic Killing vec:

$$\bar{F}_I(y, z) = y \bar{F}_I(z)$$

• If  $\exists F(z)$  is a prepotential:

$$\bar{F}_I(y, z) = \bar{F}_I(X(y, z)) = \bar{F}_I(y z^F(z)) = y \bar{F}_I(z(z))$$

Owing to Homogeneity:

$$F(X) = y^2 F(z)$$

$$\Rightarrow \bar{F}_I(z) = \partial F(z) / \partial z^I$$

Again, Recall  $N = -a$

$$N = x^I G_{I\bar{J}} x^{\bar{J}} = -a$$

$$y \bar{y} (z^I G_{I\bar{J}} \bar{z}^{\bar{J}}) = -a$$

$$K(z, \bar{z}) = -a \ln(-a^{-1} z^I G_{I\bar{J}} \bar{z}^{\bar{J}}) = a \ln(y \bar{y})$$

\* We now take the

$$\text{U(1)-Gauge: } y = \bar{y} \quad , \quad a = K^{-2}$$

$$\therefore y = e^{\frac{(K(z, \bar{z}))}{2}}$$

$$-a \ln(-a z^I N_{I\bar{J}} \bar{z}^{\bar{J}}) = K$$

$$e^{-K^2 K} = -K^2 z^I N_{I\bar{J}} \bar{z}^{\bar{J}} \\ = -K^2$$

$$N = y^2 \langle \underline{N}, \bar{\underline{y}} \rangle \rightarrow = z^I N_{I\bar{J}} \bar{z}^{\bar{J}}$$

$$-K^2 K(z, \bar{z}) = -i(K^2 / N \bar{y})$$

$$\exp\left(\frac{K(z, \bar{z})}{2}\right) = y$$

\* Introduce Kähler transformations

$$y' = e^{f(z)/2} y \quad z'^I = z^I e^{f(z)/2}$$

and Kähler covariant derivs:

$$\begin{aligned} \nabla_\alpha N &\equiv \partial_\alpha N + K^2 (\partial_\alpha K) N \\ \bar{\nabla}_{\bar{\alpha}} \bar{N} &\equiv \partial_{\bar{\alpha}} \bar{N} + K^2 (\partial_{\bar{\alpha}} K) \bar{N} \end{aligned} \quad \begin{aligned} \bar{\nabla}_{\bar{\alpha}} N &\equiv 0 \\ \nabla_\alpha \bar{N} &\equiv 0 \end{aligned}$$

$$\nabla_\alpha y = \nabla_\alpha \bar{y} = 0$$

Similarly, we have

$$\nabla_\alpha x^I = \partial_\alpha x^I + \frac{1}{2} K^2 (\partial_\alpha K) x^I = y \nabla_\alpha z^I$$

Symplectic/Kähler covariant form of metric:

$$g_{\alpha\bar{\beta}} = i \langle \nabla_\alpha N, \bar{\nabla}_{\bar{\beta}} \bar{N} \rangle = i y \bar{y} \langle \nabla_\alpha N, \bar{\nabla}_{\bar{\beta}} \bar{N} \rangle$$

$\Rightarrow$  Note that

$$\begin{aligned} \langle \partial_\alpha N, \partial_{\bar{\beta}} \bar{N} \rangle &= \partial_\alpha x^I \partial_{\bar{\beta}} \bar{F}_I - \partial_{\bar{\beta}} \bar{F}_I \partial_\alpha x^I \\ &= \partial_\alpha x^I \partial_{\bar{\beta}} x^J F_{IJ} - \partial_{\bar{\beta}} x^J \partial_\alpha x^I F_{IJ} \\ &\equiv 0 \end{aligned}$$

Also,

$$\begin{aligned} \langle N, \nabla_\alpha N \rangle &= \langle N, \partial_\alpha N + K (\partial_\alpha K) N \rangle \\ &= \langle N, \partial_\alpha N \rangle + K (\partial_\alpha K) \langle N, N \rangle \end{aligned}$$

$$\langle N, N \rangle = x^I \bar{F}_I - \bar{F}_I x^I = 0$$

$$\begin{aligned} \langle N, \partial_\alpha N \rangle &= x^I \partial_\alpha \bar{F}_I - \bar{F}_I \partial_\alpha x^I \\ &= x^I \partial_\alpha x^J F_{IJ} - \bar{F}_I \partial_\alpha x^I \end{aligned}$$

$$\text{However } F_{IJ} x^I = F_J$$

$$\text{So } \langle N, \partial_\alpha N \rangle = 0$$

$$\therefore \langle N, \nabla_\alpha N \rangle = \langle \nabla_\alpha N, \nabla_{\bar{\beta}} \bar{N} \rangle = 0$$

Reminder: Recall, for  $\mathcal{N}=1$  Super  
Conformal Super-gravity,

$$\begin{cases} g_{\alpha\bar{\beta}} = \gamma\bar{\gamma} \nabla_{\alpha} z^{\bar{I}} G_{\bar{I}\bar{J}} \bar{\nabla}_{\bar{\beta}} \bar{z}^{\bar{J}} \text{ and} \\ \gamma\bar{\gamma} = -u [z^{\bar{I}}(z) G_{\bar{I}\bar{J}} \bar{z}^{\bar{J}}(\bar{z})]^{-1} \end{cases}$$

which combine into the Matrix equation

$$\begin{pmatrix} -u & 0 \\ 0 & g_{\alpha\bar{\beta}} \end{pmatrix} = \gamma\bar{\gamma} \begin{pmatrix} z^{\bar{I}} \\ \nabla_{\alpha} z^{\bar{I}} \end{pmatrix} G_{\bar{I}\bar{J}} \begin{pmatrix} \bar{z}^{\bar{J}} \\ \bar{\nabla}_{\bar{\beta}} \bar{z}^{\bar{J}} \end{pmatrix}$$

For  $\mathcal{N}=2$ ,  $D=4, 5, 6$ , we must have a  
symplectic completion of this form. Here  
we have: Since  $g_{\alpha\bar{\beta}}$  is invertible,  
the  $2(n_v+1) \times (n_v+1)$  matrix

$$\begin{pmatrix} N & \nabla_{\alpha} N \\ \bar{F}_{\bar{I}} & \nabla_{\alpha} \bar{F}_{\bar{I}} \end{pmatrix} = \begin{pmatrix} z^{\bar{I}}(z) & \nabla_{\alpha} z^{\bar{I}}(z) \\ \bar{F}_{\bar{I}}(z) & \nabla_{\alpha} \bar{F}_{\bar{I}}(z) \end{pmatrix}$$

→ This has Rank  $(n_v+1)$

If the upper  $(n_v+1) \times (n_v+1)$  matrix is  
invertible, then a prepotential exists.

→  $\partial_{\alpha} x^{\bar{I}}$  must be invertible.

$$\begin{matrix} [n_v+1 \\ [n_v+1 \end{matrix} \begin{pmatrix} z^{\bar{I}} & \nabla_{\alpha} z^{\bar{I}} \\ \bar{F}_{\bar{I}} & \nabla_{\alpha} \bar{F}_{\bar{I}} \end{pmatrix}$$

→

$\alpha=1, \dots, n_v$

$$(n_v+1) \times (n_v+1) \text{ Matrix} \equiv \begin{pmatrix} z^{\bar{I}} & \nabla_{\alpha} z^{\bar{I}} \end{pmatrix}$$

\* The reason for this formalism is in  
the fact that it appears in the  
expression for the kinetic matrix of  
the prepotential.

vector fields:

$$\bar{N}_{I\bar{J}} = (\bar{F}_I(\bar{z}) \nabla_{\bar{\alpha}} F_I(z)) (\bar{z}^{\bar{J}}(\bar{z}) \nabla_{\bar{\alpha}} \bar{z}^{\bar{J}}(\bar{z}))^{-1}$$

\* If  $F_I$  is derivable from a pre-potential, then the kinetic matrix takes the form:

$$N_{I\bar{J}}(z, \bar{z}) = \bar{F}_{I\bar{J}}(\bar{z}) + i \frac{N_{IK} z^K(z) N_{J\bar{L}} \bar{z}^{\bar{L}}(\bar{z})}{N_{MN} z^M(z) \bar{z}^{\bar{N}}(\bar{z})}$$

$$= \bar{F}_{I\bar{J}} + i \frac{N_{IK} X^K N_{J\bar{L}} X^{\bar{L}}}{N_{MN} X^M X^{\bar{N}}}$$

\* Recall:

$$N_{I\bar{J}} X^I X^{\bar{J}} T_{ab}^+ = 2 N_{I\bar{J}} X^I \hat{F}_{ab}^{+J} - \frac{i}{4} \bar{F}_{I\bar{J}} \bar{\Omega}^{i\bar{J}} \gamma_{ab} \Omega^{iJ} X^K \xi_{Kj} - \bar{J}_{\bar{A}} \gamma_{ab} J_{\bar{B}} \bar{C}^{\bar{A}\bar{B}}$$

$\xi_i$  its c.c.

From this we extract the Bosonic Part  
 $(N_{I\bar{J}} X^I X^{\bar{J}} T_{ab}^+)_B = 2 N_{I\bar{J}} X^I \hat{F}_{ab}^{+J}$

Gauge Mult. Action:

$$e^{\int \mathcal{L}_g} = \dots - \frac{2}{16} N_{I\bar{J}} X^I X^{\bar{J}} T_{ab}^+ T^{+ab} + \frac{i}{4} N_{I\bar{J}} X^I \hat{F}_{ab}^{+J} T^{+ab}$$

$$e^{\int \mathcal{L}_g} = \dots - \frac{i}{8} N_{I\bar{J}} X^I \hat{F}_{ab}^{+J} T^{+ab} + \frac{1}{4} (\dots) + \dots = \dots \frac{1}{8} N_{I\bar{J}} X^I \hat{F}_{ab}^{+J} T^{+ab}$$

Recall,

$$N \equiv N_{I\bar{J}} X^I X^{\bar{J}}$$

$$\hat{F}_{ab}^{+I} = F_{ab}^{+I} + (\text{fermions})$$

$$e^{\int \mathcal{L}_g} = \dots \frac{1}{8} N_{I\bar{J}} X^I F_{ab}^{+J} T^{+ab} + \frac{i}{4} \bar{F}_{I\bar{J}} \hat{F}_{ab}^{+J} \bar{F}^{+ab I}$$

$$T_{ab}^+ = \frac{2 N_{I\bar{J}} X^I \hat{F}_{ab}^{+J}}{X^2} + (\text{fermions}) \leftarrow \text{Ignore}$$

$$-2\gamma = \frac{1}{N} \left( \sqrt{F F^{+J}} / (2 N_{I\bar{J}} X^I F^{+J}) \right)^2 / N$$

$$= \left( \frac{1}{4} \frac{N_{I\bar{J}} X^I N_{KL}}{N} - \frac{1}{4} i \bar{F}_{\bar{J}L} \right) \left( F_{ab}^J F_{cd}^L - \frac{i}{2} \epsilon^{abcd} F_{ab}^I F_{cd}^J \right)$$

\* These are the only combinations for which these terms are real

$$= \frac{1}{4} \frac{N_{I\bar{J}} X^I N_{KL}}{N} X^K (F_{ab}^J F_{cd}^L)$$

$$+ - \frac{1}{8} \bar{F}_{\bar{I}\bar{J}} \epsilon^{abcd} F_{ab}^I F_{cd}^J$$

$$\mathcal{L}_g e^{\bar{\psi}} = \frac{1}{4} \text{Im}(N_{I\bar{J}}) F_{ab}^I F^{ab\bar{J}} - \frac{1}{8} \text{Re}(N_{I\bar{J}}) \epsilon^{abcd} F_{ab}^I F_{cd}^J$$

$$\text{Re}(N_{I\bar{J}}) = \bar{F}_{\bar{I}\bar{J}}$$

$$\text{Im}(N_{I\bar{J}}) = (N_{I\bar{K}} X^K N_{J\bar{L}} X^L) / N$$

\* Note/Q: What happens to the imaginary terms? Do these correspond to the u.c. of this term?

→ The positivity of the K.E. term of the vectors follows from the  $(n_v+2) \times (n_v+2)$  matrix eq:

$$\begin{pmatrix} K^{-2} & 0 \\ 0 & g_{\alpha\bar{\beta}} \end{pmatrix} = i(y\bar{y}) \begin{pmatrix} \langle \tilde{V}, N \rangle & \langle \tilde{N}, \bar{\nabla}_{\bar{B}} N \rangle \\ \langle \nabla_{\alpha} N, N \rangle & \langle \nabla_{\alpha} N, \bar{\nabla}_{\bar{B}} \tilde{N} \rangle \end{pmatrix} \\ = -2(y\bar{y}) \begin{pmatrix} \bar{Z}^I \\ \nabla_{\alpha} Z^I \end{pmatrix} \text{Im}(N_{I\bar{J}}) \begin{pmatrix} Z^J & \bar{\nabla}_{\bar{B}} \bar{Z}^J \end{pmatrix}$$

→ Note, the Kinetics come from

$F_{ab}^I$  ...

derl. {Q: What do we gain from the Dual?} Self coupling?...

→  $\text{Im}(N_{IS}) \equiv (\text{Negative definite if the net is positive definite.})$

Proof:

$$\langle N, \nabla_\alpha N \rangle = \langle \bar{N}, \bar{\nabla}_\alpha \bar{N} \rangle = 0$$

$$\langle \bar{\nabla}_\alpha \bar{N}, \bar{\nabla}_\beta \bar{N} \rangle = \langle \nabla_\alpha N, \nabla_\beta N \rangle = 0$$

So the off diagonal parts are trivially satisfied:

\* Recall  $N = -K^2$

$$N = N_{IS} X^I \bar{X}^{\bar{I}}$$

$$\langle \bar{N}, N \rangle = \bar{Z}^I F_I^2 - \bar{F}_I^2 Z^I$$

Now,

$$N_{IS} = -i F_{IS} + i \bar{F}_{IS}$$

$$\Rightarrow N_{IS} i = F_{IS} - \bar{F}_{IS}$$

$$X^I \bar{X}^{\bar{J}} N_{IS} i = X^I \bar{X}^{\bar{J}} F_{IS} - \bar{F}_{IS} X^I \bar{X}^{\bar{J}}$$

$$-K^{-2} i = \bar{X}^{\bar{J}} F_J - \bar{F}_J X^I$$

$$K^{-2} = i \gamma \bar{\gamma} (\bar{Z}^J F_J - \bar{F}_J Z^I)$$

• The  $g_{\alpha\bar{\beta}} = i \gamma \bar{\gamma} \langle \nabla_\alpha N, \bar{\nabla}_{\bar{\beta}} \bar{N} \rangle$

~ Expand & compare: ....

$$-2 \gamma \bar{\gamma} \nabla_\alpha Z^I \text{Im}(N_{IS}) \bar{\nabla}_{\bar{\beta}} \bar{Z}^{\bar{J}}$$

$$= -2 \nabla_\alpha X^I \bar{\nabla}_{\bar{\beta}} X^{\bar{J}} (N_{IK} X^K N_{JL} \bar{X}^{\bar{L}}) / (N_{MN} X^M \bar{X}^{\bar{N}})$$

$$= -2 \nabla_\alpha X^I \bar{\nabla}_{\bar{\beta}} X^{\bar{J}} ( ) / ( )$$

$$= (-2) (\partial_\alpha X^I + \frac{1}{2} K^2 (\partial_\alpha K) X^I) (\partial_{\bar{\beta}} \bar{X}^{\bar{J}} + \frac{1}{2} K^2 (\partial_{\bar{\beta}} K) \bar{X}^{\bar{J}}) ( ) / ( )$$

\*The curvature of a special Kähler mfd

$$R_{\alpha\bar{\alpha}\beta\bar{\beta}} = 2\kappa^2 g_{\alpha\bar{\alpha}}(g_{\beta\bar{\beta}}) = C_{\alpha\beta\gamma} g^{\gamma\bar{\gamma}} C_{\bar{\alpha}\bar{\beta}\bar{\gamma}}$$

$$R^{\delta}_{\beta\alpha\bar{\gamma}}(\Gamma) = C_{\alpha\beta\epsilon} g^{\epsilon\bar{\epsilon}} C_{\bar{\epsilon}\bar{\beta}\bar{\delta}} g^{\bar{\delta}\delta}$$

The tensor  $C_{\alpha\beta\gamma}$  is given by the connection coeff

$$A^{\alpha}_{\beta\gamma} = \Gamma^{\alpha}_{\beta\gamma}$$

← Kähler

$$A^{\bar{\delta}}_{\bar{\alpha}\bar{\beta}} = g^{\bar{\delta}\gamma} C_{\alpha\beta\gamma}$$

$$C_{\alpha\beta\gamma} = i \langle \nabla_{\alpha} \nabla_{\beta} V, \nabla_{\gamma} V \rangle = i F_{IJK} \nabla_{\alpha} X^I \nabla_{\beta} X^J \nabla_{\gamma} X^K$$

where the second equality comes from the existence of a pre-potential  $F(X)$

$$\nabla_{\alpha} \nabla_{\beta} \nabla_{\gamma} F(X)$$

$$F(Z) = -i\kappa^2 Z^0 Z^1 \leftarrow$$

ex 1 Normalize  $F(X) \rightarrow F(X) \kappa^{-2}$

$$F(X) = -i\kappa^{-2} X^0 X^1$$

→ Here, this model corresponds to only one vac. mult.  $n_v = 1 : Z = Z^1$

• We must choose a parametrization of the upper part of  $V = y V(Z), N(Z) = \begin{pmatrix} Z^I(Z) \\ F_I(Z) \end{pmatrix}$

Choice:  $Z^0 = 1, Z^1 = -iz$

$$\therefore N = \begin{pmatrix} Z^0 \\ Z^1 \\ F_0 \\ F_1 \end{pmatrix} = \begin{pmatrix} 1 \\ -iz \\ -\kappa^{-2} z \\ -i\kappa^{-2} \end{pmatrix}$$

Recall:  $y = e^{\kappa^2 K/2}, e^{-\kappa^2 K(Z, \bar{Z})} = -i\kappa^2 \langle N, \bar{N} \rangle$

$$\langle N, \bar{N} \rangle = Z^I \bar{F}_I - F_I \bar{Z}^I$$

$$= -\kappa^{-2} \bar{z} + (-iz)(i\kappa^{-2})$$

$$= \kappa^{-2} (1 - z\bar{z}) = \kappa^{-2} (1 - |z|^2)$$

$$\begin{aligned}
 &= -K^2 \bar{z} + z K^{-2} + K^{-2} z - K^{-2} \bar{z} \\
 &= K^{-2} 2(z - \bar{z}) \\
 &= 4i K^{-2} \text{Im}(z)
 \end{aligned}$$

$$\rightarrow -i K^2 (4i K^{-2} \text{Im}(z)) = 4 \text{Im}(z)$$

$$\therefore \boxed{e^{-K^2 K(z, \bar{z})} = 4 \text{Im}(z)} \quad \checkmark$$

\*  $g_{\alpha\bar{\beta}}$  : Metric of Scalar Sector  
 $N$  : Metric of Vector Sector

$$-K^2 K(z, \bar{z}) = \ln(4 \text{Im}(z)) \quad (\text{Kähler Potential})$$

$$\boxed{K(z, \bar{z}) = -K^{-2} \ln[4 \text{Im}(z)]}$$

$$\text{Im}(z) = \frac{1}{2i}(z - \bar{z})$$

$$\partial_z K = -K^{-2} \left( \frac{1}{4 \text{Im}(z)} \right) \left( \frac{1}{2i} \right)$$

$$\begin{aligned}
 \partial_{\bar{z}} \partial_z K &= -K^{-2} \left( \frac{1}{4 \text{Im}(z)^2} \right) \left( \frac{1}{2i} \right) \left( \frac{1}{2i} \right) (-1)(4) \\
 &= (2K \text{Im}(z))^{-2}
 \end{aligned}$$

$$\therefore \boxed{g_{z\bar{z}} = \partial_z \partial_{\bar{z}} K(z, \bar{z}) = (2K \text{Im}(z))^{-2}}$$

Domain of Positivity:  $g_{z\bar{z}} \geq 0$  only if  $\text{Im}(z) > 0$

$$\begin{aligned}
 \text{Recall, } g_{\alpha\bar{\beta}} &= N_{\pm 5} \partial_z (y z^{\pm}) \partial_{\bar{z}} (\bar{y} \bar{z}^{\pm}) \\
 &= N_{\pm 5} ((\partial_z y) z^{\pm} + y \partial_z z^{\pm}) ((\partial_{\bar{z}} \bar{y}) \bar{z}^{\pm} + \bar{y} \partial_{\bar{z}} \bar{z}^{\pm})
 \end{aligned}$$

~~⊗~~ Come back to this to show that

$$\boxed{N = K^{-2} \begin{pmatrix} -\bar{z} & 0 \\ 0 & z \end{pmatrix}} \quad (\text{vector metric})$$

Symplectic Trans.:

Where  $S = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -k^2 \\ 0 & 0 & 1 & 0 \\ 0 & k^2 & 0 & 0 \end{pmatrix} \in \text{Sp}(2n, \mathbb{R})$

$$V' = \gamma \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -k^2 \\ 0 & 0 & 1 & 0 \\ 0 & k^2 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ -iz \\ -k^2 z \\ -ik^2 z \end{pmatrix} = \gamma \begin{pmatrix} 1 \\ i \\ -k^{-2} z \\ -ik^2 z \end{pmatrix}$$

Note: here  $(X^{10}, X^{12})$  is independent of  $z$ .

$$\Rightarrow \partial_\alpha X^I = 0$$

and hence, the symplectic vector cannot be obtained from a prepotential.

The Kähler metric  $g_{\alpha\bar{\beta}}$  is a symplectic invariant

$$g(SV, S\bar{V}) = g(V, \bar{V})$$

\* Compute  $N^I$ :

$$(i) \bar{N}^I = (\bar{F}_I(\bar{z}) \quad \nabla_\alpha \bar{F}_I(z)) (\bar{z}^J(\bar{z}) \quad \nabla_\alpha \bar{z}^J(z))^{-I}$$

$$\nabla_\alpha \bar{F}_I(z) = \partial_\alpha \bar{F}_I(z)$$

$$(\bar{F}_I(\bar{z}) \quad \nabla_\alpha \bar{F}_I(z)) = \begin{pmatrix} -k^{-2} \bar{z} & -k^2 \\ ik^{-2} \bar{z} & -ik^2 \end{pmatrix}$$

$$\nabla_\alpha \bar{z}^I = \frac{1}{2} k^2 (\partial_\alpha k) \bar{z}^I$$

$$(\bar{z}^J(\bar{z}) \quad \nabla_\alpha \bar{z}^J(z)) = \begin{pmatrix} 1 & \frac{1}{2} k^2 \left[ \frac{ik^{-2}}{2} \frac{1}{\text{Im}(z)} \right] \\ -i & -\frac{1}{2} k^2 \left[ \frac{k^{-2}}{2} \frac{1}{\text{Im}(z)} \right] \end{pmatrix}$$

$$= \begin{pmatrix} 1 & \left( \frac{1}{4} \right) i \frac{1}{\text{Im}(z)} \\ -i & -\frac{1}{4} \frac{1}{\text{Im}(z)} \end{pmatrix}$$

$$\det(\bar{z}^J(\bar{z}) \quad \nabla_\alpha \bar{z}^J(z)) = -\frac{1}{2 \text{Im}(z)}$$

$$\begin{pmatrix} 1 & \left( \frac{1}{4} \right) i \frac{1}{\text{Im}(z)} \\ -i & -\frac{1}{4} \frac{1}{\text{Im}(z)} \end{pmatrix} \begin{pmatrix} 1 \\ -i \end{pmatrix} = \begin{pmatrix} 1 \\ -\frac{1}{4} \frac{1}{\text{Im}(z)} \end{pmatrix}$$

$$= \begin{pmatrix} \frac{1}{2} & \frac{i}{2} \\ -2i\text{Im}(z) & -2\text{Im}(z) \end{pmatrix}$$

$$\begin{aligned} \bar{N} &= \begin{pmatrix} -K^{-2}\bar{z} & -K^2 \\ iK^{-2}\bar{z} & -iK^{-2} \end{pmatrix} \begin{pmatrix} \frac{1}{2} & \frac{i}{2} \\ 2iI & -2I \end{pmatrix} \\ &= \begin{pmatrix} -\frac{1}{2}K^{-2}\bar{z} + 2iK^2\text{Im}(z) & (-\frac{i}{2})K^{-2}\bar{z} + K^2 2\text{Im}(z) \\ \frac{i}{2}K^{-2}\bar{z} + 2K^{-2}\text{Im}(z) & \end{pmatrix} \end{aligned}$$

Went wrong somewhere...

$$(ii) N' = (C + DN)(A + BN)^{-1}$$

$$\begin{aligned} u &= K^{-2}z \\ \tilde{u} &= K^{-2}\bar{z} \end{aligned}$$

$$S = \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \left( \begin{array}{cc|cc} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -K^2 \\ \hline 0 & 0 & 1 & 0 \\ 0 & K^2 & 0 & 0 \end{array} \right)$$

$$\begin{aligned} (C + DN) &= \begin{pmatrix} 0 & 0 \\ 0 & K^{-2} \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} a & 0 \\ 0 & \tilde{u} \end{pmatrix} \\ &= \begin{pmatrix} 0 & 0 \\ 0 & K^{-2} \end{pmatrix} + \begin{pmatrix} a & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} a & 0 \\ 0 & K^{-2} \end{pmatrix} \end{aligned}$$

$$\begin{aligned} (A + BN) &= \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & -K^2 \end{pmatrix} \begin{pmatrix} u & 0 \\ 0 & \tilde{u} \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & -\tilde{u}K^2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -\tilde{u}K^2 \end{pmatrix} \end{aligned}$$

$$(A + BN)^{-1} = \begin{pmatrix} 1 & 0 \\ 0 & -K^4\tilde{z} \end{pmatrix}$$

$$N' = \begin{pmatrix} K^{-2}z & 0 \\ 0 & K^{-2} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -K^4\tilde{z} \end{pmatrix} = \begin{pmatrix} K^{-2}z & 0 \\ 0 & -K^2\tilde{z} \end{pmatrix}$$

$$\therefore \boxed{N' = -K^{-2}z \mathbb{I}}$$

# Symmetries of Special Kähler mfb's

→ The symmetries of the Special Kähler mfb's of  $N=2, D=4, 5$  sugras are subsets of the symplectic  $Sp(2n_V, \mathbb{R})$  symmetries of the theory

Review of symplectic, generalized, symmetries of

$D=4$  SYM:

\* Kinetic term for vector fields:

$$\mathcal{L}_2 = \frac{1}{2} \text{Im} (N_{IJ} F_{\mu\nu}^+ F^{\mu\nu} + 5_{\mu\nu}) = \frac{1}{2} F_{\mu\nu} (F_{\mu\nu}^+ G_{+I}^{\mu\nu})$$

where  $\mathcal{L}_1 = \mathcal{L}_2(F^+, F^-)$ .

$$G_{+I}^{\mu\nu} \equiv 2i \frac{\partial \mathcal{L}_1}{\partial F_{\mu\nu}^+} = N_{IJ} F^{\mu\nu} + 5_{\mu\nu}$$

• Note: We are setting  $f_{IJ}^R = 0$  (i.e. the theory is abelian) for Gauge group  $G = U(1)^{n+1}$

$$N_{\pm IJ} X^{\pm J} = (\bar{F}_{\pm IJ} + i N_{\pm IJ}) X^{\pm J} = \bar{F}_{\pm IJ} X^{\pm J} = \bar{F}_I$$

$$\text{or } N_{\pm IJ} \nabla_\alpha X^{\pm J} = \bar{F}_{\pm IJ} \nabla_\alpha X^{\pm J} = \nabla_\alpha \bar{F}_I$$

Define:  $\nabla_\alpha \equiv \begin{pmatrix} \nabla_\alpha X^I \\ \nabla_\alpha \bar{F}_I \end{pmatrix} \in Sp(2(n+2), \mathbb{R})$

A symplectic transformation  $v' = SV$ , induces a coordinate transformation on the scalars

$$X^I \mapsto X'^I \equiv A^I_J X^J + B^I_J F_J(X) \quad (\star)$$

$$F_I' = C_{IJ} X^J + D_I^J F_J(X)$$

→ If  $(\star)$  is invertible for  $X^{\pm}(X')$  then

$$\text{we can define } \frac{\partial F_I'}{\partial X'^J} = \frac{\partial X^K}{\partial X'^J} \frac{\partial F_I'}{\partial X^K}$$

$$\frac{\partial F_I'}{\partial X'^J} = \dots \frac{\partial F_I'}{\partial X^J}$$

$$\Rightarrow \partial F / \partial x^s = C \mp J, \quad S = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in \text{Sp}(2n, \mathbb{R})$$

$$S = M = e^T \quad (\text{Lie Alg. of } \text{Sp}(2n, \mathbb{R}))$$

$$S^T \Omega S = \Omega$$

$$(1 + \epsilon T^T) \Omega (1 + \epsilon T) = \Omega$$

$$\Omega + \epsilon T^T \Omega + \Omega \epsilon T = \Omega$$

$$\therefore T^T \Omega + \Omega T = 0$$

$$\text{Let } T = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \Rightarrow T^T = \begin{pmatrix} a^T & c^T \\ b^T & d^T \end{pmatrix}$$

$$\begin{pmatrix} a^T & c^T \\ b^T & d^T \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = 0$$

$$\begin{pmatrix} -c^T & a^T \\ -d^T & b^T \end{pmatrix} + \begin{pmatrix} c & d \\ -a & -b \end{pmatrix} = 0 \Rightarrow c = c^T, \quad c \mp J = c J \mp J \\ a^T = -d, \quad b = b^T$$

$$T = \begin{pmatrix} a & b \\ c & -a^T \end{pmatrix} \text{ w/ } b = b^T, \quad c = c^T, \quad e^T = S \in \text{Sp}(2n, \mathbb{R})$$

With the Lie Alg. in mind, we can look for transformations  $\delta x$  and  $\delta z^\alpha$  consistent w/  $V = y \nabla(z)$

Recall:  $x^F = y z^F$ , and  $h \equiv 2x^F \frac{\partial}{\partial x^F}$

where  $h$  is a Killing vector associated to the homomorphisms (diffeomorphisms) of the theory

$h$ : Defines the action of the Lie group  $\oplus$  dilations  $\oplus$   $T_n$ -transformations.

Hence

$$[h, x^F] \sim \delta x^F$$

$$\begin{aligned} \delta_p z^\alpha &= \theta^A K_A^\alpha(z) \\ \delta y &= a^r \theta^A y r_A(z) \end{aligned}$$

From this we take  $\delta y = K^2 y r(z)$

$$V' = V + \delta V$$

$$\delta V = T V$$

$$\delta V = \delta y w(z) + y \delta w(z) = T(y w(z))$$

$$K^2 y r(z) w(z) + y \delta w(z) = y T w(z)$$

$$\therefore \delta w(z) = T w(z) - K^2 r(z) w(z)$$

Application:

(i) Assume arbitrary  $\delta z^\alpha$

$\rightarrow$  Then, for an arbitrary Sp transf

$$z^0 = 1, z^1 = -iz$$

Then  $N \mapsto N'$

$$N = \begin{pmatrix} 1 \\ -iz \\ -K^2 z \\ -iK^2 \end{pmatrix} \Rightarrow \delta N = \begin{pmatrix} 0 \\ -i\delta z \\ -K^2 \delta z \\ 0 \end{pmatrix} = T N - \underline{K^2 r(z) w(z)}$$

$$\begin{aligned} T N &= \begin{pmatrix} a & b \\ c & a^T \end{pmatrix} \begin{pmatrix} 1 \\ -iz \\ -K^2 z \\ -iK^2 \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} & b_{11} & b_{12} \\ a_{21} & a_{22} & b_{21} & b_{22} \\ c_{11} & c_{12} & a_{11} & a_{21} \\ c_{21} & c_{22} & a_{12} & a_{22} \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{pmatrix} \\ &= \begin{pmatrix} a_{11}v_1 + v_2 a_{12} + b_{11}v_3 + b_{12}v_4 \\ a_{21}v_1 + \dots \\ \dots \end{pmatrix}, T_{ij} \in \mathbb{R} \end{aligned}$$

$$K^2 r(z) w(z) = K^2 r(z) \begin{pmatrix} 1 \\ -iz \\ -K^2 z \\ -iK^2 \end{pmatrix}$$

$$(ii) V' = \delta V = y \begin{pmatrix} 1 \\ -iz \\ -K^2 z \\ -iK^2 \end{pmatrix} = V + \delta V$$

$$\begin{pmatrix} -K^{-2}z \\ -iK^{-2}z \end{pmatrix}$$

$$\therefore \mathcal{N} + \delta \mathcal{N} = \mathcal{N} + T\mathcal{N} = K^{-2}r(z)\mathcal{N}(z)$$

$\Rightarrow$  The symmetry parameters

$$r \equiv \theta^\Sigma r_\Sigma = K^{-2} (i\theta^0 - \frac{1}{2}\theta^2 - \theta^3 z)$$

$$\delta z \equiv \theta^\Sigma K_\Sigma^z = \theta^1 + \theta^2 z + \theta^3 z^2$$

(Killing vectors of  $SU(2,2)$ )

$\Rightarrow$  These are symmetries of the Poincaré Plane

$\theta^0$ : Parameter of a commuting abelian R-supersymmetry generator

\* The symplectic embeddings (via  $Sp(2n+1, \mathbb{R})$ ) are different in the two cases above

i.e.  $\mathcal{N}$  v.c.  $\mathcal{N}' = T\mathcal{N} + \dots$

case (i):  $\mathcal{N}$  has only one perturbative symmetry

case (ii):  $\mathcal{N}'$  has three perturbative symmetries.

$\rightarrow$  These are a reduction from  $D=4, N=4$  Supergravity down to  $N=2$

" $SO(4)$  formulation" [case (i)]

" $SU(4)$  formulation" [case (ii)]

$\rightarrow$  For the  $SU(4)$  formulation we have

$$T = \begin{pmatrix} -\frac{1}{2}\theta^2 & \theta^0 & K^2\theta^3 & 0 \\ -\theta^0 & -\frac{1}{2}\theta^2 & 0 & K^2\theta^3 \\ -K^2\theta^1 & 0 & \frac{1}{2}\theta^2 & \theta^0 \\ 0 & -K^2\theta^1 & -\theta^0 & \frac{1}{2}\theta^2 \end{pmatrix}$$

Claim: For  $N=1, D=4$  super gravity on a Kähler manifold: the Killing vectors take the form  $K_A^I = \gamma [K_A^\alpha \nabla_\alpha \bar{z}^I + i a^{-1} P_A^I \bar{z}^I]$  for moment maps

$$P_A(z, \bar{z}) = i [K_A^\alpha \partial_\alpha K - r(z)]$$

The analogous form for  $N=2$  supergravity is given by

$$T_\Sigma N = K_\Sigma^\alpha \nabla_\alpha N + i K^2 P_\Sigma^0 N, \quad P_\Sigma^0 \equiv -e^{K\bar{K}} \langle T_\Sigma N, \bar{N} \rangle$$

Proof: Here we can take  $\delta N$  to be  $K^\alpha \partial_\alpha N = \delta N$  (as it is a Killing symm)

as well as the  $SU(4)$   $T = \theta^\Sigma T_\Sigma$

$$\therefore K^\alpha \partial_\alpha N = \theta^\Sigma T_\Sigma N - K^2 r(z) N(z)$$

$$K^\alpha \partial_\alpha N = \theta^\Sigma T_\Sigma N - K^2 \theta^\Sigma r_\Sigma N(z)$$

$$K^\alpha = \theta^\Sigma K_\Sigma^\alpha$$

$$K_\Sigma^\alpha \partial_\alpha N = T_\Sigma N - K^2 r_\Sigma N(z)$$

$$\Rightarrow T_\Sigma N = K_\Sigma^\alpha \partial_\alpha N + K^2 r_\Sigma N(z)$$

Note  $K^{-2} = i(\gamma \bar{\gamma}) \langle \bar{N}, N \rangle \leftarrow Sp(2(n+1), \mathbb{R})$  inner product

$$\langle T_\Sigma N, \bar{N} \rangle = K_\Sigma^\alpha \langle \partial_\alpha N, \bar{N} \rangle + K^2 r_\Sigma \langle N, \bar{N} \rangle$$

For a  $U(1)$  gauge condition  $(\gamma = \bar{\gamma})$

$$\gamma = e^{K^2 K/2}, \quad e^{-K^2 K} = -i K^2 \langle N, \bar{N} \rangle$$

$$\therefore \langle N, \bar{N} \rangle = i K^{-2} e^{K^2 K}$$

$$\langle T_\Sigma N, \bar{N} \rangle = K_\Sigma^\alpha \langle \partial_\alpha N, \bar{N} \rangle + K^2 r_\Sigma (i K^{-2} e^{-K^2 K})$$

$$= K_{\Sigma}^{\alpha} \langle \partial_{\alpha} N, N \rangle + i \Gamma_{\Sigma} e^{K^2 K}$$

$$\begin{aligned} \partial_{\alpha} \langle \bar{N}, \bar{N} \rangle &= \langle \partial_{\alpha} N, \bar{N} \rangle = -i e^{K^2 K} \partial_{\alpha} K \frac{K^2}{K^2} \\ &= -K^2 \partial_{\alpha} K \langle N, \bar{N} \rangle \end{aligned}$$

$$\begin{aligned} &= K_{\Sigma}^{\alpha} (-K^2 \partial_{\alpha} K) (i K^{-2} e^{-K^2 K}) + i \Gamma_{\Sigma} e^{-K^2 K} \\ &= (K_{\Sigma}^{\alpha} (\partial_{\alpha} K) + \Gamma_{\Sigma}) i e^{-K^2 K} \\ &\Rightarrow e^{K^2 K} \langle T_{\Sigma} N, \bar{N} \rangle = i [K_{\Sigma}^{\alpha} \partial_{\alpha} K - \Gamma_{\Sigma}] \equiv P_{\Sigma}^0 \quad \checkmark \end{aligned}$$

• Inserting  into  we find,

$$4i P_{\Sigma}^0 + K_{\Sigma}^{\alpha} \partial_{\alpha} K = \Gamma_{\Sigma}$$

$$\begin{aligned} T_{\Sigma} N &= K_{\Sigma}^{\alpha} \partial_{\alpha} N + K^2 [i P_{\Sigma}^0 + K_{\Sigma}^{\alpha} \partial_{\alpha} K] N(z) \\ &= K_{\Sigma}^{\alpha} \partial_{\alpha} N + i K^2 P_{\Sigma}^0 N(z) + K^2 K_{\Sigma}^{\alpha} \partial_{\alpha} K N(z) \\ &= K_{\Sigma}^{\alpha} (\partial_{\alpha} N + K^2 \partial_{\alpha} K N(z)) + i K^2 P_{\Sigma}^0 N(z) \end{aligned}$$

$$\therefore T_{\Sigma} N = K_{\Sigma}^{\alpha} \partial_{\alpha} N + i K^2 P_{\Sigma}^0 N(z)$$

as claimed

\* Special Kähler manifolds which are the dim reduction of (VSRK) have a pre-potential

$$F(x) \equiv \tilde{C}_{ABC} x^A x^B x^C / x^0$$



where  $\{x^{\pm}\} = \{x^0, x^A\}$ , and  $\tilde{C}_{ABC}$  are the coeff. s.t.

$$g_{xy} = -3 (\partial_x h^{\pm}) (\partial_y h^{\mp}) \tilde{C}_{\mp 5 \mu} h^{\mu}$$

$$\text{and } h^{\pm} \equiv K \sqrt{\frac{2}{3}} \sigma^{\pm}$$

$$\tilde{C}_{ABC} h^A h^B h^C = 1$$

• This pre potential describes all Kähler

manifolds of interest.

→ The only other special Kähler mfd's are the dim reduction of pure D=5 supergravity: leading to  $SU(1,1)/U(1)$  or, the non-compact projective versions:

$$\frac{SU(n_v, 1)}{SU(n_v) \times U(1)}$$

→ Obtained by a pre potential  
 $F(x) = i(-x^0 x^0 + x^A x^A)$

\* The non-compact Kähler mfd's which are symmetric spaces are products of irreducible symmetric Kähler spaces:

$$\begin{array}{ccc} \frac{SU(p, q)}{SO(p) \times SO(q)} \times \frac{SU(q)}{U(1)} \times U(1), & \frac{SO^*(2n)}{U(n)}, & \frac{Sp(2n)}{U(n)} \\ \frac{SO(n, 2)}{SO(n) \times SO(2)}, & \frac{E_{6,-14}}{SO(10) \times U(1)}, & \frac{E_{7,-25}}{E_6 \times U(1)} \end{array}$$

\* The only Special Kähler mfd's which are products are those in  $(0, p)$  Kähler ~~entry~~

$$\frac{SU(1, 1)}{U(1)} \otimes \frac{SO(n_v - 1, 1)}{SO(n_v - 1) \times SO(1)}$$

→ These manifolds occur in the classical limit of the compactified Heterotic string. There, the  $\frac{SU(1, 1)}{U(1)}$ : comes from the dilaton-axion

$$Ex: (VSIM) \quad d_{ABC} X^A X^B X^C = X^2 X^a M_{ab} X^b$$

where  $a, b = 2, \dots, n_V$ ,  $M_{ab} = \text{diag}(t, -, \dots, -)$

• In this case, we have only part of the  $SO(n_V - 1, 2)$  belongs to the perturbative symm. (i.e.  $S \in e^T$ )

→ Require  $B \neq 0$  in the duality group to generate the full  $SO(2, n_V - 1)$

→ Find:  $X^I M_{IJ} X^J = 0$ ,  $F_I = S M_{IJ} X^J$

$M_{IJ}$  is a metric for  $SO(2, n_V)$

$S$ : Dilaton

More explicit form:

Special Kähler models of type  $L(0, n_V - 3)$  can be described by the prepotential

$$F(X) = \left( \frac{1}{2X^0} \right) X^2 X^r X^t M_{rt}$$

$$\{X^I\} = \{X^0, X^2, X^r\}, \quad r = 2, \dots, n_V, \quad t = 2, \dots, n_V$$

$$M_{rt} = \text{diag}(t, -, \dots, -)$$

Claim: By choosing suitable coordinates  $S, z^r$  we obtain a symplectic section

$$\mathcal{N} = \begin{pmatrix} 1 \\ S \\ z^r \\ -\frac{1}{2} S M_{rs} z^r z^s \\ \frac{1}{2} M_{rs} z^r z^s \\ S M_{rs} z^s \end{pmatrix}$$

Proof:

$$\{X^I\} = \{X^0, X^2, X^r\}$$

$$X^I = Y Z^I$$

choose:  $Z^0 \equiv 1, Z^1 \equiv S, Z^r \equiv Z^r$

$$F(x) = y^2 F(z) = \left(\frac{1}{2y}\right) (Sy) (yZ^r) (yZ^t) \Omega_{tr} \\ \Rightarrow F(z) = \frac{1}{2} S Z^r Z^t \Omega_{tr}$$

$$\therefore \mathcal{V} = \begin{pmatrix} 1 \\ S \\ Z^r \\ F_0 \\ F_1 \\ F_r \end{pmatrix}$$

$$F_I = \frac{\partial}{\partial Z^I} \left( \frac{1}{2} S Z^r Z^t \Omega_{tr} \right)$$

$$F_0 = \frac{\partial}{\partial Z^0} \left( \frac{1}{2} S Z^r Z^t \Omega_{tr} \right) = -\frac{1}{2} \frac{1}{Z^{0^2}} S Z^r Z^t \Omega_{tr} \\ = -\frac{1}{2} S Z^r Z^t \Omega_{tr}$$

$$F_1 = \frac{1}{2} Z^r Z^t \Omega_{tr}$$

$$F_r = S Z^t \Omega_{tr}$$

$$\Rightarrow \mathcal{V} = \begin{pmatrix} 1 \\ S \\ Z^r \\ -\frac{1}{2} S Z^r Z^t \Omega_{tr} \\ \frac{1}{2} Z^r Z^t \Omega_{tr} \\ S Z^t \Omega_{tr} \end{pmatrix}$$



In this parametrization only  $SO(n_v - 2) \subset SO(n_v + 1, 1)$  is linearly realised (acting on  $x^\mu$ ).

so we can obtain a linear  $SO(n_v - 1, 2)$  symm after a symplectic transf.

\* Consider the symplectic transf. on the components  $\{Z^0, Z^1, F_0, F_1\}$  (leaving  $Z^r$  and  $F_r$  invariant) i.e.  $Sp(2n_v)$   $n_v = 2$

$$\begin{pmatrix} -\frac{1}{2} & 0 & 0 & 1 \\ 1/2 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} Z^0 \\ Z^1 \\ Z^r \\ Z^t \end{pmatrix} = \begin{pmatrix} -\frac{1}{2} Z^0 + F_1 \\ Z^0 \\ Z^r \\ Z^t \end{pmatrix}$$

$$- \begin{pmatrix} 0 & -1/2 & -1 & 0 \\ 0 & -1/2 & 1 & 0 \end{pmatrix} \begin{pmatrix} F_0 \\ F_1 \end{pmatrix} = \begin{pmatrix} z^T + F_1 \\ -\frac{1}{2}z^T - F_0 \\ -\frac{1}{2}z^T + F_0 \end{pmatrix}$$

$$Z'^0 = -\frac{1}{2} + \frac{1}{2} \mathcal{M}_{rs} Z^r Z^s = -\frac{1}{2} (1 - \mathcal{M}_{ij} Z^i Z^j)$$

$$Z'^2 = \frac{1}{2} (1 + \mathcal{M}_{ij} Z^i Z^j)$$

$$\begin{cases} F'_0 = -\frac{1}{2} S - (-\frac{1}{2} S \mathcal{M}_{ij} Z^i Z^j) = -\frac{1}{2} S (1 - \mathcal{M}_{ij} Z^i Z^j) \\ F'_1 = -\frac{1}{2} S - \frac{1}{2} S \mathcal{M}_{ij} Z^i Z^j = -\frac{1}{2} S (1 + \mathcal{M}_{ij} Z^i Z^j) \end{cases}$$

$$F'_0 = S Z'^0 = S \mathcal{M}_{00} Z'^0$$

$$F'_1 = -S Z'^2 = S \mathcal{M}_{11} Z'^2$$

$$\text{where } \mathcal{M}_{IJ} \equiv \text{diag}(+, -, +, -, \dots, -)$$

$$Z'^0 = -\frac{1}{2} (1 - \mathcal{M}_{ij} Z^i Z^j)$$

$$i = 1, \dots, n_V$$

$$Z'^1 = +\frac{1}{2} (1 + \mathcal{M}_{ij} Z^i Z^j)$$

$$Z'^i = Z^i$$

$$F'_I = S \mathcal{M}_{IJ} Z'^J$$

\* This symplectic section is consistent w/ symm transf.  $\delta \mathcal{H}(z) = T \mathcal{H}(z) - K^2 r(z) \mathcal{H}(z)$

$$\left\{ \begin{aligned} a^I_J &= -d^I_J = \Lambda^I_J \\ b &= c = 0 \end{aligned} \right.$$

$$b = c = 0$$

$$K^2 r(z) = (\Lambda^0_J - \Lambda^1_J) Z'^J$$

where  $\Lambda^I_J$  are parameters of  $SO(2, n_V - 2)$

$$\text{i.e. } \mathcal{M}_{IJ} \Lambda^J_K = -\mathcal{M}_{KI} \Lambda^I_J$$

$$\begin{pmatrix} \Lambda & 0 \\ 0 & -\Lambda^T \end{pmatrix} \begin{pmatrix} Z^I \\ F_I \end{pmatrix} = \begin{pmatrix} \Lambda^I_J Z^J \\ -\Lambda^J_I F_J \end{pmatrix}$$

$$-K^2 r(z) = (\Lambda^0_J - \Lambda^1_J) Z'^J \equiv \alpha$$

$$\therefore \delta N = \begin{pmatrix} \Lambda^1_5 \bar{z}^0 + \alpha \bar{z}^1 \\ -\Lambda^0_5 \bar{F}_5 + \alpha \bar{F}_1 \end{pmatrix}$$

$$\bullet \Lambda^0_5 \bar{z}^5 + ((\Lambda^0_5 - \Lambda^1_5) \bar{z}^1) \bar{z}^0$$

$$\Lambda^0_5 \bar{z}^5 + (\Lambda^0_5 - \Lambda^1_5) \bar{z}^1 =$$

$$\Rightarrow \Lambda^0_0 + \Lambda^0_1 S + \Lambda^0_0 \bar{z}^0 - \Lambda^1_2 \bar{z}^1$$

Note:  $N' - N = \begin{pmatrix} -\frac{1}{2}(1 - \eta_{ij} \bar{z}^i \bar{z}^j) \\ \frac{1}{2}(1 + \eta_{ij} \bar{z}^i \bar{z}^j) \\ \bar{z}^i \\ S \bar{z}^0 \\ -S \bar{z}^1 \end{pmatrix} = \begin{pmatrix} 1 \\ S \\ \bar{z}^i \\ -\frac{1}{2} S \eta_{ij} \bar{z}^i \bar{z}^j \\ \frac{1}{2} \dots \end{pmatrix}$

$$\Rightarrow \Lambda^0_0 + \Lambda^0_1 S + \Lambda^0_0 (-\frac{1}{2})(1 - \eta_{ij} \bar{z}^i \bar{z}^j) - \Lambda^1_2 \frac{1}{2} ( )$$

$$* \eta_{\pm 5} \Lambda^\pm_k = -\eta_{k5} \Lambda^\pm_5$$

$$\eta_{\pm 0} \Lambda^0_k + \eta_{\pm 1} \Lambda^1_k = -\eta_{k0} \Lambda^0_\pm - \eta_{k1} \Lambda^1_\pm$$

$$\underline{F=0}: \Lambda^0_k = -\eta_{k0} \Lambda^0_0 - \eta_{k1} \Lambda^1_0$$

$$(i) \underline{k=0}: \Lambda^0_0 = -\Lambda^0_0$$

$$\Rightarrow \Lambda^0_0 = 0$$

$$(ii) \underline{k=1}: \Lambda^0_1 = +\Lambda^1_0$$

$$\underline{I=2}: \Lambda^2_k = \eta_{k1} \Lambda^1_0$$

$$\Lambda^2_1 = \Lambda^1_0$$

$$\Lambda^0_1 S + \Lambda^1_0 (\frac{1}{2})(1 - \eta_{ij} \bar{z}^i \bar{z}^j)$$

$\Rightarrow$  Not sure where  $\Lambda^0_1 S$  vanishes:

$$S = \mathbb{1} + (S - \mathbb{1}) = \mathbb{1} + \begin{pmatrix} 1/2 & 0 & 0 & 1 \\ \eta_2 & 0 & 0 & 1 \\ 0 & -1/2 & -1 & 0 \\ 0 & -\eta_2 & 2 & 0 \end{pmatrix} = \begin{pmatrix} 1 & \infty & 0 \\ 0 & 1 & 0 & 0 \\ \infty & 1 & 0 \\ \infty & \infty & 1 \end{pmatrix}$$

$$= \mathbb{1} + \begin{pmatrix} -3/2 & 0 & 0 & 1 \\ 1/2 & -1 & 0 & 1 \\ \dots & \dots & \dots & \dots \end{pmatrix} = \mathbb{1} + T$$

$$\begin{pmatrix} 0 & -1/2 & -2 & 0 \\ 0 & -1/2 & 2 & -1 \end{pmatrix}$$

$$TN = \begin{pmatrix} \quad \quad \quad \end{pmatrix} \begin{pmatrix} z^0 \\ z^1 \\ F_0 \\ F_1 \end{pmatrix} = \begin{pmatrix} -\frac{3}{2}z^0 + F_2 \\ 1/2 \\ \vdots \end{pmatrix}$$

$$-\frac{3}{2} + \frac{1}{2}g_{ij}z^iz^j + (\Lambda_5^0 - \Lambda_5^1)z^1$$

$$= -3/2 + \frac{1}{2}g_{ij}z^iz^j (\Lambda_0^{z^0} - \Lambda_0^{z^1}) + \Lambda_4^0 z^1$$

$$- \Lambda_1^1 z^1$$

$$= ( ) - \Lambda_0^1 \dots$$

\* For  $n_v = 3$ , there are special properties:

Since  $SO(2,2) = SU(1,1) \times SU(1,1)$ ; this model

$$is \left( \frac{SU(1,1)}{U(1)} \right)^3 \quad (STU\text{-model})$$

$$z^0 = S, z^1 = T, z^2 = U$$